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SUPERSEDING
MIL-HDBK-539A
12 March 2026

DEPARTMENT OF WAR HANDBOOK

DIGITAL ENGINEERING AND MODELING PRACTICES

This handbook is for guidance only.
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FOREWORD

1. This handbook is approved for use by all Departments and Agencies of the Department of War (DoW).

2. This handbook supports DoW's digital engineering (DE) strategy (Figure 1).

a. The initial version contained information on DE practices with terms, definitions, and descriptions to provide guidance for program offices and other interested organizations.

b. This second version contains expanded DE content and incorporates lessons learned by experienced DE organizations and practitioners.

c. Future iterations of the handbook will incorporate new and updated content based on the continued maturation of DE and modeling practices. DoW Components are encouraged to participate in future handbook development.

3. Comments, suggestions, or questions on this publication should be emailed to osd-sea@mail.mil, with the subject line "DE MIL-HDBK 539 Comment."

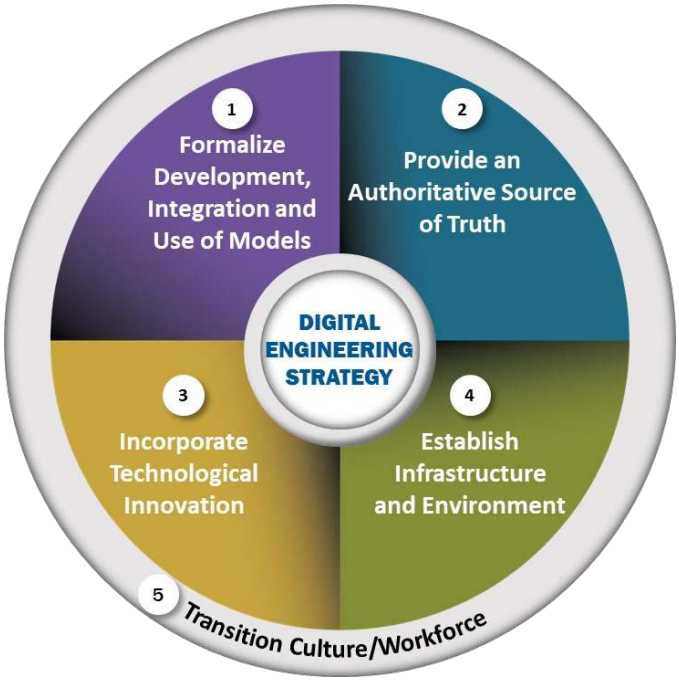


FIGURE 1. DoW Digital Engineering Strategy.

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SUMMARY OF CHANGE 1 MODIFICATIONS

1. Changed Defense Acquisition University (DAU) to Warfighting Acquisition University (WarU) throughout this document.

2. Changed the Uniform Resource Locator (URL) from Defense Acquisition University (dau.edu) to Warfighting Acquisition University (waru.edu) throughout this document.

3. Appendix I, paragraph 3.2 was changed to reference the WarU Digital Acquisition page.

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1. SCOPE

1.1. Scope. This handbook serves the DoW community, including DoW support contractors. This handbook is for guidance only and cannot be cited as a requirement.

1.2. Handbook purpose. This handbook identifies and discusses how digital engineering (DE) and modeling activities can be considered for all disciplines and functional areas within an acquisition life cycle. The handbook also provides guidance to assist users in transitioning systems engineering and modeling activities from document-based activities to digital-based processes.

2. APPLICABLE PUBLICATIONS

2.1. General. The Secretary of War's October 10, 2025, memorandum "Department of War Secondary Titles," directs the use of "Department of War" in lieu of "Department of Defense" unless required by statute. The updated DoW Issuance Style Guide stipulates that current documents using the term "Department of Defense" or "DoD" in their title or text will not be changed until they are updated. The publications listed below are not necessarily all the publications referenced herein but are those needed to understand the information provided by this handbook.

2.2. Government publications.

2.2.1. Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein.

DEPARTMENT OF WAR STANDARDS

MIL-STD-963 - Data Item Descriptions (DIDs)

MIL-STD-31000 - Technical Data Packages (TDPs)

DEPARTMENT OF WAR HANDBOOKS

MIL-HDBK-61 - Configuration Management Guidance

MIL-HDBK-516C - Airworthiness Certification Criteria

(Copies of these publications are available online at https://quicksearch.dla.mil/.)

2.2.2. Other Government publications and drawings. The following other Government publications, drawings, and publications form a part of this document to the extent specified herein.

DEPARTMENT OF WAR DIRECTIVES

DoDD 5000.01 - The Defense Acquisition System

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DEPARTMENT OF WAR INSTRUCTIONS

DoDI 5000.02 - Operation of the Adaptive Acquisition Framework

DoDI 5000.75 - Business Systems Requirements and Acquisition

DoDI 5000.86 - Acquisition Intelligence

DoDI 5000.88 - Engineering of Defense Systems

DoDI 5000.89 - Test and Evaluation

DoDI 5000.91 - Product Support Management for the Adaptive Acquisition Framework

DoDI 5000.93 - Use of Additive Manufacturing in the DoD

DoDI 5000.97 - Digital Engineering

DoDI 5010.44 - Intellectual Property (IP) Acquisition and Licensing

DoDI 8500.01 - Cybersecurity

DEPARTMENT OF WAR MANUALS

DoDM 4120.24 - Defense Standardization Program (DSP) Procedures

DoDM 5010.12 - Acquisition and Management of Contractor-Prepared Data

(Copies of these publications are available online at https://www.esd.whs.mil/DD/DoD-Issuances/

DEPARTMENT OF WAR PUBLICATIONS

Department of Defense Digital Engineering Strategy

(Copies of this publication are available online at https://www.cto.mil/wp-content/uploads/2023/06/Dig-Eng-Strat-2018.pdf

Department of Defense Data, Analytics, and Artificial Intelligence Adoption Strategy

(Copies of this publication are available online at https://media.defense.gov/2023/Nov/02/2003333300/-1/-1/1/DOD_DATA_ANALYTICS_AI_ADOPTION_STRATEGY.PDF

Department of Defense Systems Engineering Guidebook

Systems Engineering Plan (SEP) Outline, Version 4.1

Mission Engineering Guide

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Implementing a Modular Open Systems Approach in DoD Acquisition Programs (MOSA Implementation Guidebook)

Human Systems Integration (HSI) Guidebook

(Copies of these publications are available online at https://www.cto.mil/sea/pg)

Department of Defense Product Support Manager (PSM) Guidebook

Life Cycle Sustainment Plan (LCSP) Outline, Version 3.0

DoD Program Manager's Guidebook for Integrating the Cybersecurity Risk Management Framework (RMF) into the System Acquisition Lifecycle

(Copies of these publications are available online at https://www.waru.edu)

Department of Defense Architecture Framework

(Copies of this publication are available online at https://dodcio.defense.gov/library/dod-architecture-framework/)

Adaptive Acquisition Framework Website at https://aaf.waru.edu

DEPARTMENT OF WAR WEBSITE

Digital Engineering Body of Knowledge (DEBoK) at https://www.de-bok.org/

Warfighting Acquisition University at https://www.waru.edu/

UNITED STATES AIR FORCE PUBLICATIONS

Guidance for e-Program Designations

(Copies of this publication are available online at https://software.af.mil/.)

Air Force Digital Guide

(Copies of this publication are available online with an account at https://guide.dafdto.com/)

UNITED STATES ARMY PUBLICATIONS

Digital Engineering Templates, Best Practices, and Workflows

(Copies of these publications are available online, registration is required at https://armyeitaas.sharepoint.mil.us/sites/ASA-ALT-DASA-DESplaybooks)

UNITED STATES NAVY PUBLICATIONS

US Naval & Marine Corps Digital Systems Engineering Transformation Strategy

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(Copies of this publication are available online at https://nps.edu/documents/112507827/0/2020+Dist+A+DON+Digital+Sys+Eng+Transformation+Strategy+2+Jun+2020.pdf/) or at the Department of the Navy Chief Engineer Bridge (registration required) https://wiki.lift.mhpcc.hpc.mil/confluence/x/_GZMCw

UNITED STATES SPACE FORCE (USSF) PUBLICATIONS

USSF Vision for a Digital Service

(Copies of this publication are available online at https://media.defense.gov/2021/May/06/2002635623/-1/-1/1/USSF%20VISION%20FOR%20A%20DIGITAL%20SERVICE%202021%20(2).PDF.)

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION

NASA Digital Engineering Acquisition Framework Handbook

(Copies of this publication are available online at https://standards.nasa.gov/standard/nasa/nasa-hdbk-1004)

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY PUBLICATIONS

NIST SP 800-154 - Guide to Data-Centric System Threat Modeling

NIST SP 800-53 - Security and Privacy Controls for Information Systems and Organizations

(Copies of these publication are available online at https://www.nist.gov/)

2.3. Non-Government publications. The following publications form a part of this document to the extent specified herein.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Y14.5 - Dimensioning and Tolerancing

ASME Y14.34 - Associated Lists

ASME Y14.41 - Digital Product Definition Data Practices

ASME Y14.47 - Model Organization Practices

ASME Y14.100 - Engineering Drawing Practices

(Copies of these publications are available online at https://www.asme.org.)

ECMA INTERNATIONAL

ECMA-363 - Universal 3D File Format (U3D)

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ECMA-376 - Office Open XML File Formats

(Copies of these publications are available online at https://www.ecma-international.org/.)

INTERNATIONAL COMMITTEE FOR INFORMATION TECHNOLOGY STANDARDS (INCITS)

INCITS/ISO/IEC 26300 - Open Document Format for Office Applications

(Copies of this publication are available online at https://standards.incits.org/.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO) PUBLICATIONS

ISO 10303-1 - Industrial Automation Systems and Integration – Product Data Representation and Exchange

ISO/IEC 12207 - Systems and Software Engineering - Software Lifecycle Processes

ISO/IEC/IEEE 15288 - Systems and Software Engineering – System Life Cycle Processes

ISO/IEC/IEEE 24748-7 Systems and Software Engineering – Life Cycle Management – Part 7: Application of Systems Engineering on Defense Programs.
Note: This formerly was IEEE 15288.1

ISO/IEC/IEEE 24748-8 Systems and software engineering – Life cycle management – Part 8: Technical reviews and audits on defense programs.
Note: This formerly was IEEE 15288.2

ISO 32000-1 - Document Management - Portable Document Format (PDF)

ISO/IEC 19775 - Information Technology - Computer Graphics, Image Processing and Environmental Data Representation - Extensible 3D (X3D)

ISO 24517-1 - Document Management - Engineering Document Format using PDF

ISO/IEC 29500 - Information Technology - Document Description and Processing Languages - Office Open XML File Formats

ISO/IEC 19540-1 - Information Technology – Object Management Group® Unified Architecture Framework (OMG® UAF®) – Part 1: Domain Metamodel (DMM)

ISO/IEC 19450-2- Information Technology – Object Management Group® Unified Architecture Framework® (OMG® UAF®) – Part 2: Unified Architecture Framework Profile (UAFP)

(Copies of these publications are available online at https://www.iso.org.)

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NATIONAL INFORMATION STANDARDS ORGANIZATION (NISO)

ANSI/NISO Z39.18 - Scientific and Technical Reports - Preparation, Presentation, and Preservation

(Copies of this publication are available online at https://www.niso.org/.)

NATIONAL SHIPBUILDING RESEARCH PROGRAM (NSRP) ADVANCED SHIPBUILDING ENTERPRISE (ASE)

NSRP ASE 2009-350 - 2008 Systems Technology Panel Project: Interim SCIM and STEP Implementation

(Copies of this publication are available online at https://www.nsrp.org.)

OBJECT MANAGEMENT GROUP® (OMG®)

OMG® Unified Modeling Language® (UML®)

OMG® Systems Modeling Language™ (SysML®)

OMG® Unified Architecture Framework® (UAF®)

(Copies of these publications are available online at https://www.omg.org/spec/.)

SAE INTERNATIONAL

SAE AS-6969 - Data Dictionary for Quantities Used in Cyber-Physical Systems

SAE EIA-649C - Configuration Management Standard

SAE EIA-649-1 - Configuration Management Requirements for Defense Contracts

SAE GEIA-HB-649A - Configuration Management Standard Implementation Guide

SAE-GEIA-STD-0007 - Logistics Product Data

(Copies of these publications are available online at https://www.sae.org.)

U.S. PRODUCT DATA ASSOCIATION (US PRO)

Initial Graphics Exchange Specification (IGES) 5.3

(Copies of this publication are available online at https://www.intertekinform.com/en-us/.)

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3. DEFINITIONS

3.1. Acronyms.

Acronym	Term
2D	2-Dimensional
3D	3-Dimensional
3Di	3-Dimensional Intelligent
AAF	Adaptive Acquisition Framework
AM	Additive Manufacturing
AP	Application Protocol
ASME	American Society of Mechanical Engineers
BOM	Bill of Materials
ASoT	Authoritative Source of Truth
CAD	Computer-Aided Design
CAGE	Commercial and Government Entity
CCB	Configuration Control Board
CDRL	Contract Data Requirements List
CM	Configuration Management
COTS	Commercial Off the Shelf
DASO	Digital Acquisition and Sustainment Operations
DE	Digital Engineering
DEE	Digital Engineering Ecosystem
DID	Data Item Description
DLA	Defense Logistics Agency
DoD	Department of Defense
DoDAF	Department of Defense Architecture Framework
DoW	Department of War
DSM	Digital System Model
DT&E	Developmental Test and Evaluation
GRA	Government Reference Architecture
HWIL	Hardware-in-the-loop
IDE	Integrated Digital Environment
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
IGES	Initial Graphics Exchange Specification
INCITS	International Committee For Information Technology Standards

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IP	Intellectual Property
IPT	Integrated Product Team
ISO	International Standards Organization
IT	Information Technology
JAMMEX	Joint Additive Manufacturing Model Exchange
JT	Jupiter Tessellation
LCSP	Life Cycle Sustainment Plan
MBD	Model-Based Definition
MBE	Model-Based Enterprise
MBPS	Model-Based Product Support
MBSE	Model-Based Systems Engineering
MBTE	Model-Based Test Engineering
MBT&E	Model-Based Test and Evaluation
MBx	Model-Based, “x” representing for any type of model category
MOSA	Modular Open Systems Approach
MS&A	Modeling, Simulation, and Analysis
NASA	National Aeronautics and Space Administration
NAVAIR	Naval Air Systems Command
NAVSEA	Naval Sea Systems Command
NISO	National Information Standards Organization
OAD	Object Architecture Description
OMG®	Object Management Group®
PDF	Portable Document Format
PDM	Product Data Management
PLM	Product Lifecycle Management
PM	Program Manager
RA	Reference Architecture
RFP	Request for Proposal
RMF	Risk Management Framework
SAD	Solution Architecture Description
SAE	Society of Automotive Engineers
SEP	Systems Engineering Plan
SETR	Systems Engineering Technical Review
SoS	System of Systems
STEP	Standard for the Exchange of Product Model Data
SWIL	Software-in-the-loop

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SysML®	Systems Modeling Language™
T&E	Test and Evaluation
TDP	Technical Data Package
TRL	Technology Readiness Level
UAF®	Unified Architecture Framework
UML®	Unified Modeling Language
USMC	United States Marine Corps
USSF	United States Space Force
UxS	Unmanned Systems
WarU	Warfighting Acquisition University
XMI	XML Metadata Exchange

3.2. Definitions. Master terms and definitions are as specified in DoDD 5000.01. For the purposes of this handbook, the following definitions apply:

3.2.1. 3-dimensional intelligent (3Di) technical data. A 3-dimensional (3D) viewable representation of an item provided in a widely available software format (e.g., ISO 32000-1 Portable Document Format (PDF)). This representation details the physical or functional requirements of the required design configuration to include but not limited to geometry, topology, relationships, tolerances, attributes, metadata, and other features necessary to define a component or assembly.

3.2.2. 3D model. A digital representation describing the form of a physical object, displayed via specialized software.

3.2.3. 3Di format. The standard arrangement and organization of information within a 3Di viewable representation of an item. This includes features such as the size and arrangement of information blocks (e.g., title blocks), notes, lists, revision information, view states, restriction notices, and the use of optional or supplemental blocks, as specified in MIL-STD-31000.

3.2.4. Annotation. Visible dimensions, tolerances, notes, text, or symbols, as specified in ASME Y14.41 or ASME Y14.5.

3.2.5. Architecture. The structure of components, their relationships, and the principles and guidelines governing their design and evolution over time. Types of architecture include:

3.2.5.1. Conceptual architecture. The high-level vision and blueprint of a system, solution, or structure. The first step in the design process, focusing on what and why before diving into the how. Defines the essential components, their relationships, and the overall principles guiding the design.

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3.2.5.2. Logical architecture. The second level of abstraction in system architecture, building upon the foundation laid by the conceptual architecture. Describes *how* the system will be structured and how its components will interact to meet the requirements, but *without* specifying the exact technologies or physical infrastructure that will be used. A more detailed representation than the conceptual architecture, but still technology-agnostic.

3.2.5.3. Physical architecture. The most concrete and detailed level of system architecture. It describes how the system will be implemented, specifying the exact hardware, software, network, and infrastructure components that will be used to realize the logical architecture. It is the final step in the architecture design process, translating the logical design into a tangible, deployable system.

3.2.5.4. Requirements architecture. A structured approach to managing, organizing, and relating requirements in a way that facilitates the design and development of a system or solution.

3.2.5.5. Enterprise architecture (EA). A holistic discipline that provides a blueprint and framework for aligning an organization's business and operations strategy, IT capabilities, and technology investments. It is about understanding the big picture – how all the parts of the organization fit together and how they can work together more effectively to achieve strategic goals. EA helps organizations optimize their IT infrastructure, processes, and resources to improve efficiency, agility, and innovation.

3.2.5.6. Reference architecture. A documented set of best practices, patterns, and reusable assets that provides a template for designing and implementing a specific type of solution or system. It is not a complete, out-of-the-box solution, but rather a blueprint or guide that helps organizations accelerate development, reduce risk, and improve consistency across multiple projects.

3.2.5.7. Data architecture. The design, creation, deployment, and management of an organization's data assets. A blueprint that outlines how data will be collected, stored, processed, and used to meet the organization's strategic goals, business needs, and operational needs. It encompasses both the technical aspects of data management and the business context in which data is used.

3.2.5.8. Application architecture. The high-level blueprint that defines the structure, behavior, and interactions of an application or a set of applications. It provides a roadmap for building, deploying, and maintaining applications in a way that aligns with business and operational needs, technical constraints, and architectural principles.

3.2.5.9. Software architecture. The fundamental organization of a software system, embodied in its components, their relationships with each other and to the environment, and the principles governing its design and evolution. It is the high-level blueprint that provides a structured approach to building, deploying, and maintaining a software application.

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3.2.5.10. Hardware architecture. The design and organization of physical components within a computer system, electronic device, or other hardware-based system. It defines the structure, functionality, and interconnections of hardware elements such as processors, memory, storage, input/output devices, and communication interfaces. It is the blueprint for how these physical elements are arranged and how they interact to achieve a specific purpose.

3.2.5.11. Electrical architecture. The design and organization of electrical and electronic systems within a device, vehicle, building, or other complex system. It encompasses the hardware, software, and communication networks that work together to provide electrical power, control, and functionality. It defines how electrical components are connected, how power is distributed, how data is transmitted, and how the system is controlled.

3.2.6. Architecture framework. A framework that establishes a customary practice for creating, interpreting, analyzing, and using architecture descriptions within a particular domain of application or stakeholder community.

3.2.7. Attribute. A dimension, tolerance, note, text, or symbol required to complete the product definition or feature of the product that is not visible but is available upon interrogation of the annotated model, as specified in ASME Y14.41 or ASME Y14.5.

3.2.8. Authoritative. Able to be trusted as being accurate and true.

3.2.9. Authoritative source of truth (ASoT). The reference point for models and data across the system life cycle. The authoritative source of truth provides traceability as the system evolves, capturing historical knowledge and connecting configuration-controlled versions of models and data.

3.2.10. Bill of materials (BOM). A listing of the parts, components, and sub-assemblies; and the quantities of each; needed to define, manufacture, or support an end product. BOMs typically are identified by nomenclature and product identifiers.

3.2.10.1. BOM, structured. A listing of components, typically represented by nomenclature and product identifier(s), that make up a product, arranged in a hierarchy such that product structure and component quantities can be determined for the end item itself and for all assemblies and subassemblies contained in the end item.

3.2.10.2. As-built BOM. A structured BOM that reflects status of the system as delivered.

3.2.10.3. As-maintained BOM. A structured BOM that reflects the status of the system as it has been maintained in the operation of the system.

3.2.10.4. Engineering BOM (E-BOM). A structured BOM that defines a product as developed by engineers.

3.2.10.5. Manufacturing BOM (M-BOM). A structured BOM that identifies components and materials necessary to manufacture a product.

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3.2.10.6. Provisioning BOM (P-BOM). A structured BOM that identifies components necessary to logistically support a product by identification of spare and repair components.

3.2.10.7. Service BOM. Defines the components needed to service or repair a product in the field.

3.2.10.8. Software BOM (S-BOM). A structured identification and list of software components and information about these components. An S-BOM aids in software license management, increases software transparency, and can help to increase cybersecurity responsiveness.

3.2.11. Company standard. A company document that establishes engineering and technical limitations and applications for items, materials, processes, methods, designs, and engineering practices unique to that company.

3.2.12. Computer software. Computer programs, source code, source code listings, object code listings, design details, algorithms, processes, flow charts, formulae, and related material that would enable the software to be reproduced, re-created, or recompiled. Computer software does not include computer databases or computer software documentation, as specified in DFARS Clause 252.227-7014.

3.2.13. Computer software documentation. Owner's manuals, user's manuals, installation instructions, operating instructions, and other similar items, regardless of storage medium, which explain the capabilities of the computer software or provide instructions for using the software, as specified in DFARS Clause 252.227-7014. ISO/IEC Standard 12207 uses the term "software lifecycle data" to address software documentation.

3.2.14. Contract Data Requirements List (CDRL). A list of authorized data requirements for a specific procurement that forms part of a contract. It is composed of a single DD Form 1423, or a series of DD Form 1423s containing data requirements and delivery information. The CDRL is the standard delivery format for identifying potential data requirements in a solicitation and deliverable data requirements in a contract and identifies what data products are to be formally delivered to the Government by a contractor, as well as when and possibly how (e.g., format and quantity) they are to be delivered.

3.2.15. Data access. A user's ability to effectively and securely create, manipulate, deposit or upload, retrieve or download, store, and exchange all data residing within a structured electronic collaborative database or other repository, as specified in MIL-HDBK-61.

3.2.16. Data delivery. The physical delivery or transmission of information and files from one source or ownership to another.

3.2.17. Data dictionary. A central repository that defines and describes data elements used in a system, model, or database. A data dictionary ensures data is consistently defined and consistently used across systems and models. For example, Battery Level: integer, % from power bus.

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3.2.18. Data model. The definition of all data elements (and their relationships) to be maintained within a product data management (PDM) system. The data model defines the database schema for the PDM system.

3.2.19. Digital artifact. A product or output, in computer (i.e., digital) format, created within or generated from the digital engineering ecosystem. Digital artifacts provide data for alternative views to visualize, communicate, and deliver data, information, and knowledge to stakeholders.

3.2.20. Digital acquisition. Acquisition life cycle modernization by leveraging digital methods and digital engineering outputs to acquire, field, and sustain systems and services.

3.2.21. Digital engineering (DE). A means of using and integrating digital models and the underlying data to support the development, test and evaluation, and sustainment of a system.

3.2.22. Digital engineering ecosystem (DEE). The interconnected infrastructure, environment, trained workforce, and approaches (process, practices, methods, and tools) used to store, access, analyze, and visualize system data and models to address the needs of the stakeholders. The Integrated Digital Environment (IDE) is a digital workspace within the larger DEE.

3.2.23. Digital model. A digital (i.e., in an electronic form, able to be read and manipulated by computer) representation of an object, phenomenon, process, or system. The representation can include form, attributes, and functions and may be depicted visually or described via mathematical or logical expressions.

3.2.24. Digital product support. The use of DE methods and digital data and system models to implement the Product Support Strategy, enable data-driven decision making, and deliver effective and efficient product support outcomes throughout the system life cycle. (DoD PSM Guidebook)

3.2.25. Digital system model (DSM). A digital representation of a system, which integrates the technical data and associated digital artifacts and defines multiple aspects of the system for specific activities throughout the system's life cycle.

3.2.26. Digital Technical Review (DTR). The execution of technical reviews leveraging model-based engineering tools and methodologies.

3.2.27. Digital thread. An extensible and configurable analytical framework that seamlessly expedites the controlled interplay of technical data, software, information, and knowledge in the digital engineering ecosystem, based on the established requirements, architectures, formats, and rules for building digital models. It is used to inform decision makers throughout a system's life cycle by providing the capability to access, integrate, and transform data into actionable information.

3.2.28. Digital thread (Alternate definition 1). A communication framework that connects a system's data across functional perspectives.

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3.2.29. Digital thread (Alternate definition 2). A collection of linked authoritative digital information pertaining to a process, product, or system, whose consistency is actively managed throughout the life cycle. This enables accessibility, traceability, currency, applicability, and credibility of information, thus facilitating the capture, communication, and use and reuse of knowledge to efficiently inform decisions that realize value.

3.2.30. Digital transformation. The integration of digital technology into all areas of an organization, fundamentally changing how the organization operates and delivers value. Digital transformation is more than implementing new technology, it includes rethinking and reengineering processes, culture, customer service and product delivery to fully leverage the power of digital tools. Digital transformation within DoW is focused on leveraging technology and process changes, including digital acquisition, to increase agility, resiliency, and the effectiveness of warfighting capabilities.

3.2.31. Digital twin. A computerized representation (integrated set of models) that serves as the real-time digital counterpart of a physical object or process.

3.2.32. DoD Architecture Framework (DoDAF). The DoDAF provides the structure for organizing architecture concepts, principles, assumptions, and terminology about operations and solutions into meaningful patterns to satisfy specific DoW purposes. It enables the sharing and reuse of architectural data.

3.2.33. Engineering design data. Engineering drawings; 3Di viewable, native computer-aided design (CAD) models; neutral CAD models; or a combination of these, that define an item by means of graphic and textual presentations, the physical and functional requirements of an item, sufficient to fulfill its technical data package (TDP) element and level requirements, as specified in MIL-STD-31000. Engineering design data comes in three levels corresponding to its TDP level requirements as defined in the following paragraphs.

3.2.33.1. Conceptual design data. Engineering design data that describes the engineering concepts on which a proposed technology or design approach is based.

3.2.33.2. Developmental design data. Engineering design data that describes the physical and performance characteristics of a specific design approach to the extent necessary to permit the analytical evaluation of its ability to meet specified requirements and enable the development, manufacture, and testing of prototype or experimental materiel.

3.2.33.3. Product design data. Engineering design data that describes the complete physical and performance characteristics of an item or component in sufficient detail to ensure that an item or component produced in accordance with this data will be essentially identical to the original item or component.

3.2.34. Entrance criteria. Artifacts and other review or audit elements that must be completed before the review or audit can be conducted.

3.2.35. Exit criteria. Review or audit elements that must be assessed, completed, and action items closed before successful completion of the technical review or audit can be declared.

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3.2.36. Fully annotated model. A 3D CAD dataset in which all necessary features to fully define the item (full design disclosure including dimensions, tolerances, materials, notes, surface finishes, etc.) are included in a readily viewable form, as specified in MIL-STD-31000, ASME Y14.41, ASME Y14.41.1, or ASME Y14.5.

3.2.37. Integrated Digital Environment (IDE). A set of connected digital tools, data repositories, models, and interfaces which form a secure, collaborative, and authoritative digital workspace extending across the acquisition life cycle. IDEs operate as a node within the larger enterprise DEE.

3.2.38. Jupiter tessellation (JT) file. A 3D data format used for product visualization, collaboration, CAD data exchange, and long-term data retention. It contains a combination of 3D geometry, product and manufacturing information, and metadata that is a compression of a native CAD file.

3.2.39. Logical data model. A data model of a specific problem domain expressed independently of a particular database management product or storage technology (physical data model) but in terms of data structures such as relational tables and columns, object-oriented classes, or extensible markup language (XML) tags. This is as opposed to a conceptual data model, which describes the semantics of an organization without reference to technology.

3.2.40. Metadata. Generally defined to be data about data. For the purpose of this handbook, metadata is defined as data about design data and its defining engineering design data documents. Metadata is used by the procuring activity to store, manage, and provide access to TDP elements.

3.2.41. Mission engineering (ME). An interdisciplinary process encompassing the entire technical effort to analyze, design, and integrate current and emerging operational needs and capabilities to achieve desired mission outcomes.

3.2.42. Model. A representation of an actual or conceptual system that involves mathematics, logical expressions, and/or computer simulations that can help predict how the system might perform or survive under various conditions or in a range of environments.

3.2.43. Modeling and simulation. As a noun, the discipline that comprises the development and use of models and simulations. As a verb, the use of models and simulations, either statically or over time, to develop data as a basis for making managerial or technical decisions.

3.2.44. Model-based definition (MBD). A dataset containing the exact solid representation of a product, its associated 3D geometry, and the 3D annotation of its dimensions, tolerances, materials, finishes, and other notes to specify a complete product definition. Additional definitions are as specified in ASME Y14.41.1, ASME Y14.47, or ASME Y14.5.

3.2.45. Model-based enterprise (MBE). An organization that uses model-based definitions and other types of models for the purpose of design, commission, operation, service, and decommission of a product (Ref ASME Y14.47).

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3.2.46. Model-based product support (MBPS). A NAVSEA-branded concept implementing principles of Digital Product Support and Model-Based Sustainment focused on three distinct capabilities: (1) Navy Product Data Management; (2) Navy Common Readiness Model; and (3) Navy Data Acquisition Requirements Tool.

3.2.47. Model-based systems engineering (MBSE). The formalized application of modeling to support system requirements, design, analysis, verification, and validation activities beginning in the conceptual design phase and continuing throughout development and later life-cycle phases.

3.2.48. Model-based test engineering (MBTE). An engineering method using digital models to guide both hardware and software testing. For hardware, this includes the use of digital twins to perform virtual simulations. MBTE also encompasses automated activities to generate test routines and conduct testing of physical systems and hardware-in-the-loop performance. For software, models are used to automatically generate test cases and procedures to improve efficiency, test coverage, and enhance system validation. MBTE shifts testing activities to the earliest stages of the development life cycle and reduces costs, increases productivity, and provides deeper insights into system performance.

3.2.49. Model-based test and evaluation (MBT&E). A MBSE-based approach used by developmental and operational test organizations, program offices, and independent evaluators. MBT&E uses models representing the system and its mission environment to generate test cases and support system verification and validation.

3.2.50. Model-based x (MBx). A general term used to describe a category of model-based approaches.

3.2.51. Model types:

3.2.51.1. Mathematical model. A symbolic model whose properties are expressed in mathematical symbols and relationships.

3.2.51.2. Analytical model. A method of mathematical modeling that frames a process as an equation with many parameters and calculates an exact solution without computer simulation.

3.2.51.3. Numerical model. A method of mathematical modeling that runs a set of boundary and initial conditions with varying parameters and approximates a solution.

3.2.51.4. Simulation model. A model that emulates a real-life system through computer software.

3.2.51.5. Discrete event simulation. A method of simulating a system as a sequence of events that occur at specific points in time and change the state of the system.

3.2.51.6. Physical model. A representation of an actual system, showing what parts exist, how they connect, and how functions map to real-world components.

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3.2.51.7. Physical data model. A representation of a data design as implemented, or intended to be implemented, in a database management system. In the life cycle of a project, it typically derives from a logical data model, though it may be reverse engineered from a given database implementation. A complete physical data model will include all the database artifacts required to create relationships between tables or to achieve performance goals, such as indexes, constraint definitions, linking tables, partitioned tables, or clusters. Analysts can usually use a physical data model to calculate storage estimates; it may include specific storage allocation details for a given database system.

3.2.52. Modular open systems approach (MOSA). An acquisition and design strategy consisting of a technical architecture that uses system interfaces compliant with widely supported and consensus-based standards (if available and suitable). The strategy supports a modular, loosely coupled and highly cohesive system structure that allows severable system components at the appropriate level to be incrementally added, removed, or replaced throughout the life cycle of a system platform to afford opportunities for enhanced competition and innovation.

3.2.53. Ontology. A formal representation of concepts, their properties, and relationships. In DE, ontology is a framework defining concepts such as system, subsystem, requirement, function, interface, behavior, architecture, and their relationships. It is composed of formal vocabulary and a relationship map, ensuring consistency across models. For example, satellite function, satellite subsystem function, function which satisfies the requirement.

3.2.54. Product data management (PDM). The business function, often within product lifecycle management (PLM), that is responsible for the management and publication of product data.

3.2.55. Product lifecycle management (PLM). A strategic business approach that applies a consistent set of business solutions in support of the collaborative creation, management, dissemination, and use of product definition information (product data) across the extended enterprise, and spanning from product concept to end of life, integrating people, processes, business systems, and information.

3.2.56. Systems Engineering Technical Review (SETR). A structured systems engineering event-driven process that consists of program-independent reviews with defined entrance and exit criteria. The SETR process provides an assessment of technical health, requirements, design maturity, testing effectiveness, and sustainment over a program's life cycle to support acquisition milestone gate decisions. This contrasts with a general technical review, which is a less formal and more narrowly focused review of a specific aspect of a project or system.

3.2.57. Taxonomy. A structured way of categorizing and grouping entities into a hierarchy of categories and subcategories. For example, system, subsystem, and component.

3.2.58. Technical data. Recorded information, regardless of the form or method of the recording, of a scientific or technical nature (including computer software documentation). The term does not include computer software or data incidental to contract administration, such as financial or management information, as specified in DFARS Clause 252.227-7013.

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3.2.59. Technical Data Package (TDP). The authoritative technical description of an item. This technical description supports the acquisition, production, inspection, engineering, and logistics support of the item (MIL-STD-31000).

3.2.60. Telemaintenance. A means of communication enabled by DE that allows technicians and engineers to share equipment condition, troubleshooting, and repair information from diverse locations in real time.

3.2.61. Threat modeling. A form of risk assessment that models aspects of the attack and defense sides of a logical entity, such as a piece of data, an application, a host, a system, or an environment. Additional guidance is specified in NIST SP 800-154 and NIST SP 800-53.

3.2.62. Unified Architecture Framework (UAF)®. The UAF is an OMG standard for describing and analyzing complex systems, particularly in the context of defense, aerospace, and other industries. It is designed to support the development of enterprise architectures, system architectures, and system of systems (SoS) architecture.

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4. GENERAL GUIDANCE

4.1. Digital engineering and modeling overview. The goal of digital engineering (DE) is to accelerate the development, testing, and evaluation of products, systems, and subsystems, to reduce risk, and to improve life-cycle support. DE is an integrated approach that leverages data and models starting with mission engineering and extending throughout a system's development and life cycle. It makes use of computational technology, analytics, data sciences, artificial intelligence, and modeling to design, build, deliver, and sustain warfighter capabilities. The DoD Digital Engineering Strategy lists five elements necessary for a DE ecosystem (DEE) to thrive:

(1) Formalize the use of models for program and enterprise decision making.

(2) Establish an authoritative source of truth (ASoT).

(3) Incorporate technological innovation.

(4) Establish a supporting infrastructure and environment for collaboration and communication across stakeholders.

(5) Change the culture and workforce to support DE throughout the system's life cycle.

Figure 2 illustrates DE throughout the acquisition life cycle.

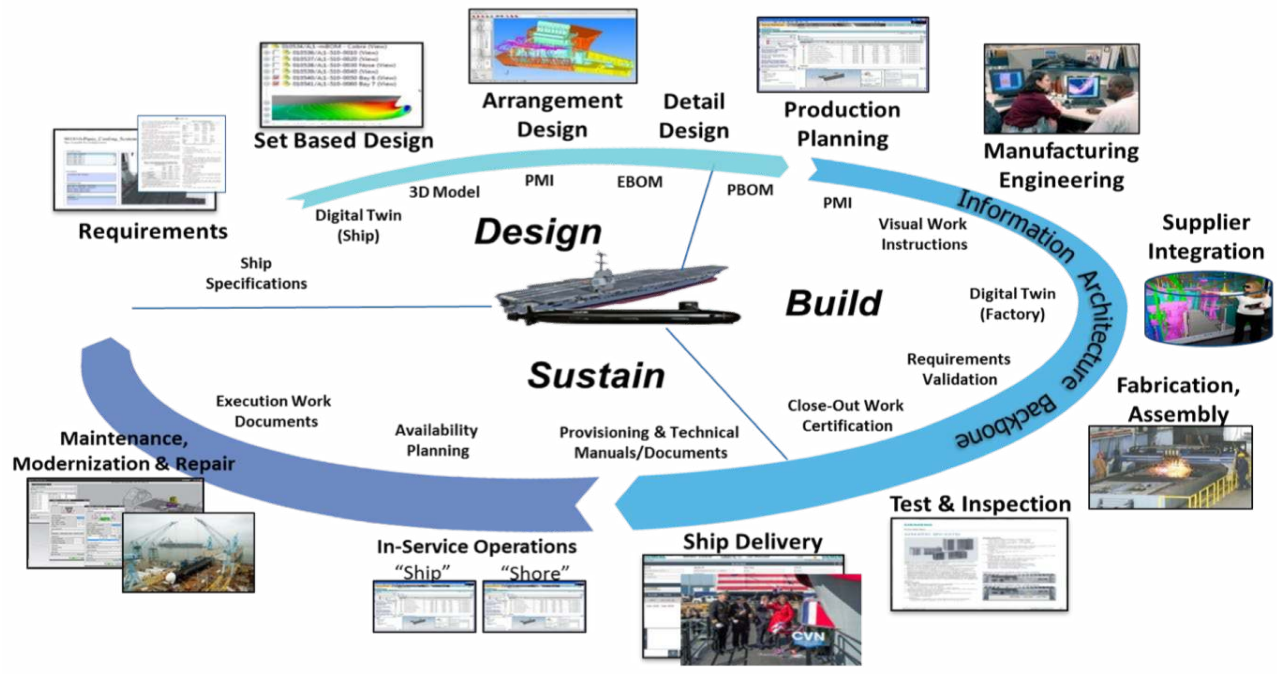
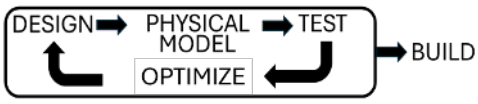
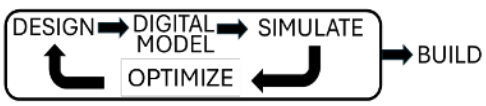


FIGURE 2. Navy example of DE throughout the life cycle.

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4.2. DE benefits. DE combines model-based techniques, digital practices, and computing infrastructure to enable delivery of highly effective and timely solutions to the warfighter. It modernizes how we design, operate, and sustain capabilities to outpace adversaries. The Digital Enterprise connects people, processes, data, and capabilities to improve technical, contract, and business practices through an ASoT and digital artifacts. Table I highlights some of the differences between a traditional enterprise approach and a DE-based approach to product development and acquisition.

TABLE I. Traditional enterprise versus digital engineering and model-based approach

Category	Traditional Enterprise Approach	DE and Model-Based Approach
Product Development	Design requirements established early in the design process and fixed, presenting challenges adjusting to operational or technical changes	System functional architecture updated as knowledge is developed. Design requirements modeled and constantly reviewed and adjusted throughout the design process
	Design trade-offs may be less well understood or considered	Design trade-offs well understood, modeled, simulated, and constantly re-examined
	Design Process 	Design Process 
	Modeling capabilities ad hoc and not well integrated	Constantly improving modeling and integration capabilities within each engineering area. This includes consistent examination of what modeling capabilities are needed, are the best models being used, does sufficient expertise in those models exist, are models applied consistently across programs, do models drive decision making, and is the information provided by the models stored and managed as an ASoT
Product Definition	Paper-based	Digital
	2D drawings represent the authoritative definition of the product	Complete, accurate & robust 3D models represent the authoritative definition of the product
	Focus on human-readable product data only	Focus on human- and machine-readable product data
	Engineering changes are slow and inefficient	Engineering changes are fast, efficient, and tailored to the context of the approved change
	Product definition changes done with unincorporated engineering actions that require the user of the data to have	Product definition information is online, on-demand, in the right format and always up to date (i.e., product definition information is accessible to authorized

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Category	Traditional Enterprise Approach	DE and Model-Based Approach
	drawing and hanging changes to have the most up-to-date product definition	users when needed, in the best format to fit their need, the information is current, and approved changes are quickly incorporated)
Project Information	Information travels as email attachments or hand carried hardcopies	Established digital workflows for standard processes
	Multiple copies of information	Information exists as a single authoritative master accessible by all authorized users
	Information recreated and reentered for downstream applications	Information created once and accessed/linked by downstream applications
	Significant time spent looking for routine project information. Local copies of information stored because official repositories are slow or unreliable	Project and product information can be accessed and retrieved rapidly from the ASoT in a digital environment.
	Key project and product information cannot always be relied on to be accurate or up-to-date	Key project and product information managed to ensure Truth, Trust and Traceability. (i.e., the data is accurate and updated, people know it to be accurate/current, and the information is traceable and controlled over its life cycle.)
Build Information	Design information can be separate from manufacturing information	Seamless flow from design to manufacturing
	Additive and advanced manufacturing are not always considered in initial design	Additive and advanced manufacturing built into the design process
	M-BOM created separately from E-BOM	M-BOM generated from and connected to E-BOM
Operations & Sustainment	Operations and sustainment information tends to be managed separately from design information.	Model-Based Definition (MBD) and other design information seamlessly flows into operations and sustainment information
	P-BOM created and maintained independently	P-BOM generated from and connected to E-BOM
	Field feedback on deficiencies, defects, or needed design improvements not easily correlated or trended to determine the need for fleet-wide changes.	Established and robust processes to obtain, track and trend field feedback on product deficiencies, defects, and possible design improvements
	Product operational information (e.g., operator manuals and technical manuals/orders) consists of paper documents and black line art	Product operational information is digital, constantly up-to-date, and contains dynamic, interactive 3D models, how-to videos, and other multimedia presentations
	Reactive maintenance, Reliability Centered Maintenance, Condition Based Maintenance	Digital twin/digital thread enhances ability to predict and avoid failures and maintenance issues, along with enabling

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Category	Traditional Enterprise Approach	DE and Model-Based Approach
		telemaintenance assistance from engineers.
	Inventory based, centralized supply strategy	Maximize use of advanced manufacturing (including additive) to supply on demand/on location

4.3. Model-based enterprise (MBE) overview. DE relies on models for the technical baseline of a capability, system, and product across the acquisition life cycle. An MBE is necessary to enable the development, integration, and use of models in enterprise and program decision making. Figure 3 illustrates the various major modeling categories that can be used in an MBE (the subcategory of modeling types referred to generically as MBx's). The MBx's presented are not intended to be all encompassing; other MBx's may be added to fit each organization's overall mission. This is further shown in Appendix F.

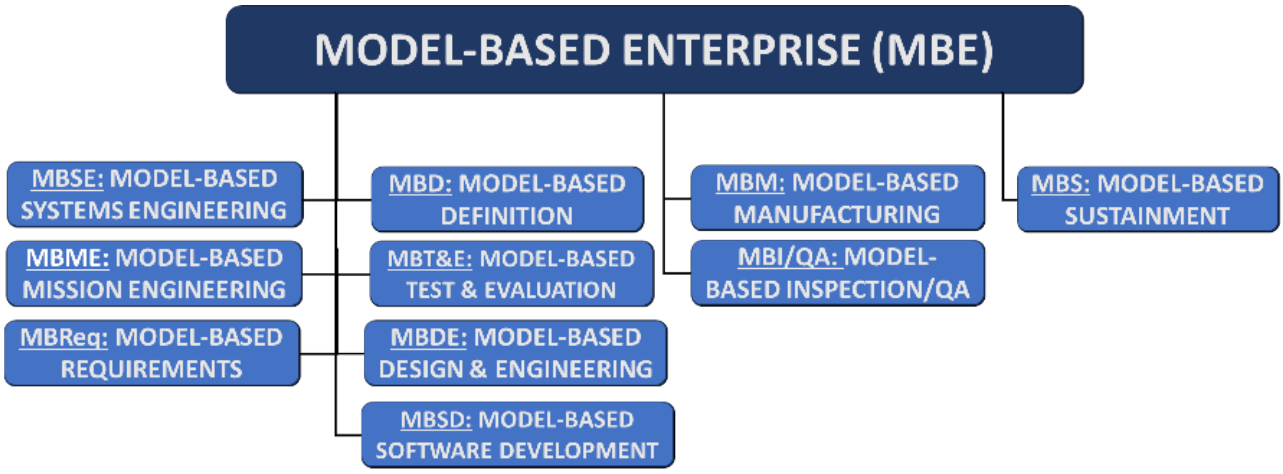


FIGURE 3. Model-based enterprise.

4.4. Model-based definition (MBD) overview. The move from traditional 2D product data to 3D model-based data is an essential part of DE and the MBE. 3D model-based data is more complex than 2D data and requires advanced software, hardware, and training to manage effectively. If constructed correctly, 3D model-based data can provide complete and accurate product design information, which can improve effectiveness, efficiency, and the ability to meet necessary product requirements. Managing 3D data requires implementing a PLM capability to make data easily accessible when and where it is needed.

4.5. DE data overview. DE, at its core, is about the use of data to create information and knowledge; information and knowledge to drive faster, more effective decisions; and better decisions in designing, building, and sustaining warfighter equipment. As discussed in Appendix A of the 2023 DoD Data, Analytics, and Artificial Intelligence Adoption Strategy, programs should start with the VAULTIS framework, that is, make data:

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a. **V**isible. Consumers can locate the needed data.

b. **A**ccessible. Consumers can retrieve the data.

c. **U**nderstandable. Consumers can recognize the content, context, and applicability.

d. **L**inked. Consumers can exploit data elements through innate relationships.

e. **T**rustworthy. Consumers can be confident in all aspects of data for decision-making.

f. **I**nteroperable. Consumers have a common representation/comprehension of data.

g. **S**ecure. Consumer data is protected from unauthorized use/manipulation.

4.6. DE data perspectives. In planning a data strategy, each program should address data from different perspectives:

a. Life cycle. What data is needed in each phase of the life cycle to support DE?

b. Functional. What data is needed for each functional area such as engineering, quality, configuration management, manufacturing, logistics, finance, program management, etc.?

c. MBx. What MBx areas will be used, what data is needed to input to those models, what data is expected to derive from those models, and how will this data be managed and interconnected?

d. Information technology. What IT infrastructure is necessary to support DE-related data?

e. PLM/PDM. What PLM/PDM tools are necessary to support the data infrastructure?

f. Data planning. How will the program go through the process of planning for data to support DE?

4.7. Integrated Digital Environment (IDE). The IDE is the "digital engineering workbench" technical framework that enables DE practices at the program or project level. The IDE operates as an interconnected node within the context of the larger enterprise-wide DEE.

4.8. Digital engineering ecosystem. A conceptual DEE is shown below in figure 4. This diagram represents a notional, fully implemented enterprise DEE encompassing many interconnected IDE nodes and operating over the entire physical product or services development and sustainment life cycle. The DEE has two parts: environment and activities. This DEE produces views and digital artifacts for the IDE nodes supporting engineering, program management, and sustainment processes.

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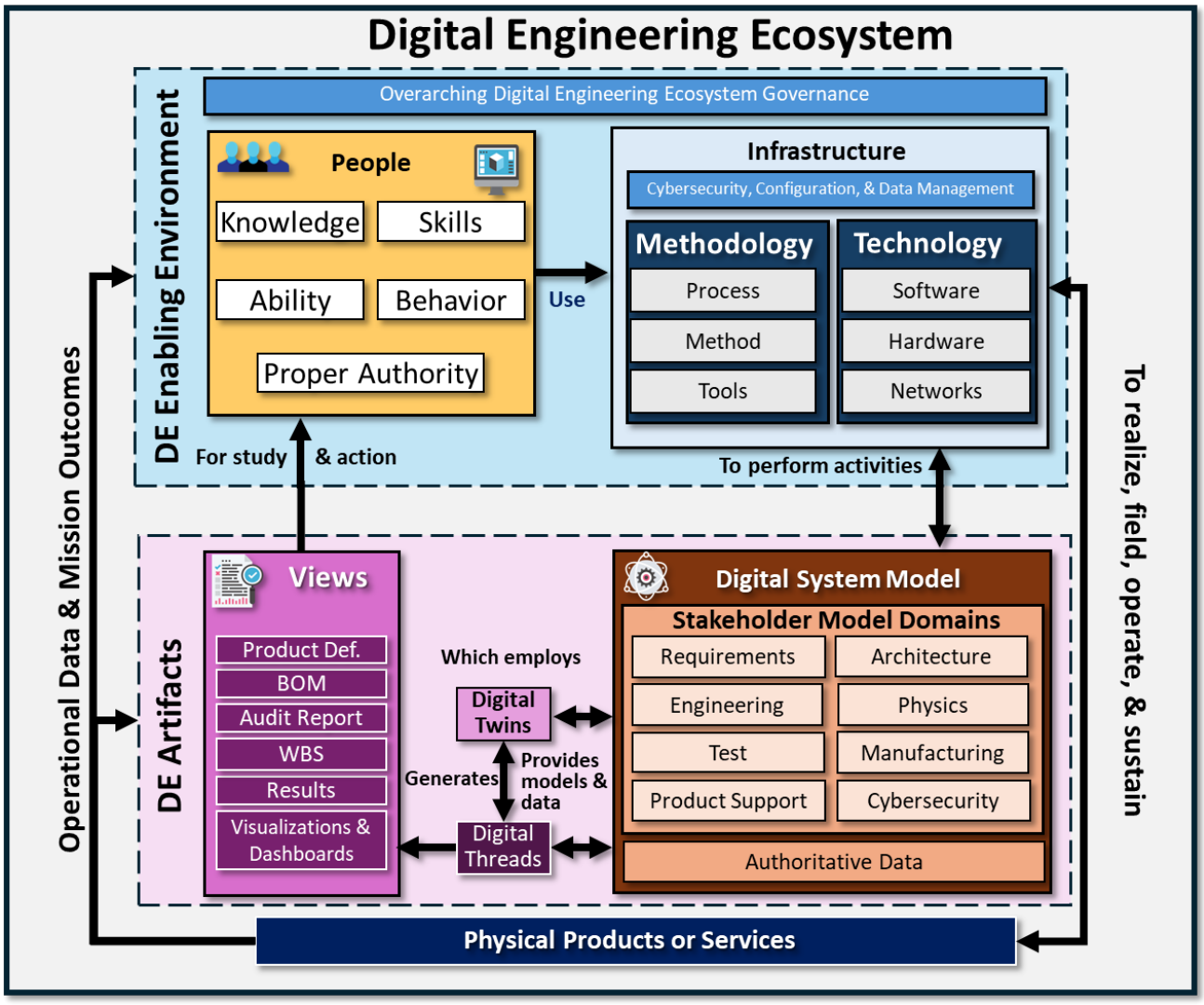


FIGURE 4. Conceptual Digital Engineering Ecosystem.

4.8.1. DEE enabling environment. The DEE enabling environment consists of people and infrastructure.

4.8.1.1. People. People with the requisite knowledge, skill, ability, behavior and proper authority use the DEE infrastructure. These members of Government and industry stakeholder groups collaboratively work together to develop, produce, and sustain products and services over the acquisition life cycle. These ecosystem members provide inputs to the system infrastructure and evaluate resulting views and artifacts.

4.8.1.2. Infrastructure. DEE infrastructure assets may be provided by DoW Components, contractors, or a combination of both.

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4.8.1.3. Overarching DEE governance. Effective governance is required to ensure the long-term success and sustainability of a digital ecosystem. Governance encompasses stakeholder management, defining stakeholder goals and objectives, setting DEE policy and standards, system management, and montioring system performance. Governance also includes planning and budgeting for DEE sustainment, architecture changes, and system upgrades.

4.8.1.4. Infrastructure cybersecurity, configuration, and data management. DEE governance is responsible for establishing and montioring compliance with cybersecurity requirements to ensure data integrity and prevent breaches. Configuration and data management standards, established by stakeholder agreement, are also enforced by the governance structure to ensure reliable system operation and system evolution, data integrity, and confidence in DEE outputs.

4.8.1.5. Methodology. The DEE incorporates stakeholder approaches to engineering activities, which include modeling languages and processes, methods, and tools. Some of these digital activities will differ from those employed in legacy SE activities.

4.8.1.6. Technology. Along with methodology, DE is realized through the integration of hardware, software, and networks. DE uses applications and databases designed for digital engineering and modeling activities, which increases efficiency and provides new capabilities. DEE technology can be provided by DoW Components, contractors, or a combination of the two.

4.8.2. DE activities. DE activities are performed by people that use the DE infrastructure. These activities generate digital artifacts such as models, data, digital twins, digital threads, and views.

4.8.2.1. Digital system model. The digital system model (DSM) is a representation of a system or a service, and is based on the combined effort of systems engineering and other engineering disciplines. The DSM is composed of various stakeholder model domain areas, and will grow in scope and complexity over the acquisition life cycle. The DSM also contains authoritative data.

4.8.2.1.1. Stakeholder model domains. Stakeholder model domains develop, collaborate, and excange data on their part of the DSM using the DE infrastructure. These domains represent key aspects of an acquisition development and life cycle. They are not limited to representing just physical items, they can represent services processes along with product support in the operations and maintenance phase.

4.8.2.1.2. Data. Data is consumed and produced by the DSM, Digital Twins, and or Digital Threads. Data is managed and controlled using the DEE Governance methodology and technology in the DE Infrastructure. The data is easily accessible and available to stakeholders via digital threads.

4.8.2.1.3. Authoritative data. Data contained witin the DSM is authoritative because database configuration, data capture, data creation, and data maintenance processes are well documented and subject to quality verification. The data is easily accessible and available to stakeholders via digital threads.

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4.8.2.2. Digital twins. A digital twin is a virtual representation of a physical asset, system, or service that mirrors its real-world counterpart. Digital twins are used for analysis, simulation, and optimization to support development of products and services. Digital twins use models from the DSM and data that are connected via digital threads and are used to produce views and artifacts. Digital twins also exchange data with external physical products or services that flows through the DEE infrastructure and is used to support operations and sustainment activities.

4.8.2.3. Digital threads. Digital threads transport a continuous and synchronized flow of data and information. Data flows via digital thread among models, simulations, engineering domains, stakeholder groups, and across all stages of the acquisition life cycle. Digital threads facilitate efficient communication, collaboration, and the production of views and artifacts.

4.8.2.4. Views and artifacts. Views and artifacts are interactively generated or tailored perspectives on the system, product, or process for a particular purpose or stakeholder. Views and artifacts are generated from digital models and processes, at a specific point in time, to visualize, communicate, and deliver data, information, and knowledge to stakeholders.

4.8.3. Realization and operation of physical product or service. The DE ecosystem infrastructure extends throughout the entire product or service life cycle. People use the digital engineering products and other outputs from the development cycle as inputs to the methodology and technology of product and service realization. Data from prototypes and testing is fed back into the DEE to refine the DSM and adjust the corresponding digital twins, producing new views and artifacts for further product or service refinement. Data from fielded systems or services, such as maintenance issues, repair part consumption, and operations performance measurements, can be ingested via the DE infrastructure for use by product support stakeholders. The data also can be used for incremental design improvements.

4.9. DE planning during the acquisition life cycle. Life-cycle planning is critical to implement and sustain DE. Each program office should: (1) Assess the digital maturity and current state of the program, (2) identify the available DE ecosystem, (3) plan for DE implementation within a system acquisition context, and (4) continuously improve their DE capabilities.

4.9.1. DE planning strategy example. Figure 5 is a conceptual illustration of DE as applied across the Major Capability Acquisition life cycle. Tools and activities within Increment 1 establish a communication framework that connects program data and digital models/digital twins via a digital thread running from post-Milestone A through the Production and Deployment phase of Milestone C. Increment 2 expands the DE thread to include Materiel Solution Analysis and Increment 3 encompasses the Operations and Support life-cycle phase. Within this framework, program management offices tailor their approaches to implementing or expanding the scope of DE based on their DE maturity level and unique program requirements.

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- **Increment 1 (Initial Digital Thread and Twin)**
  - Establish a single 3D TDP format and single (federated) repository to form Authoritative Source of Truth (ASoT)
  - Integrate systems models, requirements to provide functional context
  - Integrate physics models to establish engineering linkage to system performance
  - Integrate manufacturing tools to facilitate prototyping and transition to production
  - Integrate inspection data to increase fidelity of as-built digital twin
- **Increment 2 (Extend Digital Thread and Twin)**
  - Integrate test and evaluation data to train and validate models
  - Integrate operational models to establish linkage to military utility
  - Integrate decision analytics and AI to improve decision making and system design and development
  - Integrate experimentation tabs to enable virtual constructive capabilities
  - Support digital twins for chemical, material, and manufacturing processes
- **Increment 3 (Digital Sustainment)**
  - Integrate product support tools to more efficiently create and update logistics products
  - Integrate product support model tools for optimized sustainment
  - Integrate system telemetry and diagnostics data to realize true Reliability Centered Maintenance and Condition Based Maintenance (RCM & CBM)

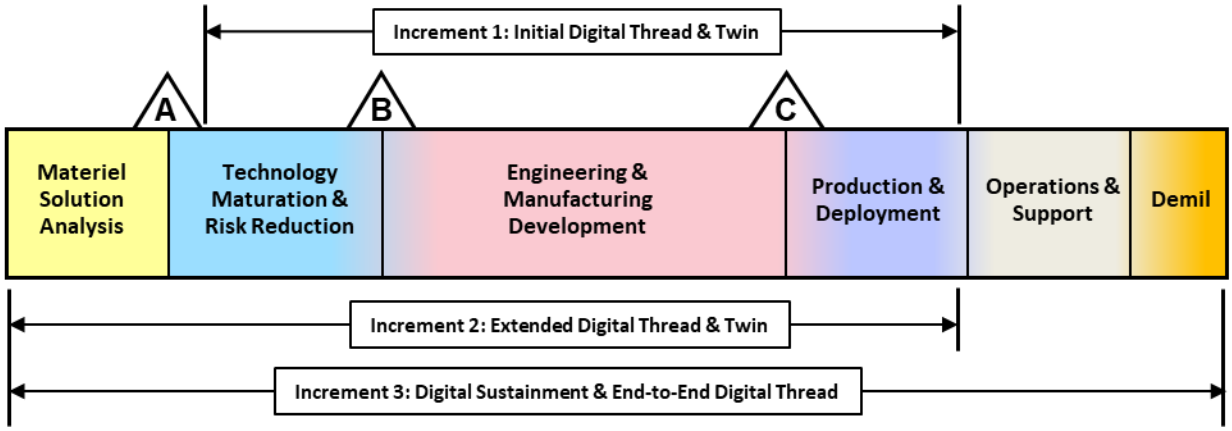


FIGURE 5. DE life-cycle planning strategy example.

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5. DETAILED GUIDANCE

5.1. DE and the Adaptive Acquisition Framework (AAF) pathways. DE supports all AAF pathways as specified in DoDI 5000.02 and DoDI 5000.97. Additional guidance is listed in DoDI 5000.70 and DoDI 5000.88 for topics covered in this handbook. This handbook focuses primarily on DE as applied to Major Capability Acquisitions, but the concepts also apply in a tailored manner to other acquisition pathways.

5.2. DE acquisition planning guidance. Effective acquisition requires proper planning, which includes planning for digital systems engineering activities throughout the acquisition life cycle. As shown in figure 6, common DE elements are applicable to all acquisition pathways, such as planning, mission engineering, MOSA, and other considerations.

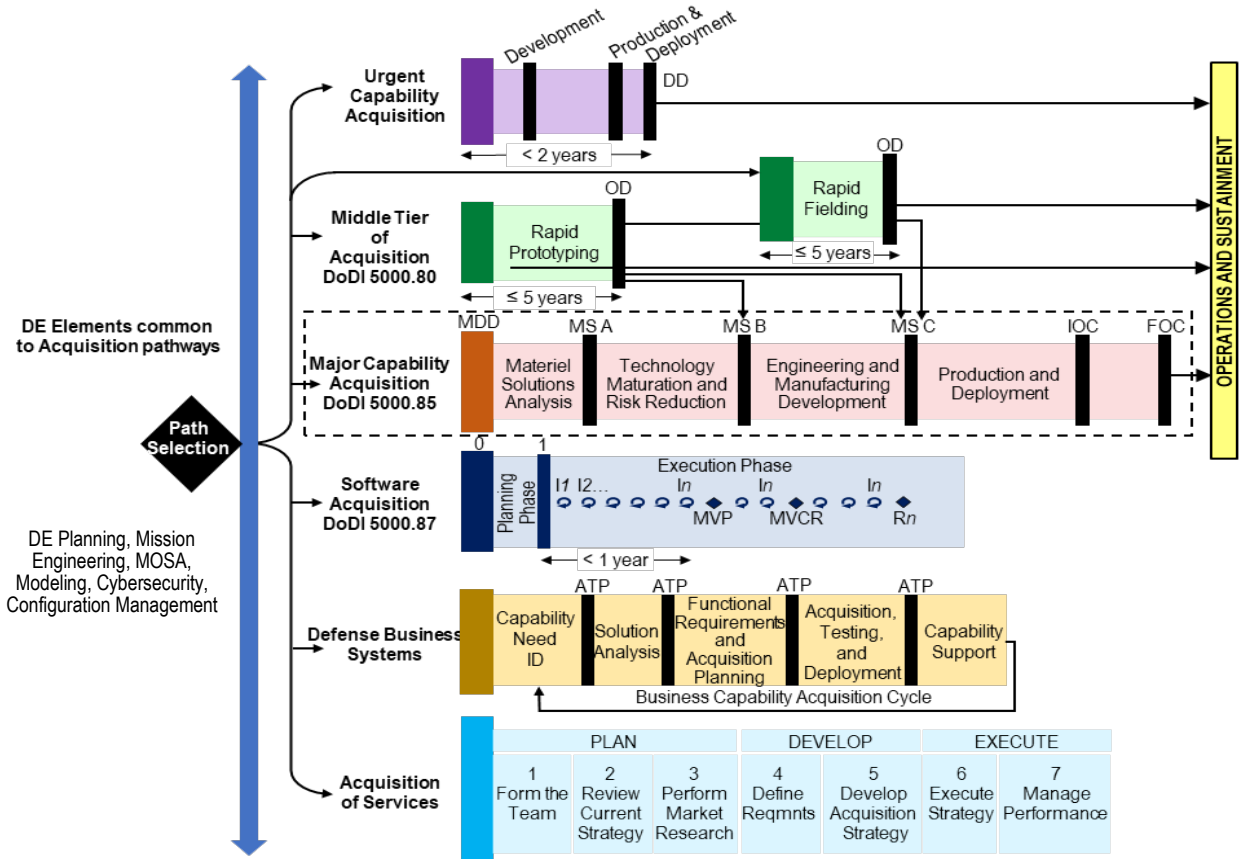


FIGURE 6. Common DE elements to the AAF pathways.

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5.2.1. DE planning. DoDI 5000.97 requires project and program managers (PMs) to address DE engineering in their Acquisition Strategy and to implement DE procedures as early as possible in the planning process and across the acquisition life cycle. This effort is to ensure sufficient thought, planning, training, and resources are applied to DE early in the program to maximize the benefit and minimize downstream costs and rework. In addition, contractor DE requirements and deliverables need to be clearly spelled out in proposal requests to ensure the best possible responses and associated contractor performance. Any exceptions to the use of DE for capability development should be granted by the program's decision authority and documented in the Acquisition Strategy (DoDI 5000.97 1.2c(1); 2.73(3); 3.4c).

5.2.1.1. DE considerations for Middle Tier and Urgent Capability. The planning and reporting requirements for each acquisition pathway are somewhat different and are listed in section 3.4 of DoDI 5000.97. Middle Tier and Urgent Capability acquisitions need to ensure their DE processes, data formats, and DEE will be as compatible as possible with any planned eventual transition to a major capability pathway or managed portfolio. This compatibility will help mitigate the risk of the acquisition becoming a "digital island" that may require significant DE rework in the sustainment phase.

5.2.2. Mission engineering (ME). Missions are tasks and actions undertaken to achieve a specific military objective. In contrast to the conventional focus on individual programs and systems, ME views the entire mission as a single system. ME is closely associated with a system of systems as most military missions are undertaken through the coordination of and interoperability of multiple systems. ME applies systems engineering to study and identify the right technologies, systems, systems of systems, processes, logistics, and organization structure to achieve intended mission outcomes. This analysis process uses digital models, data, and artifacts. ME study results may generate new programs and may motivate modifications to existing programs. The ME Guidebook and other documents listed in the section 2.2.2 references should be consulted for detail.

5.2.3. Modular open systems approach (MOSA). Both statute and DoW policy require defense acquisition programs to use MOSA, tailored appropriately to the acquisition. MOSA is a set of principles and practices that guide the design and development of systems with the aim of enhancing interoperability, flexibility, and adaptability. Complex systems are broken down into smaller, self-contained modules or components incorporating consensus-based standards for interfaces, communication protocols, and data formats. MOSA enhances component interoperability and reusability, system scalability, vendor independence, technology insertion, life-cycle cost savings, and cybersecurity.

5.2.3.1. Implementing MOSA. The program office will need to establish tailored requirements, business practices, development, acquisition, test and evaluation, and other strategies that underpin MOSA. DoW and industry DE practices, including MBSE, support development and delivery of MOSA throughout the acquisition life cycle. The guide Implementing a MOSA into DoW Acquisition Programs and other references listed in section 2.2.2 provide detailed information on MOSA requirements and implementation.

5.2.4. Modeling considerations. MBSE models support systems engineering efforts across the acquisition life cycle of a product or a system.

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5.2.4.1. Modeling tools. Several digital modeling tools provide a platform for research, design, development, and verification methods to improve cost, schedule, and performance during the acquisition of a product and to manage the fielded product across its life cycle. Every effort should be made to conduct an exhaustive review of vendors and their digital modeling tools to ensure those selected best meet the program's needs. These tools also form the basis to support more efficient sustainment and modification of acquisition items addressed in other chapters of this handbook. Some tools can share data better than others. Tools based on different standards or design criteria will make translated data files larger and may result in a loss of data fidelity and continuity. Programs should establish a clear, criteria-based tool selection process that prioritizes tools supporting open standards, interoperability, and life-cycle integration. The goal is to reduce tool proliferation, minimize data translation effort, and ensure consistent, reliable access to authoritative data across the digital ecosystem.

5.2.4.2. DoW standards. Standards also can be included within digital environments. For example, the DoW is developing a standardized approach for modeling unmanned systems (UxS) using the information exchange standards shown in figure 7 to support DE and MOSA.

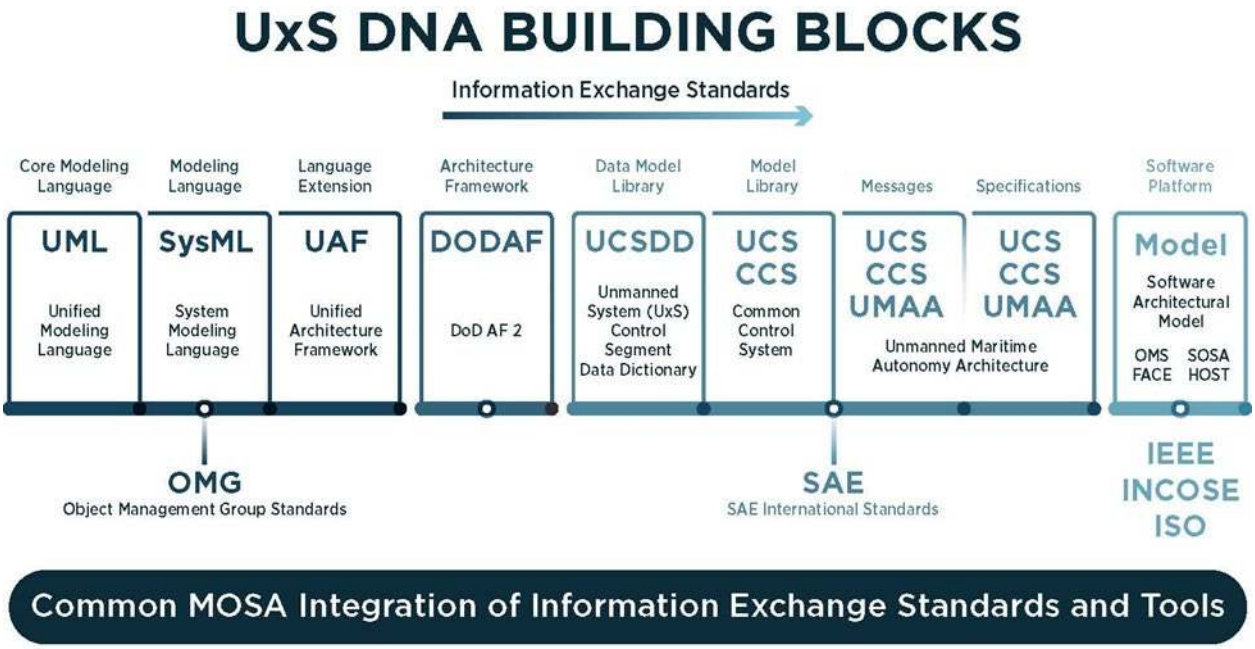


FIGURE 7. UxS modeling building blocks.

5.2.4.3. Object Management Group®. The initial modeling languages shown in figure 7 are specified in the Object Management Group® (OMG®) unified modeling language (UML®) and OMG® Systems Modeling Language (SysML)™ standards. Standardizing on a modeling language leads to improved interoperability because when a model is delivered in other formats it will need to be converted, which can be time consuming and costly.

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5.2.4.4. DoD Architecture Framework (DoDAF). The DoDAF is a model description that allows DoW to share information more effectively. DoDAF focuses on architectural data and does not define a physical model. DoDAF data can consist of documents, spreadsheets, dashboards, or other graphical representation templates and is defined as a "view." The DoDAF standard was last updated in 2011.

5.2.4.5. Unified Architecture Framework (UAF)®. The UAF is an OMG standard for describing and analyzing complex systems, particularly in the context of defense, aerospace, and other industries. It is designed to support the development of enterprise architectures, system architectures, and system of systems (SoS) architecture. The UAF is an evolution of the DoDAF with a broader scope and applicability. It is intended to provide a standardized, vendor-agnostic approach to architecture development, enabling better communication, collaboration, and integration among stakeholders. More specifically, key features of the UAF include:

a. Domain-agnostic. The UAF is designed to be applicable across various domains, including defense and aerospace.

b. Architecture views. The UAF provides a set of predefined architecture views, such as the Capability View, Operational View, Systems View, User View (also known as a human viewpoint per DoDI 5000.95), and Technical Standards View, among others. These views help architects and analysts describe and analyze different aspects of the system or enterprise.

c. Metamodel. The UAF has a well-defined metamodel, which provides a common language and structure for representing architecture elements, relationships, and concepts.

d. Integration with other frameworks. The UAF is designed to be compatible with other architecture frameworks, such as The Open Group Architectural Framework (TOGAF), Zachman, and DoDAF, allowing for easier integration and transition.

e. Support for modeling and simulation. The UAF provides a foundation for modeling and simulation, enabling the analysis and evaluation of complex systems and architectures.

5.2.4.6. UAF and SysML. The UAF is tightly coupled with SysML and more easily integrates program requirements with MBSE processes. A more detailed overview of MBSE (UML, SysML, UAF) is provided in section 6.

5.2.4.7. DoDAF to UAF transition. Government and industry are beginning the transition to using UAF, which allows for the creation of DoDAF-compliant views; however, many legacy and long-term program acquisitions and contract documents will continue to contain DoDAF information because of the cost associated with conversion and transition.

5.2.4.8. Data dictionary. One of the standards used across multiple programs is a data dictionary, which can be applied to all DoW systems. The data dictionary is a glossary of terms designed to support all DoW acquisitions. The use of digital standards provides efficiency in the acquisition process and improved integration among multiple activities. As an example, a low-level data dictionary used across DoW is specified in SAE AS-6969.

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5.2.5. DE models. When determining what standard or specially negotiated license the Government needs to acquire contractor-delivered models, the Government should contractually define the following:

a. All entities that need to receive copies of the model and by what means (e.g., physical, electronic).

b. How those entities will use those models.

c. Model delivery time frames.

d. How long those entities will use those models.

e. What organization or entity owns the models, and what does ownership mean.

5.2.6. Scope of intellectual property license. DoDI 5010.44 IP Acquisition and Licensing, DoDI 5000.86 Acquisition Intelligence, and DoDI 5000.91 Product Support Management for the Adaptive Acquisition Framework specify parameters that establish the scope of any intellectual property (IP) license to satisfy the program's intelligence threat inputs and supportability considerations.

5.2.7. Cybersecurity considerations.

5.2.7.1. Cybersecurity planning. DoDI 5000.02 and DoDI 8500.01 indicate acquisition programs should establish cybersecurity requirements early in the acquisition process and integrate the requirements throughout the life cycle. See the DoD Program Manager's Guidebook for Integrating the Cybersecurity Risk Management Framework (RMF) into the System Acquisition Lifecycle for general guidance on integrating cybersecurity into the program life cycle.

5.2.7.2. Engagement of cybersecurity professionals. Cybersecurity professionals are an essential part of the systems engineering design team and should be included early in the development of a new capability. All design decisions pertaining to cybersecurity should be made in consultation with cybersecurity professionals, including the program's RMF activities of security categorization, security control selection and implementation, security control assessment, information system authorization, and security control monitoring, as well as system security engineering tasks such as cybersecurity risk assessments, tabletop events, and test procedures.

5.2.7.3. DE development environment. Both the Government and contractor should assess the DE development environment at the beginning of any new major project. System security engineers need to be consulted because DE aggregates data that has previously been disparate. A data classification study, including inference and aggregation analysis, should be conducted to assess all systems, not just those with multi-security classification levels. System security engineers should use guidance for classifying databases, rather than documents, when determining the appropriate development environment and access controls for DE artifacts.

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5.2.7.4. Modeling cybersecurity. Because cybersecurity can be costly to visualize and test, MBSE is a beneficial methodology for representing and analyzing the cybersecurity of a system. A detailed discussion of MBSE, including cybersecurity considerations, is contained in section 6.

5.2.8. Configuration management considerations for DE.

5.2.8.1. CM background. Applying the CM principles contained in MIL-HDBK 61B, along with DoW's adopted non-government standards SAE-EIA-649 series and SAE GEIA-HB-649, will ensure consistent documentation and tracking of a product's performance, functional, and physical attributes with its requirements, design, and operational information.

5.2.8.2. CM transition. The transition from legacy processes to digital methodologies significantly increases the CM complexity of the organization's supporting IT infrastructure and software. Likewise, code production, modeling, simulation, and artifact generation require a robust CM system to maintain the traceability and the ASoT of DE products. Management emphasis on CM as a critical planning and life-cycle management activity for the DEE, DE processes, and DE outputs will produce long-term efficiencies and help avoid potential problems that in some cases could persist throughout the system life cycle. MIL-HDBK-61B is an excellent overall CM resource, and the sections on release management, digital engineering, and MOSA should be reviewed.

5.2.8.3. CM domains and DE. The DoW has three CM domains to consider when planning and implementing CM for DE:

a. Enterprise. The top-level CM effort is typically at the program office or similar level governing the infrastructure and services required to maintain the DEE throughout the life cycle. The DoW continues to mature the DoD Enterprise Service Management Framework (DESMF) as an accessible resource to understand more about CM and integrated processes. The DEE infrastructure is in this domain.

b. Manufactured product. CM is the technical and management process applying appropriate resources, processes, and tools to establish and maintain consistency among the product requirements, the product, and associated product configuration information. CM is an umbrella activity that crosses multiple functional domains, including but not limited to program management, systems engineering, production, life cycle logistics, test and evaluation (T&E), and contracting. Digital artifacts as a contract deliverable fall under this domain.

c. Cross-continuum systems and software engineering. CM is not a single acquisition phase-specific activity. Rather, CM is a continuous process spanning the entire product life cycle beginning with concept and running through system development and demonstration, operations and maintenance, and disposal. Employing a cross-continuum CM approach ensures consistency, stability, and security across the system life cycle.

5.2.8.4. DE CM terminology. DE facilitates transitioning from the traditional product-driven CM to a full life cycle CM orientation. This "shift left" cross-continuum systems and software engineering approach has generated new terminology.

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5.2.8.5. Digital Asset Configuration Item (DACI).

a. An aggregation of hardware, software, data, or all that is designated for CM and treated as a single digital artifact entity in the CM process.

b. The digital artifact entity within a configuration that satisfies an end use function and that can be uniquely identified at a given reference point.

5.2.8.6. Digital asset.

a. Embodies computer-based economic benefits or services that can be used in the future. The DoW controls access to the computer-based economic benefits or services and, therefore, can obtain them or deny or regulate the access of other entities. [1]

b. For purposes of configuration management processes and procedures within DoW DE, a digital asset is a term delineating the complete organizational unit defining a configuration item at the appropriate level to manage and reproduce the CI. Examples include but are not limited to: DEE; form, fit, and function data of a model; digital artifacts; DE capabilities; digital instance; digital thread; digital twin; interface intellectual property; models; simulations; or any required data about data or digital artifacts encapsulated within the digital element.

5.2.8.7. Digital product. The result of research, development, test, and evaluation in terms of computer-based digital artifacts being developed (engineered). Also known as a computer-based end item. The item stipulated in a contract to be delivered under the contract (i.e., model, simulation, data, software, service, study, or hardware).

5.2.8.8. DE version control. Software version control tools are used to track, document, and store individual configuration items such as source code, scripts, and external documentation. Integrated Modeling Environment (IME) software providers have similar model management versioning but typically work only with the digital assets created within the IME. DE will have these combined features plus digital assets designed to interface and interoperate while maturing throughout the acquisition life cycle. These terms are critical to understand and consider for DE version control:

a. Baseline. Defined quantity or quality used as a starting point for subsequent efforts and progress measurement; it can be a technical, cost, or schedule baseline.

b. Version. The numbering scheme used to identify sets of digital files that are tagged and named at a certain point in time.

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[1] Adapted from section 2.2 of DoD 7000.14-R Financial Management Regulation Volume 4, Chapter 1 October 2024

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c. Revision. A numbered draft of specific updates to individual digital files. Each time a user edits a file and commits it back to the repository, the file's revision number increases.

d. Merging. Combining multiple changes made to different working copies of the same digital files in the source repository. Merging is a strategy for managing conflicts by letting multiple developers work at the same time (with no locks on files) and then incorporating their work into one combined version. Merging works well when two sets of changes are made to different lines in a file and can be easily combined. When changes to a file are made on the same line or lines, conflicts occur, requiring someone to edit the file manually before the changes can be committed to the source repository successfully.

e. Trunk. Trunk/Main: The primary location for code in the repository. Using a family tree analogy, the trunk is the main line.

f. Branch. To isolate changes onto a separate line of development. When files are changed on a branch, those changes do not appear on the main trunk or other branches. Branch, rather than "copy," should be a CM reserved word used in version control. Use of a branch version should be documented so a program can compare a derivative or referenced work against the "as is" version within the ASoT. The phrase "branch model" implies the model will merge back to the ASoT trunk later.

5.2.8.9. CM and the complexity of large DEEs. Figure 8 illustrates the complexities of a fully integrated cloud-based DEE required to create and use digital assets. A strong and actively managed CM program is vital to maintaining the system architecture, the DE processes, and the integrity of DE outputs.

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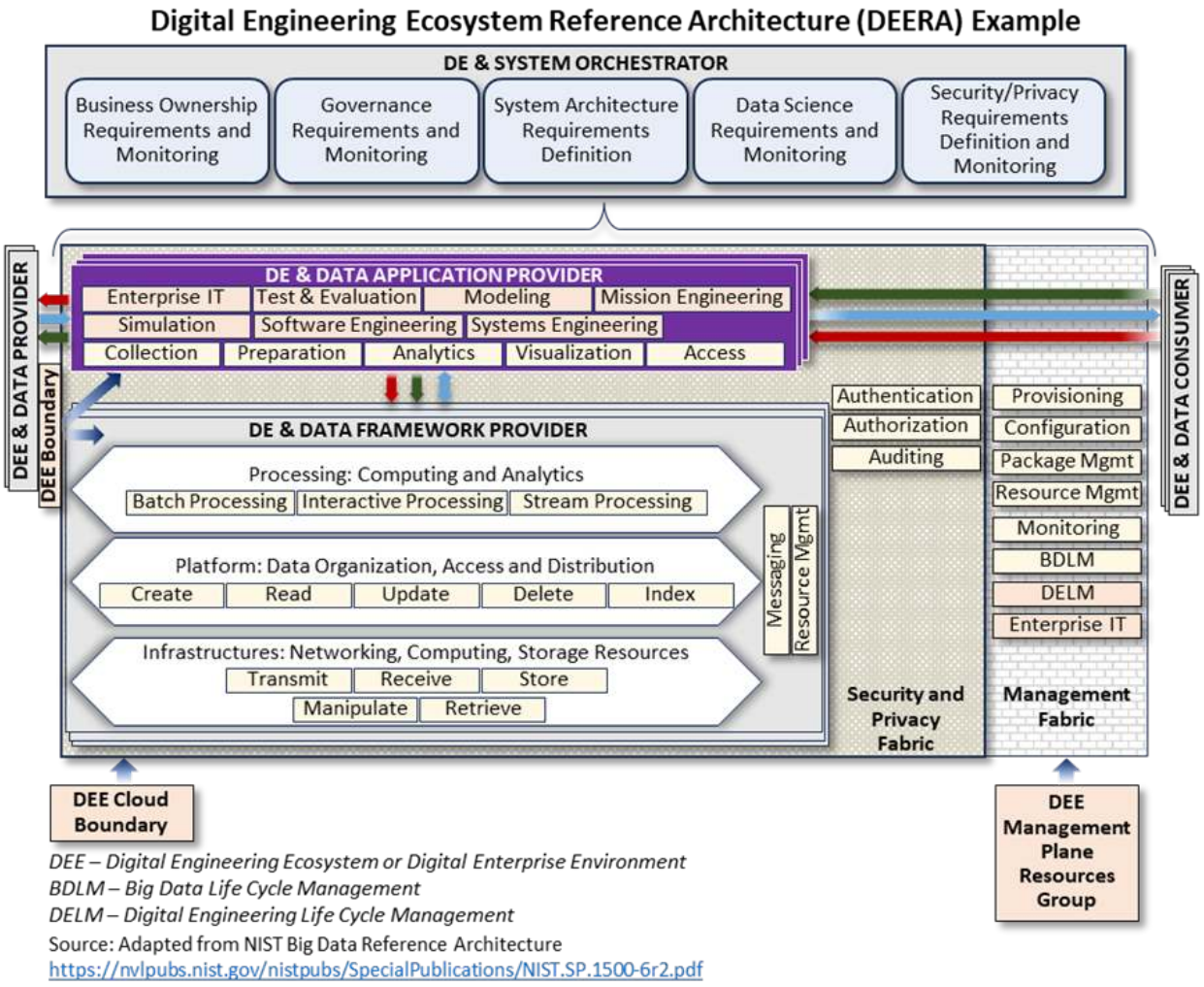


FIGURE 8. DoW Test Resource Management Center DE ecosystem.

5.2.9. Contracting for DE.

5.2.9.1. Contracting language for DE implementation. Enforceable contract language is critical to successful DE. The program's DE strategy, expressed within the RFP, is essential to establishing exactly what the Government is requiring the contractor to do consistent with the program's Capability Development Document (CDD), Acquisition Strategy, Systems Engineering Plan (SEP), and Life Cycle Sustainment Plan (LCSP). The program office should engage with the contracting officer early in the RFP development process to ensure the program's DE strategy and intellectual property strategy is fully reflected in the contract requirements.

5.2.9.2. Contract deliverables. DoDI 5000.97 section 3.4b states the minimum contract deliverable requirements for digital models, data, artifacts, and intellectual property. Additional items for consideration in an RFP include the following:

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a. Clear definitions of DE terms such as digital artifacts, digital engineering, digital thread, digital twin, model, etc., to ensure mutual understanding by all parties.

b. Contractor requirements for a DE management plan and/or a contractor DE strategy. These documents will be listed in a CDRL and have an associated Data Item Description (DID) to detail the format and time frame the deliverables are to be submitted.

c. The minimum viable DE deliverable acceptance criteria and their required delivery time frames.

d. Verification, validation, and accreditation (VV&A). The who and the how need to be specified.

5.2.9.3. Example clauses. The following brief contract clause examples are intended to conceptually illustrate how an RFP could address several of these areas:

a. DE. "The Contractor shall develop and deliver a Digital Engineering Management Plan (DEMP) in accordance with (IAW) CDRL AAA1 and Attachment 00XX (Digital Engineering Strategy). The DEMP shall include the Contractor's configuration management plan for developing and maintaining model-based configuration baselines throughout the program life cycle IAW the configuration baseline definitions in SAE EIA-649. The DEMP shall include the Contractor's approach to use Modeling, Simulation, and Analysis (MS&A) to reduce risk entering physical test."

b. Modeling. "Contractors shall develop the <weapon system> using Model-Based Systems Engineering (MBSE) and within the digital domain to communicate the design's maturity and evolution via a series of interconnected digital models throughout the program lifecycle."

c. Intellectual property (IP) rights. This is usually detailed in Section L "Instructions, Conditions, and Notices to Offers." "The Government requests the intellectual property rights to the systems, sub-systems, components, processes, or software listed below to: (1) Reduce the cost and increase the speed of future incremental upgrades, (2) Reduce sustainment costs, (3) Enable organic maintenance, (4) Allow the Government to use advanced manufacturing and other innovative techniques to reduce the time for vehicle repairs, (5) Support Foreign Military Sales (FMS), and (6) Use the technical data or software on other Government programs". Note: The list following this sample language should reflect the IP and software strategies. If neither the IP nor software strategies have been developed, the list should reflect the elements of the data model essential to program goals. Specific to subparagraph (6), purchasing IP rights beyond "access" to the entire data model could be both impractical and cost prohibitive.

5.2.9.4. Data delivery examples.

a. Sample CDRL, DID table, and TDP option selection worksheet for 3D data delivery. See Appendix A for a sample CDRL (DD Form 1423), DID tables, and the TDP option selection worksheet for contract data delivery. The TDP (as specified in MIL-STD-31000) will be further managed through the program office configuration management master plan and the prime contractor configuration management plan.

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b. Sample contract clause for electronic delivery of CDRLs. See Appendix B for a sample contract clause with prioritized list of data formats for electronic delivery.

c. Sample one-time DID. See Appendix C for a sample one-time DID. Because of the complexities of specific 3D product model data attributes, a general DID may not be effective. The tailored one-time DID is referenced by the CDRL and will contain specific data requirements. Another option to avoid a complex one-time DID would be to include the DID requirements within the contract.

d. Sample TDP drawing metadata format. See Appendix D for sample TDP drawing metadata formats that will be associated with the contract DID or one-time DID.

5.3. DE and the product development and sustainment life cycle.

5.3.1. Overview. Figure 9 illustrates four end-to-end acquisition life cycle perspectives that require thoughtful consideration when planning for DE implementation or preparing for expanding the use of DE.

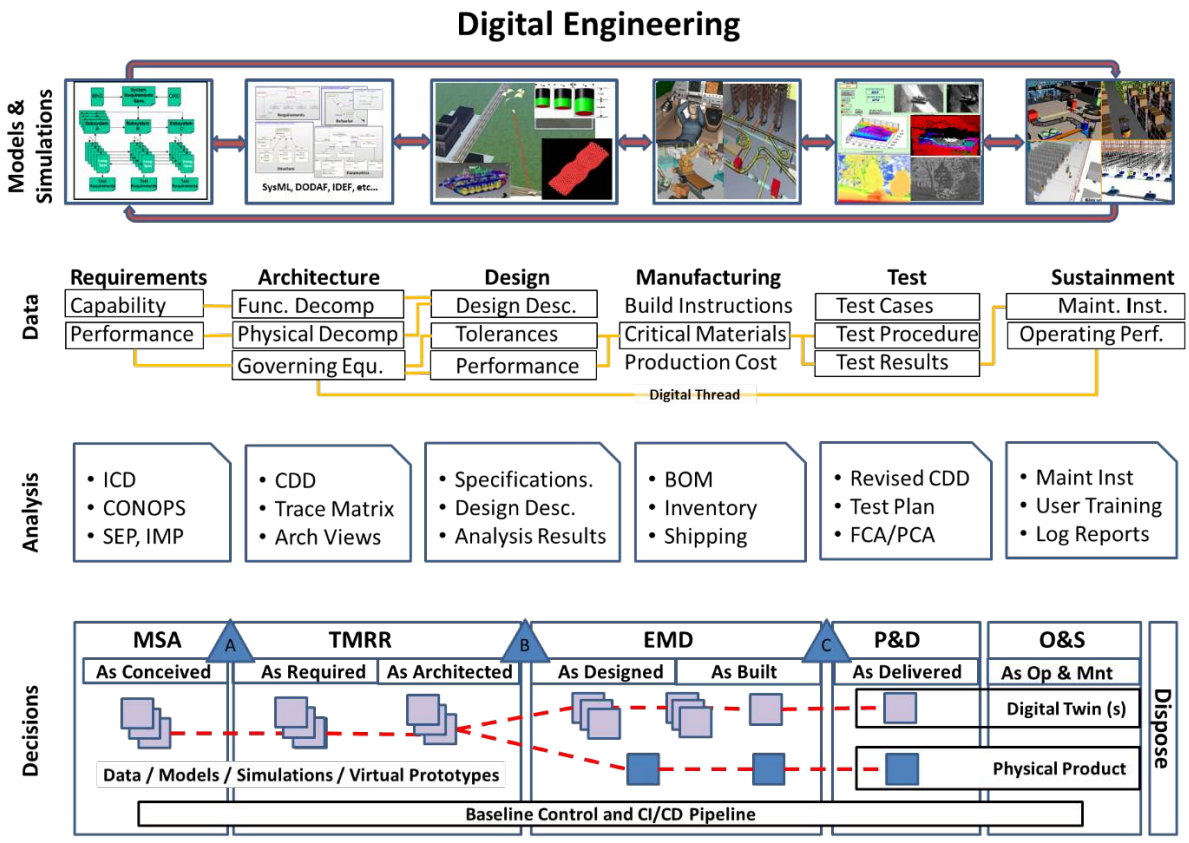


FIGURE 9. Digital engineering perspectives.

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5.3.2. Models and simulations perspective. Models and simulations are created for use in various activities such as mission and systems architecture, requirements development, mechanical design, and so on throughout the life of a program. DE planning needs to consider what models will be built and how they will be used, how they will be integrated, how they will be maintained, and what and how they will be delivered.

5.3.2.1. Data perspective. The data perspective involves knowing how the data will be used, who will need to use the data, how the data will be stored, how to protect the data, who can access the data, and how the data will be analyzed. The data, along with models and simulations, must be well planned and understood to be able to support collaboration and integration across an engineering Integrated Product Team (IPT) and the wider program organization. Further details on this perspective are provided in the data management template in Appendix H.

5.3.2.2. Analysis perspective. The analysis perspective is important for understanding, reporting, communicating status, and informing decisions. In digital engineering, the focus is shifted away from writing documents in silos and copying and pasting text from document to document. Instead, data and models are used as the authoritative source for the information that is compiled into familiar views, much like what we are used to seeing in documents. Some typical views would be a list of interfaces, interface definitions tables that describe interface inputs and outputs, performance analysis, weight budget rollup, a comparison between a design and requirements, and requirements verification trace matrix to name a few.

5.3.2.3. Decision perspective. The decision perspective involves making informed decisions based on the results of analysis and other information. In many cases, decisions that need to be made drive the models, simulations, data, and analysis. Decisions about the architecture, requirements, cost, interfaces, performance, and more are not much different from those made using conventional process. However, in a digital environment these decisions can be made incrementally and in real time with complete and current information.

5.3.2.4. Perspective strategy and management. Each of these perspectives highlights the importance of planning, integration, and governance of the program's DE system elements and processes. As such, it is important to strategize and actively manage the configuration and baselines of all four perspectives throughout the acquisition life cycle.

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6. MODEL-BASED SYSTEMS ENGINEERING

6.1. Overview. The DoW is transforming the way the Government develops, acquires, maintains, and sustains weapon systems using digital engineering tools, methods, and processes. This will lead to the establishment of a digital foundation enabling collaboration, integration, and the improvement of engineering activities to decrease the amount of time between system concept and delivery into the field.

6.1.1. Adoption of an approach. The adoption of a model-centric approach using MBSE helps facilitate this transformation. MBSE is defined by the International Council on Systems Engineering (INCOSE) as the formalized application of modeling to support system requirements, design, analysis, verification, and validation activities beginning in the conceptual design phase and continuing throughout development and later life cycle phases.[2]

6.1.2. Model-centric approach. The adoption of a model-centric approach provides additional advantages not found when using a traditional approach. When using a traditional approach, documents and discrete electronic files are used to capture a system's technical data, which is acquired from various sources, represented in various formats, and stored in a variety of locations across a system's life cycle. Artifacts such as digitized drawings, concept documents, functional and performance requirement documents, interface documents, and other forms of documentation are often stored in Microsoft Word documents, Microsoft Excel spreadsheets, and Adobe PDF documents. This can make information difficult to locate, integrate, and trace across the life cycle of the system. It also adds difficulty to maintaining version control because of the complexities of tracking changes made to sources of information maintained and stored at various locations.

6.1.3. Digital System Model (DSM). A digital model-centric approach supports systems engineering efforts across the life cycle of a product or system by maturing into a DSM that supports the design, analysis, verification, and operational activities. In addition, the integration of design, requirements, human-system interactions, structural, manufacturing, analysis, certification, cybersecurity, logistics, and financial models enables a centralized approach to locating, accessing, tracing, reusing, and integrating authoritative data from digital models and external sources across functional areas throughout the system life cycle. This leads to improved data consistency.

[2] *INCOSE Systems Engineering Vision 2035*, published 2021, available at https://www.incose.org/publications/se-vision-2035

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6.1.4. Planning. Goal 1 of the 2018 DoD Digital Engineering Strategy articulates that DoW organizations will develop formal plans for model creation, curation, integration, and related program and enterprise engineering activities throughout the life cycle. Plans will describe how models will be realized in a coherent and effective manner as work activities are performed and as analyses and decisions are supported. This promotes the use of digital representations of systems as a means of communication across all program stakeholders, moving a program away from using a traditional approach and toward a model-centric approach providing an ASoT for information for use across the life cycle. This planning also will help with establishing foundational standards and rules allowing for the integration and interoperability of models through the application of syntax, semantics, lexicons, taxonomy, and standards.

6.1.5. MBSE as the foundation of digital engineering. MBSE provides structured data in an extensible and deliberate manner, using a common language and framework, such that complex system data can be communicated across engineering communities. This commonality enables programs to leverage the potential of the underlying structured data to support the concept, requirements, structure, behaviors, and relationships of a complex system at a given point in time within the system's life cycle. MBSE provides a faster way to respond and address changing conditions, threats, and other urgent needs by providing the ability to tie, trace, and link various pieces of a system's authoritative data and information located within different data sources such as engineering data stored in a PLM system and a requirements management tool.

6.1.6. MBSE for informed decision making. Digital models and simulations are systems engineering tools that can be used by multiple functional areas during a system's life cycle. Modeling of systems is an important step in modernizing the approach used to understand and manage complex systems and system interdependencies. It also advances and improves communication and collaboration within a program team and its stakeholders across a system's life cycle. By using a model-centric approach that visually represents an overall system and how all the parts of that system interact, digital models provide a modernized approach for exploring system concepts, structure, requirements and their allocation and traceability, system performance, and alternative solutions used to support informed decision making.

6.2. Digital transformation. As digital transformation occurs within a program, the use of "authoritative and current" digital model configuration baselines will begin to supersede traditional "born analog" baselines that are maintained using documents and discrete electronic files used to capture a system's technical data acquired from various sources, represented in various formats, and stored in a variety of locations across a system's life cycle. As this shift occurs, configuration and data management teams will also need to begin planning for the integration of digital models into practice, working with contractors to modernize how they will be acquired, configuration managed, and maintained across the life cycle. Ensuring digital model configuration baselines maintain consistency with the physical counterparts they represent across the life cycle will lead to benefits that include:

a. A modernized approach to understanding and managing a complex system with large numbers of requirements, system interfaces, and interdependencies.

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b. A centralized approach to locating, accessing, tracing, reusing, and integrating of configuration baseline information from the ASoT across functional areas throughout the life cycle of a system.

c. A visual approach that can be used to explore system concepts, requirements allocation and traceability, system performance, early discovery and reduction of defects and inconsistencies before a system being manufactured, rapid prototyping, and alternative solutions, used to support informed decision making.

d. A digital approach facilitating communication and collaboration among all IPT members across multiple locations, using tailored views of system models instead of static documents.

e. A new approach that enables the Government to receive proposals using a digital format that will facilitate easier integration into an Integrated Digital Environment (IDE).

f. A business approach that reduces cost, time, and effort associated with managing a system across the life cycle.

6.2.1. Metadata. A digital model can include metadata describing one or more aspects of the model's organization, elements, views, and assists with the identification, indexing, searching, and management of objects contained within a model.

6.2.2. Modeling pillars. There are "Three Pillars" (A Modeling Language, Method, and Tool) associated with MBSE enabling the creation of a digital representation of a system:

6.2.2.1. Modeling language. Modeling language is used to define different elements that can be included within a model, allowing for the visual representation of relationships between those elements, and notations that can be used within each diagram in a model to display the elements and their relationships. It provides the syntax and rules used in communicating a model's abstract idea. Examples of semiformal graphical modeling languages used with MBSE include:

a. Unified Modeling Language (UML®). UML is defined by the Object Management Group (OMG) as "a graphical language for visualizing, specifying, constructing, and documenting the artifacts of a software-intensive system," adopted the Unified Modeling Language (OMG UML™) in 1997, providing specifications that define UML as a graphical language that provides software engineers and developers with a digital approach to visualize, develop, and document software systems, through the definition of programming language statements, database schemas, and reusable software components. UML is independent of specific programming languages and uses a standard graphical notation that leverages object-oriented concepts, providing a graphical method for visualizing, developing, and documenting the Structure, Behavior, Environment, and Implementation of a software development effort across the life cycle. Additional information on UML can be found on the OMG UML website at the following link: https://www.uml.org/

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b. Systems Modeling Language™ (SysML®). OMG defines SysML as "a general-purpose graphical modeling language for specifying, analyzing, designing, and verifying complex systems that may include hardware, software, information, personnel, procedures, and facilities." In 2006 the OMG adopted the SysML, an open source, general-purpose graphic modeling language most used with MBSE, and provides a graphical method for visualizing, developing, and documenting the Requirements, Structure, Behavior, and Parametrics of a system or process across the life cycle. The adoption of a model-centric approach using SysML provides a single ASoT for systems engineering information and data, providing the ability to generate different views and digital artifacts using the same model elements, supporting reuse and consistency. In addition, when a model-centric approach using SysML is incorporated within an IDE, engineering teams and their stakeholders can collaborate, communicate, and manage system changes in a digital manner as they occur over the system's life cycle. The OMG subsequently approved SysML v2.0 in June 2025, and information on the DoW transition will be provided in the next handbook revision. Additional information on SysML can be found on the OMG SysML website at the following link: https://www.omgsysml.org.

c. Unified Architecture Framework (UAF®). UAF is an OMG standard that supports the development of enterprise architecture and mission engineering analysis. The DoDAF artifacts required by the Joint Capabilities Integration and Development System process should be replaced with the complementary UAF artifacts. The components of UAF include: (1) view specifications organized in viewpoints and aspects (grid), (2) the UAF domain metamodel (DMM), (3) the UAF Modeling Language (UAFML based on the SysML) and the UAF enterprise architecture guide. The intent of UAF (and UAFML) is to support semantic interoperability among architectures (both enterprise and system), Government agencies, military agencies, customers, and suppliers. It provides a common standard to express architectures and support engineering and acquisition by using a Strategic layer, Operational layer, Resources/Services layer, and Design layer. Additional information on UAF can be found on the OMG UAF website at the following link: https://www.omg.org/uaf/.

6.2.2.2. Modeling method. Modeling methods are a set of related design conventions, techniques, and activities supported by a set of digital tools ensuring digital models are built and updated consistently, and modelers are working toward a common end point. Examples of well-defined modeling methods, which should not be considered DoW endorsed, include:

a. INCOSE Object-Oriented Systems Engineering Method (OOSEM).

b. IBM Telelogic Harmony-SE method (Use Case Realization Method).

c. Dassault CATIA, MagicGrid Method.

d. JPL State Analysis (SA) method.

e. Lockheed Martin Model-based Systems Engineering Method (MSEM).

f. IBM Rational Unified Process for Systems Engineering (RUP SE).

g. Weilkiens' System Modeling (SYSMOD) method.

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It is possible that the well-defined modeling methods listed above may not meet a project's specific needs and goals, so it may be necessary to include the customization of a modeling method when planning MBSE efforts.

6.2.2.3. Modeling tools. Modeling tools are software applications used to build, update, and modify digital models, which may be used in conjunction with physical model inputs, that are designed to comply with the "rules" of the modeling language used in the creation of well-formed models. For example, widely used digital modeling tools, which should not be considered DoW endorsed, include:

a. Dassault Systems *MagicDraw*, *Magic Systems of Systems Architect* (previously known as *Cameo Systems Modeler),* and *Cameo Enterprise Architecture.*

b. Sparx Systems *Enterprise Architect.*

c. IBM *Rhapsody.*

Each tool listed is compliant with the OMG SysML language specifications to varying degrees of fidelity and offer differing capabilities.

6.3. Planning for MBSE. MBSE planning helps set stakeholder expectations and increase the probability of a successful program office transition. The following items should be considered when developing transition plans.

6.3.1. Defining the purpose and scope for the implementation and use of MBSE. Defining the purpose and scope for the implementation and use of MBSE is an important consideration in planning the adoption of an MBSE approach within a program. Each program will be at different starting points. New programs will be able to start designing and implementing their approach from the ground up, with efforts moving forward being *born digital*. Existing programs will be at different points in the life cycles of the systems they manage and may have more targeted and limited needs associated with their overall planning and implementation of their MBSE approach.

6.3.2. Determining goals, objectives, and expected outcomes. It is important to define goals, objectives, and expected outcomes related to the implementation and use of an MBSE approach within a program. Clearly defined goals, objectives, and expected outcomes also will lead to a program able to better understand what needs and resources are necessary in obtaining the expected outcomes. MBSE is not an all-or-nothing effort but one in which activities build upon each other, staying aligned and current with continued advances in digital technologies supporting MBSE that are available and in use at any point in time.

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6.3.3. Program assessment. Conducting an in-depth assessment of the overall program should occur early in the transition into an MBSE environment, helping with the definition of the goals, objectives, and expected outcomes. This assessment also includes taking into consideration any necessary licensing and customization needs required to define the overall MBSE environment. It also includes performing an assessment of MBSE and DE capability maturity as a key part of the planning process. This assessment helps ensure the organization understands where it stands regarding the many synergistic capabilities that need to be instantiated and matured, enabling it to achieve its MBSE and DE objectives. A good first-pass instrument for assessment is the Air Force Digital Transformation Office's Digital Maturity Assessment (https://guide.dafdto.com/2023/01/04/digital-maturity-assessment/) while a good second-pass instrument for assessment is the INCOSE Model-Based Capability Matrix (https://portal.incose.org/commerce/store?productId=INCOSE-MBCM).

6.4. MBSE style guide. MBSE model development is often performed using a collaborative approach, with work divided among modeling team members or industry partners at various locations. A lack of definitions and constraints in how models are created and maintained can result in inconsistencies among model objects, elements, and data. This concern is amplified when different modeling styles have been applied by various sources. These inconsistencies can make a model difficult to navigate and understand and can make integrating models within a digital system model or system-of-systems model more difficult.

6.4.1. Purpose. An MBSE style guide is critical for alignment between the government reviewers and vendors who perform the modeling. A style guide provides guidelines and standards for how the program will implement MBSE modeling activities in a consistent manner. An MBSE style guide specifies rules defining how the language will be used and any constraints that go beyond the rules of the modeling language (e.g., UML, SysML). Also, the establishment of validation rules within the MBSE style guide can help reduce the number of inconsistencies, increasing the chances of achieving successful model integration. The guidebook should include specific RFP and SOW language such as mandating joint development, approval, and compliance reviews to help make the style guide an enforceable part of the deliverables. This also sets clear expectations upfront and ties adherence to review milestones or acceptance criteria.

6.4.2. Topics to include. Examples of topics to consider including in an MBSE style guide include:

a. Model techniques used with structural and behavioral modeling.

b. Naming conventions to be used within the model.

c. The application of color codes within the model.

d. How multiplicity is applied within the model.

e. Interface definitions.

f. Usage of ports.

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g. Flows and flow property usage.

6.4.3. Benefit of a style guide. Using a style guide allows modelers to focus on comprehensively representing the system of interest within the model instead of having to also focus on modeling technique at the same time.

6.5. Using MBSE to support the acquisition process. MBSE can be used to provide a technical approach to support the acquisition process using Reference Architecture (RA) models included as part of a comprehensive RFP package and provided to bidding contractors as a method of information exchange. The OMG is leading an effort, with participation from members of Government, industry, and academia, to create the standards and guidance required to support a Model-Based Acquisition (MBAcq) approach through the development and advancement of reference architectures and standardized patterns. This initiative will help advance the use of MBSE during acquisitions.

6.6. Government Reference Architecture models. A type of RA model, known as Government Reference Architecture (GRA) models, are developed using MBSE. GRA models serve as minimally restrictive model templates that define the intended modeling framework to be used by all stakeholders in designing a system. They help establish consistency in the overall characteristics used during the beginning stages of the modeling process by providing a framework that reinforces the use of a common language, rules, constraints, and interface specifications. GRA models reinforce consistency in standards, style, specifications, and patterns, ultimately providing a reusable template for building and tailoring models, reducing ambiguity, and speeding up the overall acquisition process.

6.6.1. Object Architecture Description. When used to support an acquisition, approved GRAs, which are often located in and downloaded from a Digital Reference Library or other authoritative source, are tailored to represent the Object Architecture Description (OAD) for the acquisition and are provided to bidding contractors as part of a comprehensive RFP package. OADs provide bidding contractors with a model-centric approach representing a logical system concept to expand upon, allowing proposals received during source selections to be evaluated on an even level. Along with defining requirement specifications, OADs are also used to perform trade-off analysis in determining the affordability, risk, performance, and mission effectiveness associated with the design of a new system or the upgrade of an existing system. The program should work with stakeholders early in the acquisition process to define “reference architecture” or “GRA” and reach agreement on what is in and out of scope for any RA or GRA used to support the acquisition process.

6.6.2. Solution Architecture Description (SAD). As the acquisition progresses, the OAD model also evolves into a solution architecture description (SAD) providing more fidelity of the overall system to be produced, representing the system baseline used to support the system’s development, production, fielding, and sustainment activities across the life cycle. The initial SAD is included with a bidding contractor’s proposal and evaluated against competing SADs to make the contract award determination. Once the contract award has been made, the combination of both the OAD and SAD establishes the DSM supporting the configuration baseline across the life cycle of the acquisition.

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6.7. Change Management Considerations. It is important to acquire, and manage the configuration of, digital models representing the "as-designed" configuration, the "as-built" configuration for each physical system that is produced and delivered, and the "as-maintained" configuration each time Engineering Change Proposals (ECPs) are approved across the life cycle and the changes incorporated within the system. By maintaining version control for digital models representing each of these baselines through a formal change management process, increased accuracy in the digital representation of a system at specific milestones in the life cycle will help reduce errors encountered during production, procurement, modification, and re-procurement activities.

6.8. Metadata considerations. Metadata is an important aspect of digital modeling. The OMG standard for exchanging metadata information through extensible markup language (XML) is the XML metadata interchange (XMI). XMI enables interchange of metadata between meta object facility-based metadata and UML®-based modeling tools. Vendor XMI implementations encounter technical challenges exchanging UML® model files between tools.

6.9. Tool interoperability considerations. Programs should standardize modeling tool sets early in development and carefully consider costs to move data and files among tools. When considering a new tool, they should analyze tool data interchange incompatibility. The program should analyze new tools in terms of program advantages versus the associated costs and effects to the schedule.

6.9.1. Contract language. Ensure contract language is digitally appropriate. It is important that contract language supporting an acquisition provides for the appropriate delivery of, or access to, MBSE models and the data and digital artifacts they contain. MBSE model file deliverables should be requested and captured in their Native file format which supports the lossless exchange and use of model files among various modelers who use the same modeling tool application. Examples of the native file formats associated with three frequently used SysML digital modeling tools are shown in table II.

TABLE II. Native file formats associated with frequently used SysML digital modeling tools.

Modeling Tool	Native File Extension
Cameo Systems Modeler	.mdzip, .mdxml
Sparx Systems Enterprise Architect	.eap, .eapx, .qeax
IBM Rational Rhapsody	.rpy, .rpyx

6.9.1.1. Data access vs. data delivery. According to the MIL-HDBK-61 Configuration Management Guidance, having access to data, including accessing data contained within MBSE digital models managed within a contractor's digital environment, is not a substitute for data delivery. Ownership of data requires sufficient description in the contract (SOW, CDRL, DID), Delivery, Inspection/Acceptance, and IP Rights, whereas access to data only requires IP Rights. Program teams need to ensure access to standards and data rights by:

a. Obtaining and securing available standards.

b. Gaining access to data regarding standards.

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c. Managing appropriate standards documentation and maintaining the necessary licenses.

d. Securing and managing the appropriate data rights.

e. Obtaining the appropriate access to storage, maintenance, control, use, and exchange of data.

f. Ensuring appropriate access and retrieval rights are specified in contract language.

6.9.1.2. Metadata deliverable. Metadata should also be a standard CDRL deliverable requested by the program to enable the exchange and use of models among various modelers who use the same, or different modeling tool applications capable of using .xmi files. Contract language also should specify that Conforming and Justified data markings be displayed with easy digital recognition on applicable diagrams or content stored, viewed, or generated from within digital models, including data printed or exported from digital models, associated with:

a. Distribution Statement (DoDI 5230.24).

b. Controlled Unclassified Information (CUI) Label (DoDI 5200.48_DAFI 16-1403).

c. Export-Control Warning (DoDI 5230.24).

d. REL TO Distribution Notice/Banner (DoDI 5230.24, DoDI 5200.48).

e. Restrictive Data Rights Markings (DoDI 5230.24, DFARS clauses 252.227-7013, 7014, and 7015).

f. Copyright Notices (Sections 401 and 402 of Title 17, U.S.C).

g. Configuration Identification (EIA-649-1).

h. Contract number under which the data is delivered.

6.9.2. Managing model relationships with other models. Once the goals, objectives, and expected outcomes for the program's transition to a MBSE approach are established, there is a need to identify and define how the Government and contractor models will integrate, interact, and flow data and information between each other. Establishing techniques and process flows for interoperable model management, along with the creation of a new, or update of an existing Configuration Management Plan (CMP), are necessary to define and maintain consistent communication with stakeholders and industry partners in a digital environment and maintaining model accuracy and getting the most value from all the MBSE activities. Guidance for this activity can be found in the INCOSE Model Portfolio Management Guide (https://portal.incose.org/commerce/store?productId=INCOSE-MPMG).

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6.9.3. Develop an approach for model federation. Model federation is an approach that enables the connection of different system models, such as those created and managed by industry partners and program teams, to establish a system-of-systems model approach. This contrasts with working only with stand-alone system models. Model federation further defines the expectations associated with model/data interoperability, along with models serving as the ASoT. Stand-alone models provide a single, centralized authoritative source for a system's data. Access to these models, and the data and digital artifacts contained or linked within them, can be difficult to accomplish, and common data elements may not exist. Federated system models also provide a network of decentralized and interconnected data sources that enable data and digital artifacts to flow among multiple trusted and authoritative sources supported by common data elements. Guidance can be found in the INCOSE Model Portfolio Management Guide (https://portal.incose.org/commerce/store?productId=INCOSE-MPMG).

6.9.4. Plan for model maturation. How models mature requires consideration when defining the scope of any program office digital modeling effort. As each acquisition milestone is reached, and as modifications and upgrades occur to fielded systems across the life cycle, any MBSE models representing the system baseline will evolve accordingly. This will introduce different levels of system definition. Using the MBSE Style Guide to inform the development of a model and validation framework will help ensure MBSE modeling activities are performed consistently and increases the chances of successful model integration. Guidance can be found in the INCOSE Model Portfolio Management Guide (https://portal.incose.org/commerce/store?productId=INCOSE-MPMG).

6.9.5. Stay current with digital elements found in policy and guidance. Use policy and guidance to help define, guide, and determine the digitally focused decisions and actions used in planning and implementing a program's MBSE approach.

6.10. MBSE and Cybersecurity. Section 5.2.7 provides a high-level overview of cybersecurity considerations for DE. This section covers cybersecurity as it pertains to MBSE.

6.10.1. Modeling integrated cybersecurity design. Whenever possible, cybersecurity modeling and analysis should be integrated into system design models alongside other system design requirements and decisions. Information exchange requirements, RMF technical controls, and cybersecurity-related key performance parameters and system functions should be modeled in the same fashion, and in the same model, as other system requirements and functions. This approach will help maximize collaboration between cybersecurity subject matter experts and other engineers.

6.10.2. Modeling cyber resilience. Cyber resilience is the system's capability to prevent, withstand, and recover from cyber-attacks intended to degrade or defeat the system. Modeling and simulation can examine degraded operations and provide insight into recovery path options. Some areas of modeling resilience may include modeling of the components of the networking architecture, along with modeling policy engines and trust algorithms, system responses to attack paths such as honeypot activities, or hot and cold redundancy to help understand behavior and performance.

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6.10.3. Cyber threat modeling. Threat modeling analyzes what effect a threat has on system capabilities. Designs should focus on reducing and eliminating risk elements as much as on developing capabilities. DE modeling and simulation activities can use threat modeling to improve design decisions. The selected modeling tools should permit quick add, remove, and movement operations of architectural components as more is known about the threat to optimize the technical solution. The context of each threat-modeling activity can use known tactics, techniques, and procedures to uncover unique threat vectors where safeguards will improve security posture. Cyber-actor strategies can be modeled and simulated in set phases. Each phase will allow the design team to focus on specific tactics, techniques, and procedure sets. Output from cybersecurity design provides inputs to the requirements verification traceability matrix, system requirements specification, and system/subsystem specification. Some areas of threat modeling include identifying areas of initial exploitation, persistent access points, malicious tool installation, and adversary lateral movement within the system.

6.10.4. System architectural views. DoDAF and UAF architectural views provide the system's operational concept and key parameters for analysis of cybersecurity. Service and system views, along with the UAF Security Viewpoint, help ensure that applicable cybersecurity controls and standards are integrated into the functional baseline architecture. Requirement traceability matrices demonstrate how cybersecurity requirements are allocated to, and satisfied by, system components. Mission traceability tables identify desired effects, operational and system tasks associated with those effects, and system parts that play a role in the effects, thus identifying mission-essential functions and mission criticality of system components. Implementing RMF and standardization early in the systems engineering process will ensure cybersecurity is accounted for during the System Requirements Review and System Functional Review.

6.10.5. MBSE model content for cybersecurity. As a starting point, MBSE system models should capture at least the data contained in the Cybersecurity Supply Chain Risk Management DIDs DI-SESS-82433 (SBOM) and DI-SESS-82434 (HBOM).

6.10.6. MBSE cybersecurity modeling standards. Ideally, each Military Service should standardize model formats and views for the RMF using MBSE schemas, profiles, shared model libraries, and style guides. Because many RMF mitigations are inherited from systems external to the system under evaluation, it is important to standardize the modeling pattern to ensure that program offices understand the system risk they are inheriting and that models can be read together to provide an understanding of the whole picture. At a minimum, RMF controls and control correlation identifiers should be created as a shared MBSE model to avoid rework across models. Cybersecurity modeling standards should account for the MBSE model content captured in Section 6.8.5 above.

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6.10.7. RMF officials. RMF officials should consult with their Service's MBSE standards team and vendors to ensure that the RMF is compatible with the Service's overall systems engineering modeling standards. RMF officials also should consult with qualified RMF validators, information system security managers, and information system security engineers to confirm that the MBSE model is capturing the correct data. Once the RMF MBSE modeling standard has been piloted, the Service should begin training all RMF qualified validators in reading MBSE models to understand the data contained therein. Service information system security managers, information system security engineers, and information system security officers should be trained in how to enter RMF data into the MBSE system model.

6.10.8. Service cybersecurity. Services may wish to develop additional MBSE modeling standards for other cybersecurity data, such as cyber survivability attributes, cybersecurity risk assessments, cybersecurity tabletop events, or Program Protection Plan tables. Standardizing the formatting of this data can reduce errors in system vulnerability assessments, particularly with integrated systems, and can accelerate workforce adoption of DE practices for cybersecurity. To reduce model rework, the Service MBSE standards should be invoked in contracts that require the contractor to implement DE practices.

6.10.9. Automated cybersecurity assessments. All cybersecurity assessments should be completed independently of the MBSE system model (for instance, in an external tool or in an MBSE analysis model external to the system model). This will help a program avoid overclassifying its MBSE system models, while protecting system data by keeping need-to-know data out of the system model shared by a larger audience; however, cybersecurity assessments should use the MBSE model as the primary or sole source of information to retain a single source of truth.

6.10.10. Cybersecurity assessment tools. Several existing and emerging DoW hardware and software tools can assist with automated cybersecurity assessment of a system by using system scans or MBSE system models. Because many of these tools produce different types of risk assessment, each Service should assess alternatives to determine which tools best capture system risk to meet the Service's needs. When evaluating an automated cybersecurity assessment tool that ingests MBSE system models, each Service also should measure the tool against Service MBSE modeling format standards to determine the rework necessary to use the tool. Programs should avoid having to rework an MBSE model to accommodate an assessment tool. Services are encouraged to negotiate with tool vendors about reading data encoded by the Service's MBSE standards.

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7. TECHNOLOGY MATURATION AND RISK REDUCTION

7.1. Technology maturation and risk reduction DE and modeling practices. This section focuses on the Technology Maturation and Risk Reduction (TMRR) phase of the program life cycle, meaning Technology Readiness Level (TRL) 1-6, before the technology transitions to the Engineering and Manufacturing Development (EMD) phase. Even though the focus of the TMRR phase is to "get the technology to work," adopting DE and modeling practices can be of value to early development efforts and to set up the program for long-term success.

7.2. Analysis of Alternatives/military utility analyses (System or Force-on-Force Simulations). As a materiel concept advances through the technology maturation process before a Materiel Development Decision, a significant amount of data is produced from MS&A and experimentation (including bench level, breadboard, and hardware-in-the-loop/software-in-the-loop (HWIL/SWIL). This legacy of data generation feeds what the Air Force Materiel Command calls "Digital Materiel Management" or DMM[3], which is intended to aid authorities to make decisions on technology and capability investment.

7.3. Lower technology readiness model considerations. In the context of DMM, requirements are data driven, reinforced by models, and lead to common reference architectures. However, during concept development, the content, depth, and fidelity of the data varies across maturity, organizational, and funding levels from 6.1 Basic Research through 6.4 Advanced Concept Development and Prototyping.

7.4. Maturity of models and simulations mirror technology maturity. With respect to MS&A, these levels can be represented as a pyramid with high-fidelity, discrete, physics-level computational data serving as the foundation on which each ascending level is constructed up to low-fidelity, large-scale, campaign-level simulations at the peak (figure 10). Development across the Research, Development, Test, and Evaluation (RDT&E) enterprise spans organizations, commands, and geographic locations, making a continuity of data across levels challenging. The application of DE, specifically the establishment of ASoTs and implementation of digital threads, enables decision makers from program managers to portfolio managers and acquisition decision authorities to access whatever fidelity of data is appropriate to their needs. Ideally, "Truly collaborative designs are built on shared models, shared data, and common infrastructure that set the stage for all subsequent lifecycle phases".[4]

[3] *Digital Materiel Management: An Accelerated Future State*, June 12, 2023. Available at https://media.defense.gov/2023/Jun/12/2003239595/-1/-1/0/DMM%20-%20AN%20ACCELERATED%20FUTURE%20STATE_FINAL_compliant_17AUG23.PDF

[4] *Digital Material Management: An Accelerated Future State*, p.3.

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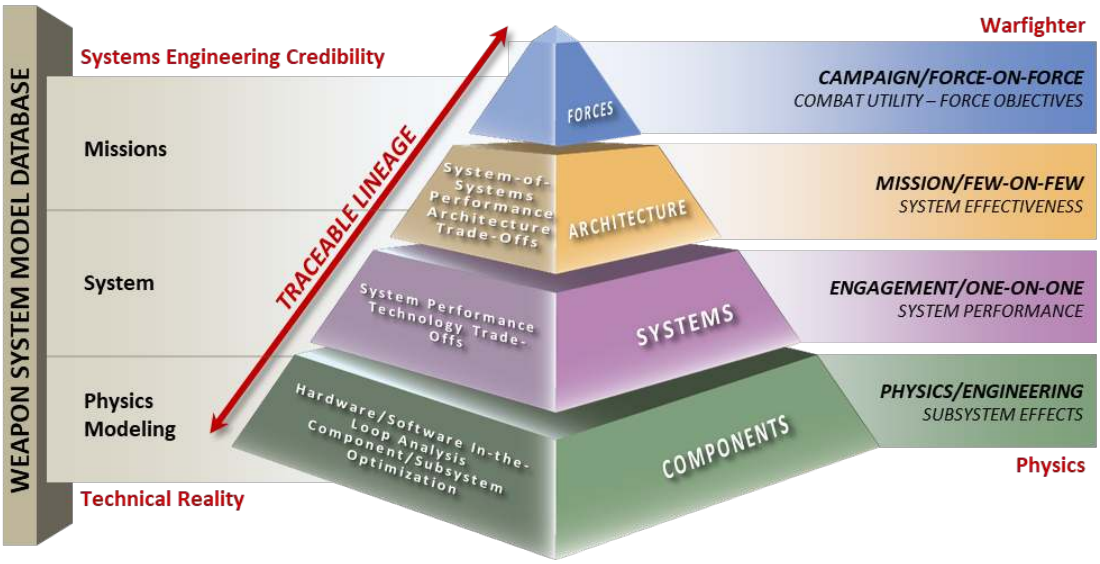


FIGURE 10. MS&A pyramid.[5]

7.5. Benefits of using Analyses of Alternatives in technology development. AoAs that take place before Milestone A are an example of how modeling and simulation can be used in the TMRR phase of a program. Physics and engineering-level MS&A inform the development of technology alternatives (governed by a Technology Alternative Working Group) as Measures of Performance, such as range or payload, are used to down-select alternatives. Then, selected alternatives are evaluated based on Measures of Effectiveness, such as probability of kill P_k or probability of arrival P_a. These values are developed by an Effectiveness Analysis Working Group through analysis at the engagement, mission, or campaign-levels and can include Military Utility Analysis or Capability-Based Analysis. Effectiveness Analysis is then combined with Cost Analysis, Operations Analysis (to include personnel and logistics), and other capability-specific analyses to make acquisition recommendations.

7.6. Networked file sharing enables more efficiencies. Before resources such as SharePoint or DoD SAFE were available, data was transferred among analysis working groups either in person or by mailing CDs or other removable media. This arduous process can be streamlined with digital threads connecting each working group and ASoT serving as the authoritative repository for data.

7.7. DE to reduce risk in test and evaluation. DE can play an important role in experimentation, prototyping, and Developmental Test and Evaluation (DT&E). This may be in terms of mathematical, probabilistic, analytical, structural, behavioral, and requirements modeling as during the technology development phase.

[5] Available via the Virtual Distributed Laboratory portal https://www.vdl.afrl.af.mil/ path: AFRL/RY - COMPASE - Modeling and Simulation > One AFRL – One Fight > Display (Note: user account required for access), also https://www.afrl.af.mil/About-Us/Fact-Sheets/Fact-Sheet-Display/Article/2331811/afrlry-compase-modeling-and-simulation/ AFRL/RY

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7.8. Using M&S to reduce risk. Although experiments often are used to verify and validate M&S, programs also can conduct M&S to apply DE principles in subscale or bench-scale experimentation. An ASoT, defined early in the technology development process, reinforced by a Design of Experiments and anchored by M&S, can be used to define experimental conditions and test envelopes in addition to serving as an ongoing data repository of material, component, or subsystem properties. To incorporate test engineering into this DE transition, we can reference the following: "As digital engineering becomes more standard in system development, the test community must be ready to integrate into digital environments. This paper presented a metamodel and methodology for Model Based Test Engineering using SysML that gives the test community a means to take advantage of MBSE principles, increases technical rigor, and provides a dynamic, data-driven linkage between the system model and the test model. The MBTE approach presented allows for fully physical tests, fully digital tests, or any hybrid approach that the testers of the System Under Test deem appropriate."[6] This ASoT can then be used in the design process for prototyping, HWIL, or SWIL.

7.9. Alignment of technology maturity to scale of experimentation. As technology matures toward acquisition decisions, the scale of experimentation grows to include DT&E, qualification testing, risk reduction testing, or any combination thereof. For hardware, the OUSD(R&E) Technology Readiness Assessment Guidebook (February 2025) defines TRL 6 as "System/subsystem model or prototype demonstrated in a relevant environment"; for software, it defines TRL 6 as "Model and/or subsystem validation in a relevant end-to-end environment. A technology must be at TRL 6 to enter Engineering and Manufacturing Development (EMD)" 10 U.S.C. § 4252(b)(2).

7.10. Demonstration and relevant environment. What constitutes "demonstration" and "relevant environment"? Does a developing technology need to be tested at full scale and for full duration to count as a "demonstration"? What is a "relevant environment" for a system that operates in space when opportunities for in-space testing are few and costly? A well-maintained ASoT, including a digital thread that can tie back to early simulations and design decisions, can be used to define test conditions intended to meet TRL definitions.

7.11. Integrated testing. When initiated early, integrated testing offers significant advantages. Identifying concerns and issues upfront leads to improved system design and more effective development efforts. Early adoption of integrated testing also enhances the overall effectiveness and efficiency of the acquisition T&E program. Successful implementation relies on collaborative planning and execution to ensure the ASoT for models, data, and test artifacts is maintained and accessible via digital threads, supporting independent analysis, evaluation, and reporting by all stakeholders.

[6] J. Coil et. al., "Model Based Test Engineering: Incorporating Test Engineering Into the Digital Engineering Transformation," 2023 Society of Flight Test Engineers International Symposium, Annapolis, MD, 16-19 Oct 2023.

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7.12. Using DE to reduce complexity risk. Before an acquisition development decision, experimentation, prototyping, and developmental testing are conducted in the TMRR phase to reduce risk to the future acquisition program. DE should be employed during all phases of experimentation and test to address the complexity of future defense systems.

7.13. Establishing digital system model confidence using uncertainty quantification. Multiple approaches may be useful for communicating system model confidence to system stakeholders. One of these methods is rigorous uncertainty quantification. In the complex world of engineering, the ability to accurately predict and model real-world phenomena has value. As engineers, we rely on models to guide decision making, design optimization, and risk management; however, it is difficult for any model to perfectly represent reality given the inherent uncertainties. For technology development, uncertainties exist in modeling and simulation predictions, experimental tests to mature technology concepts, and different levels of operational and developmental testing. For human performance as estimated into technology development, human reliability analysis may prove useful (HSI Guidebook, OUSD(R&E) 2022, Change 1 2024, section 2.4.2). Understanding the role uncertainty quantification plays for each type of component, system, or activity, and how they are connected to each other to gain confidence, requires specialized skill sets that may not be ingrained within DoW organizations. Understanding and communicating confidence in model predictions is crucial to enable accelerated product development that leverages DE best practices. Uncertainty quantification provides a framework to assess and communicate the reliability of system-oriented model predictions.

a. Engineering model uncertainty. Uncertainty in engineering models is broadly classified into two categories: aleatory and epistemic. Aleatory uncertainty, or inherent variability, results from natural randomness in the system or process being modeled. Epistemic uncertainty, on the other hand, arises from a lack of knowledge about the system. A lack of system knowledge may arise because of inadequate data, model simplifications, or unknown parameters. Uncertainty quantification aims to address both types of uncertainties, providing a comprehensive view of model confidence.

b. Uncertainty quantification methods. Statistical methods, probabilistic modeling, and sensitivity analysis are used to quantify and reduce model uncertainty.

1. *Statistical methods* involve the use of historical data and statistical techniques to estimate the uncertainty in model predictions. Confidence intervals and hypothesis testing are common tools used to assess reliability of model outputs.

2. *Probabilistic modeling* uses probability distributions to represent uncertain parameters within models. Monte Carlo simulations and Bayesian methods are widely used in probabilistic modeling, offering a way to incorporate and propagate uncertainty through models.

3. *Sensitivity analysis* evaluates how variations in model input parameters affect outputs. This method is useful for identifying parameters that contribute to output uncertainty and may be used to guide campaigns to improve model accuracy.

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c. Communication of model confidence.

1. *Effectively communicating model confidence* is as crucial as the uncertainty quantification activities used to generate the underlying statistics. As engineers, one of our roles is to convey the level of uncertainty in model predictions with stakeholders who may not have a technical background. This communication should focus on the implications of uncertainty on decision making and risk assessment.

2. *Visual representations*, such as confidence intervals, probability distributions, and sensitivity analysis plots, are powerful tools for conveying complex uncertainty information in an accessible manner. In addition, engineers should provide clear explanations of the sources of uncertainty and the measures taken to minimize their impact.

7.14. Government-led versus industry-led technology maturation and risk reduction considerations. Design and development considerations may differ depending on whether the TMRR phase of the program is being conducted by in-house Government organizations or through a contracted effort to industry or academia. Knowing those differences and being able to adjust the DE strategy for the program can help the program gain the most value from adopting DE. Considerations include:

a. Government.

1. *Developing models for the TMRR effort to support model-based acquisition*. Having models already in the format and at the appropriate level of detail will help with future engineering maturation and design, as well as with model-based acquisition. Using standard practices for the DEE, architectures, and ontology will allow for smoother adoption and easier integration.

2. The use of models and simulations for AoAs, Military Utility, or Force-on-Force effectiveness can help the program identify both high-payoff and best value technologies for the application, helping to focus TMRR efforts and down-select candidate considerations. These models should be representative of the actual technology (as is best known at the current maturity level) and must be input into simulations that fit the required mission profile(s). Expanding the basic functional simulations to military utility and force-on-force scenario-based simulations will further help the program identify the strong and weak points of the candidate technologies for a more objective and comprehensive analysis and therefore a more data-driven decision and selection.

3. *Models or simulation data contribute to the digital thread* by linking to requirements and helping build the foundation for what can be used later in the development life cycle. Models and simulation data also provide traceability for how and why decisions were made and what data a program used to make those determinations.

4. *Having Government-owned technical data in a detailed model* allows for a more competitive production environment, increased flexibility in program operations owing to full understanding of the technology, and reduced time to prepare for procurement actions because the technical data is complete and kept up to date. Government-owned technical data allows for other SE design considerations such as MOSA and SoS integration. However, the program team

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is only required to develop an IP strategy to acquire the technical data necessary to manage the program.

b. Industry and academia.

1. *When procuring technical data during the TMRR phase* from an industry or academic partner, models allow for more objective and analytical deliverables both to and from partners. This improvement in deliverables can help programs identify CDRLs and formats along with enabling a clear technical understanding from industry and academic partners through model-based acquisition. Saving time on translating text-based documents to ensure common understanding is a significant benefit.

2. Implementing DE and modeling practices also allows for early integration and interfacing between Government and industry DE tools. Ensuring the data is in the format and includes elements necessary to integrate with the DEE will reduce the time needed to ensure integration and can eliminate a need for translation into the accepted format and required data elements. The RFP should describe the expected coordination between Government and industry or academia. Failure to clearly articulate the Government's expectation for format in the RFP may result in additional costs to convert the data into a format that can be utilized by the DEE.

3. *Using DE and modeling practices permits a more objective evaluation of proposals from industry and academia* when offerors' proposed DSMs are based on the Government Reference Architecture (GRA). In this case, the Government provides the GRA in the RFP, and the offerors submit their initial proposed DSMs with more detail at lower levels of indenture.

4. *Using DE and modeling practices facilitates the source selection team's evaluation* of each offeror's proposed IP rights for CDRLs that require delivery of digital artifacts.

7.15. Model-based development to feed model-based acquisition. Using the TMRR phase to incorporate DE principles and modeling practices allows many program benefits, such as:

a. *A more objective evaluation of proposals from industry and academia.* Evaluators can trace the offeror's proposed detailed model to the Government's logical model.

b. *Easier integration and interfacing between Government and industry DE tools*. Once the data elements and format of the models are identified and agreed upon, integrating the tools using a DEE should be consistent throughout the life cycle of the program. Using a data-centric rather than a document-centric approach allows the program to maintain a complete set of data in the correct format to be used across different systems integrated into the same DEE. This greatly helps gain value from the DEE because of faster access to information, more objective decisions, and data assurance as the program is accessing the actual data rather than copying it or relying on data entry.

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c. *Earlier use of models in simulations for AoA, military utility, or force-on-force effectiveness.* This earlier use of models in a program can promote data-driven decisions and better-informed analyses.

d. *Starting to create the digital thread and linking requirements to models early in the life cycle* allows for those benefits to progress without changing the workflows later in the program. If changed later, the program would add the cost and time of transferring or translating information into the proper formats and architecture. As with many of the Systems Engineering design considerations and principles, DE and Modeling Practices are more easily and less costly incorporated into the program early on.

7.16. Verification and validation to enable model assurance. Verification and validation are critical digital TMRR activities. Verification ensures that a model aligns with design specifications. Validation ensures a model is suitable for the intended use and that it accurately represents the real-world scenario it simulates. Transitioning from document-centric to data-centric processes enables earlier opportunities for verification and validation concepts and constructs to be explored, particularly as new technologies are identified and multiple technologies are integrated to deliver new capabilities. Blending verification and validation activities within TMRR provides engineers with the ability to establish the reliability and accuracy of DSMs to accurately represent the real world and perform as expected. These activities also quantify the potential of a technology to satisfy intended goals and objectives.

7.16.1. Verification activities. Verification activities during digital concept development are required to increase confidence in models used to represent system behavior performance. Rigorous application of these techniques is necessary to uncover potential system failures resulting from subsystem interactions that are difficult to capture with document-based acquisition approaches.

a. Code review. Software review and static analysis involves detailed examination of source code to identify errors and ensure adherence to software development standards and practices without executing the software. Workflows may be implemented to automate parts of this process to identify software bugs or vulnerabilities.

b. Testing. Unit testing involves testing individual components or units of the model in isolation to ensure each part functions correctly. These activities aid with identifying errors in implementation of model algorithms. Integration testing is performed after unit testing and explores how different components of the model interact with each other. These activities are useful for identifying issues between model components and for highlighting coupling between components that may drive system response. Regression testing involves regular re-execution of simulation tests after model changes are implemented to ensure that previous functionality remains intact and new changes have not introduced new errors. Finally, benchmark testing is a formal activity to compare the model's performance against established benchmarks and reference models to verify its computational accuracy and efficiency.

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7.16.2. Validation activities. Validation should be considered in a DSM's design process to help establish and communicate credibility of an ASoT. A traditional validation approach in early concept design involves comparing model predictions with data gathered from real-world experiments or observations. This approach is critical for confirming a model's accuracy in simulating real-world phenomena. Other validation activities include:

a. Sensitivity analyses examine how variations in model input parameters affect outputs. This type of analysis increases stakeholder understanding of model robustness and may identify critical parameters that significantly affect model behavior.

b. Uncertainty quantification activities allow the model builder to assess the impact of known uncertainties in model outputs and predict a range of confidence that can be assigned by stakeholders to predicted outputs and behavior. Uncertainty qualification is mandatory for providing guiding assignment of confidence levels to model predictions.

c. Scenario testing evaluates model performance for different scenarios, including extreme or edge cases. The rationale for performing this testing is to ensure a model is robust and can handle a wide range of conditions within its intended application domain.

7.17. Peer review. An aggressive test of model confidence is provided by frequent peer review and application of expert judgment. These activities involve having the model and its results reviewed by external subject matter experts that have no conflict of interest related to a digital concept's success or failure. Peer review also can be useful for identifying oversights and providing feedback to improve the model. Engineering teams can also cross-validate with other models by comparing predictions with predictions from established models that simulate similar phenomena. Real-world testing and pilot studies provide the best test of model assurance. In this environment, teams implement models in a controlled real-world setting to assess how well the digital system model will predict or simulate actual outcomes.

7.18. Modular open systems approach (MOSA). This handbook focuses on DE and modeling practices rather than MOSA; however, these topics are complementary. DE and MOSA are based upon the GRA and the contractor's DSM properly identifying all major system components, modular systems, and modular system interfaces as such. In so doing, the DSM will allow for configuration management and communication of the approved (or authoritative) source of truth for analysis and temporary (requests for variance) and permanent changes (ECPs) to the program's technical baseline. Similarly, having MOSA defined through the GRA and the contractor's DSM for the program will permit easier analysis of replacement major system components, modular systems, modular system interfaces, and compliance with contract requirements. More detail on MOSA can be found at https://www.cto.mil/sea/mosa/.

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7.19. Meshing different levels or echelons of models to inform early development and design. Using good modeling practices early in the design and development life cycle will allow the program to more easily visualize, analyze, and assess the overall design and interfaces by meshing different levels of models or creating interfaces with existing models. Using DE principles to control the configuration through ASoT and linkages to requirements and testing will also allow for benefits in both (1) determining how the levels/echelons of technical data fit together, and (2) determining design consideration that may benefit the overall system or SoS by visualizing the tiered levels or echelons of models.

7.20. Digitizing airworthiness and seaworthiness. The Warfare Acquisition University (WarU) defines airworthiness as the property of an air vehicle/aircraft system configuration to safely attain, sustain, and terminate flight in accordance with approved usage limits. For technology development organizations, traditional airworthiness is a document-based process that relies heavily on the inputs and active feedback from subject matter experts with deep knowledge levels and experience with airworthiness reviews and feedback mechanisms to programs of record.

a. The Military Services and their programs use different approaches to achieve airworthiness, and these also differ in some fashion from commercial and Government entities such as the Federal Aviation Administration (FAA). From the perspective of the Air Force, MIL-HDBK-516C establishes the airworthiness certification criteria to be used in the determination of airworthiness of all manned and unmanned, fixed, and rotary wing air vehicle systems.

b. Best practices for digitizing the implementation of MIL-HDBK-516C are evolving and include the transition from using document-based artifacts and processes to digital systems, which can enhance efficiency, accuracy, and accessibility of the artifacts. Implementing a digital airworthiness management system requires careful consideration to ensure compliance with regulatory standards, security, and interoperability.

c. From a broad airworthiness perspective, common best practices for digitizing document-based airworthiness processes are adherence to regulatory standards, data integrity and security, interoperability, user training and support, quality control and assurance, scalability and flexibility, transition planning, stakeholder engagement, document conversion and validation, and implementing metrics to assess continuous improvement.

d. Adherence to regulatory standards ensures that the digital system complies with all relevant regulatory requirements. This approach includes regulations from other Government entities such as the FAA and NASA. At appropriate places in the digitized process, digital artifacts and processes must be recognized as legal equivalents to their document-based counterparts.

e. Programs should observe data integrity and security best practices as part of robust data management. Best practices include using secure servers, encryption, regular backups, and access controls to prevent unauthorized access and data breaches. Regular audits and updates should be conducted to maintain organization specific risk postures.

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f. To enable a digitized airworthiness process, the digital system model should be compatible with existing aviation standards for data exchange and integration to the greatest extent possible. This approach ensures that the system can communicate with other IDEs to facilitate sharing information between the operator, maintenance organizations, regulatory bodies, and manufacturers. For military, Government, and commercial entities, the stakeholders within their respective organization may follow similar but essentially independent guidelines unless interaction between these groups is considered during development of a system's digital Acquisition Strategy.

g. User training and support for a digital airworthiness process is necessary to ensure users understand how to leverage the IDE and assessment tools effectively. Continuous support should be provided to address issues that arise during transition from document-based to model-based airworthiness activities and regular use. Quality control and assurance measures are useful to ensure the accuracy and reliability of digital artifacts produced by a digitized airworthiness activity. These activities include validation checks, regular audits, and user feedback mechanisms to identify and correct inaccuracies or issues with the IDE. A digital airworthiness process should also be scalable to accommodate growth to large numbers of use cases and flexible enough to adapt to changing regulatory requirements and technological advances. This approach ensures that the IDE that hosts a digitized airworthiness process remains effective and relevant over time.

h. Transition planning from document-based to model-based airworthiness determinations should occur to communicate timelines, milestones, resource allocation, and risk management strategies to ensure smooth and efficient operations. Engaging all activity stakeholders increases confidence that a model-based airworthiness process satisfies requirements and meets the expectations of all parties involved. This stakeholder group may include regulatory authorities, system operators, system manufacturers, and other subject matter experts that collaborate within a specific DEE.

i. For legacy systems, there may be value, on a case-by-case basis, in transitioning from document-based artifacts to model-based artifacts. Several legacy systems within DoW have explored this approach, with varying levels of success. Conversion from existing document-based artifacts to new digital formats requires activities to ensure the transformation is accurate and complete. Validation processes should also be implemented to confirm the integrity of the converted artifacts.

j. For new and legacy systems that must address airworthiness concerns and issues, it is useful to establish mechanisms to allow for continuous monitoring, feedback, and improvement of the digital airworthiness process. This activity includes staying informed about technological innovations and regulatory changes that could impact the digital system model.

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k. Seaworthiness is not managed like airworthiness. Seaworthiness is generally defined as a vessel being built to maritime engineering and design standards along with having passed required tests and safety checks. Worldwide seaworthiness standards are established by the International Maritime Organization and in the United States by the American Bureau of Shipping, Coast Guard, and Navy. Within these organizations there are multiple aspects of seaworthiness handled under separate technical or certification areas. DE and ship engineering (structure and systems) modeling take place at varying levels within Coast Guard and Navy research activities, program offices, and shipyards. Some of these general points about airworthiness may apply to the transition of document-based seaworthiness processes to digital formats.

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8. DIGITAL TECHNICAL REVIEW

8.1. General. Digital technical review (DTR) refers to the process of leveraging MBSE tools and artifacts in System Engineering Technical Reviews (SETRs). Participants in the DTR process align with the acquirer-supplier agreement mode, as specified in ISO/IEC/IEEE 24748-8. SETRs are made up of review events, such as the System Requirements Review (SRR), Preliminary Design Review (PDR), and Critical Design Review (CDR). These reviews are currently implemented as events occurring at points within a product life cycle where the system's technical maturity is evaluated using contractually agreed upon metrics and artifacts. Each technical review or audit should have defined entrance and exit criteria tied to the required level of technical maturity for the product design and development to include all technical disciplines. DTRs, when successfully evaluated, demonstrate that the program is on track to achieve its final goals and is ready to proceed to the next acquisition phase. The acquirer and supplier work together to define requirements, establish agreements, and ensure that the delivered system meets the specified needs. Both the acquirer and supplier are responsible for the delivery of their required artifacts and meeting the metrics that have been contractually agreed upon. The acquirer is also responsible for formally accepting the artifacts and for overseeing, approving, and participating in the technical review process.

8.1.1. Artifacts and traceability. MBSE artifacts can be used in lieu of traditional document-based artifacts to provide decision makers with a holistic view of the system, which enables the acquirer to execute the review more effectively.[7] DTRs enable direct and robust traceability between SETR criteria and the corresponding artifacts. Mapping this traceability in a digital, interactive format significantly enhances efficiency compared with traditional document-based methods. Traceability can manifest in a variety of ways, such as with SETR checklists and traceability reports. Artifacts can be explicitly linked to entrance and exit criteria ensuring clear alignment among requirements, design elements, or test cases. This digital traceability streamlines the navigation through data to criteria to provide a holistic view of progress. DTRs also reduce the ambiguity often found in traditional SETRs, where criteria and artifacts may be distributed across multiple documents and environments. All artifacts presented should be version-controlled, and a baseline copy should be submitted prior to each review. This baseline will serve as the official record of the design at that point in time and must include sufficient metadata (e.g., version, date, description) to ensure traceability and configuration control.

8.1.2. SETR process overview. Figure 11 shows a high-level overview of the SETR process. Technical reviews are conducted within the acquisition pathways, illustrated previously in figure 6, to establish technical requirements, assess system maturity, and evaluate risks related to system performance, cost, and schedule. Figure 12 shows the relationship between technical reviews and audits and the technical baselines across the acquisition life cycle.

[7] The WarU Adaptive Acquisition Framework Document Identification webpage (https://www.waru.edu/aafdid) provides a breakout of required artifacts which program staff can review and extrapolate to a digital format.

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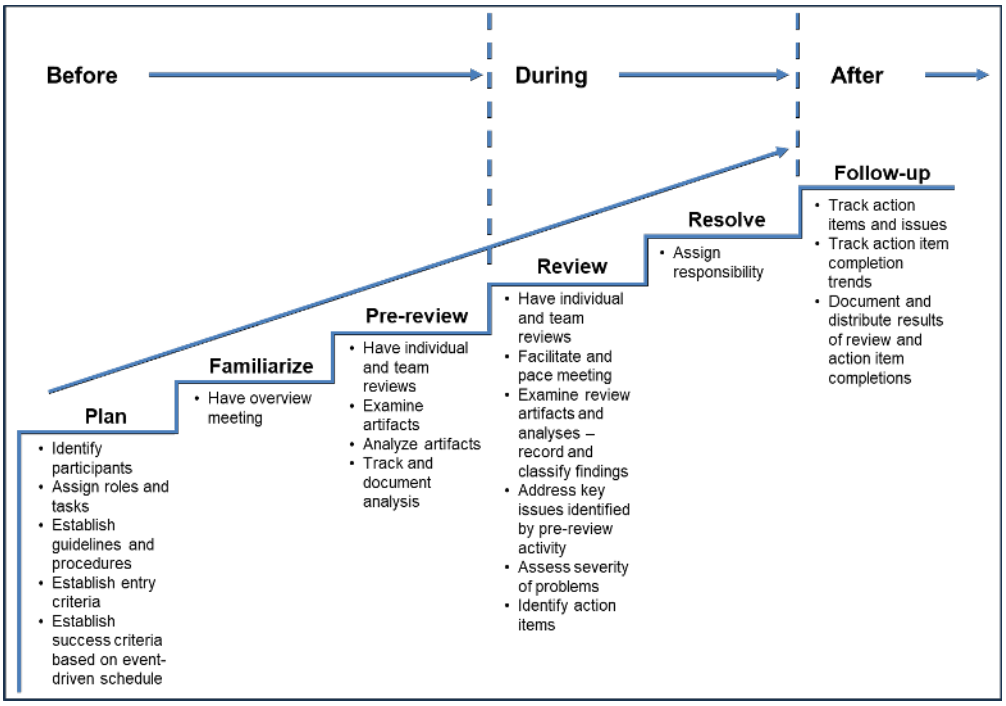


FIGURE 11. Technical review process from SE Guidebook 2022.

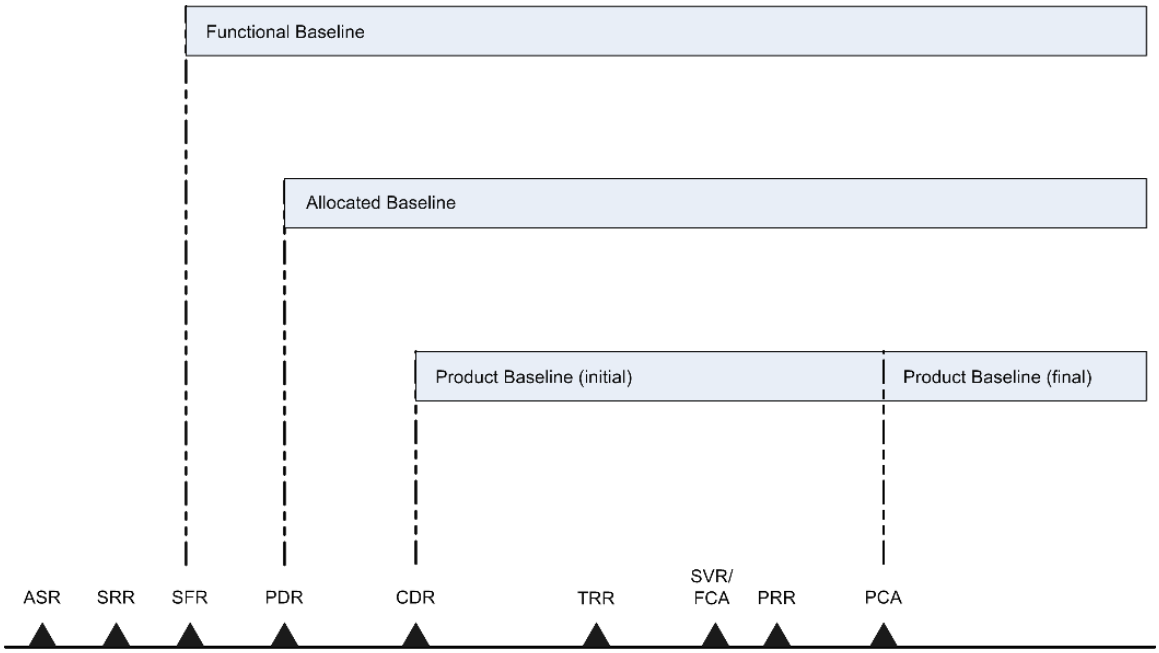


FIGURE 12. Relationship between technical reviews and audits and the technical baselines across the acquisition life cycle from ISO/IEC/IEEE 24748-8-2019.

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8.1.3. DTR tools. Various MBSE models exist within the DoW DEEs to facilitate DTRs. Figure 13 illustrates the DE framework. DTR tools are part of the DE software infrastructure, providing guidance for technical reviews and facilitating the development and delivery of digital artifacts.

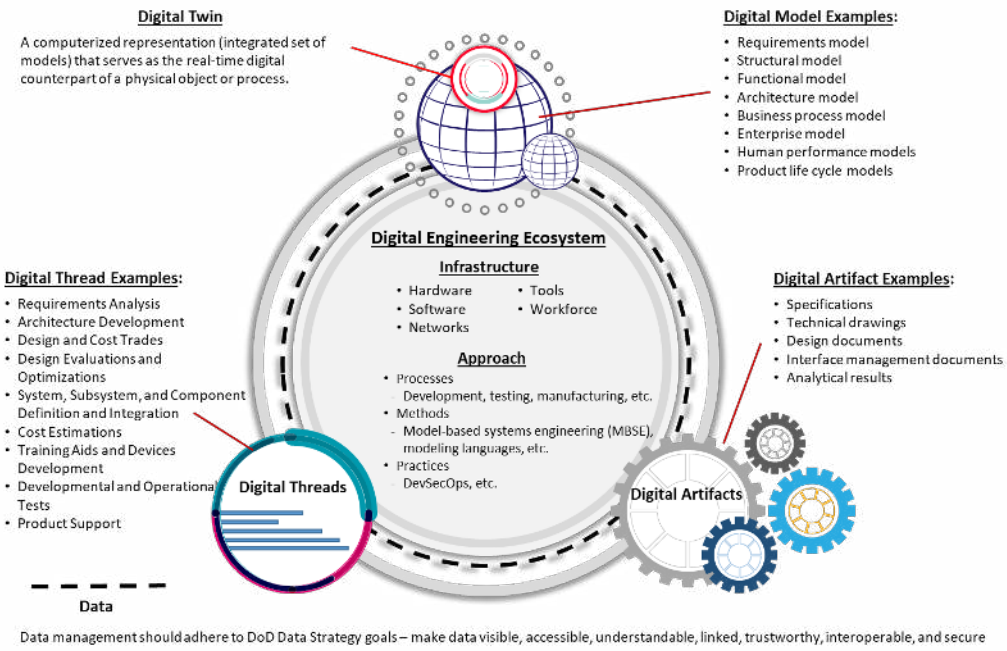


FIGURE 13. DE framework from DoD Instruction 5000.97, Digital Engineering.

8.1.4. DTR Tool Examples. Figures 14 and 15 show the navigation pages of two tools built within Cameo Systems Modeler using SysML: the NAVWAR DTR toolkit and the Air Force Digital Acquisition and Sustainment Operations (DASO) and GRA SETR. Figure 16 shows an example of an Integrated Systems Engineering Environment (ISEE). ISEEs provide collaborative capabilities that directly assist in the execution, tracking, configuration management, and reporting of SETRs and SETR events. These types of tools help to generate artifacts, guide users, and track execution status in various milestone reviews and associated DTR events such as PDR, CDR, etc. Programs are encouraged to explore available tools, which are commonly found in a Service's DEE. These tools can provide a standardized and repeatable foundation for executing DTRs with standardized views, standardized model assessments, and read-only access for all stakeholders; however, as reviews differ depending on their tailored criteria, it is important to prioritize review execution based on established entrance and exit criteria, using these tools as appropriate to meet review objectives. DTR models must satisfy the required criteria and enable access to essential review artifacts, regardless of the use of supplementary DTR tools.

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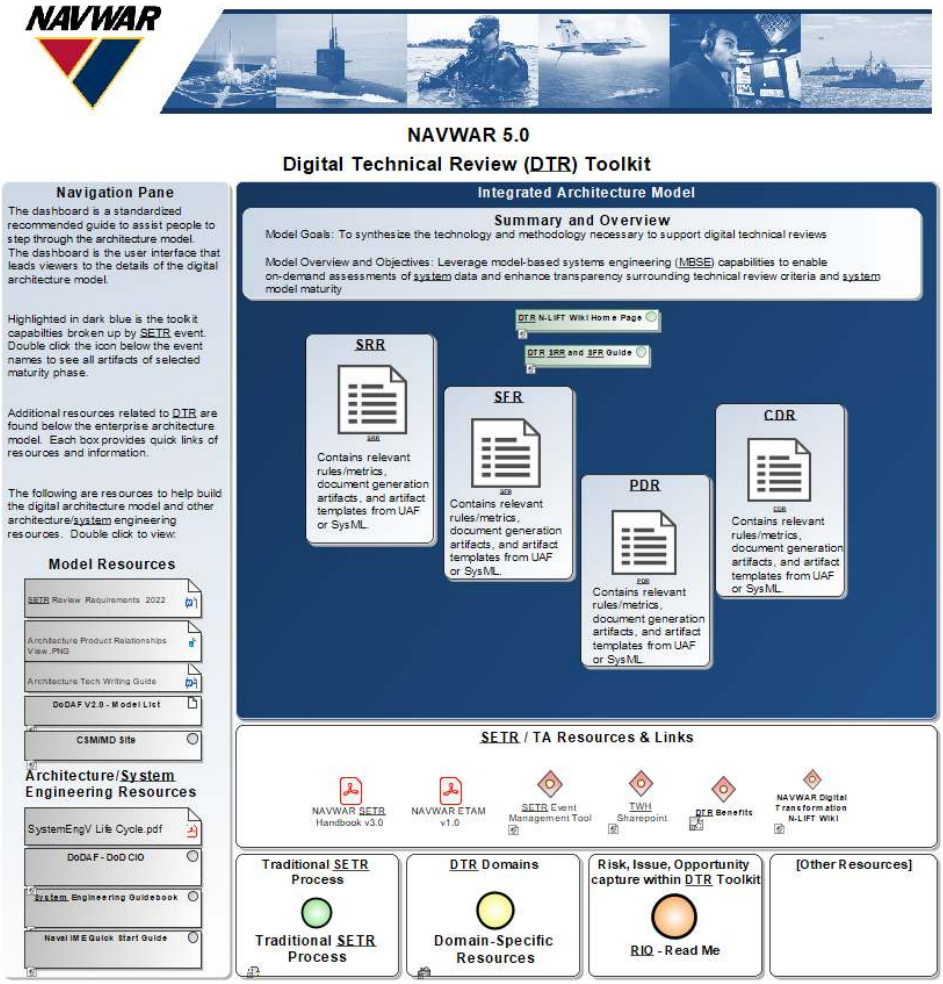


FIGURE 14. NAVWAR DTR toolkit home page.

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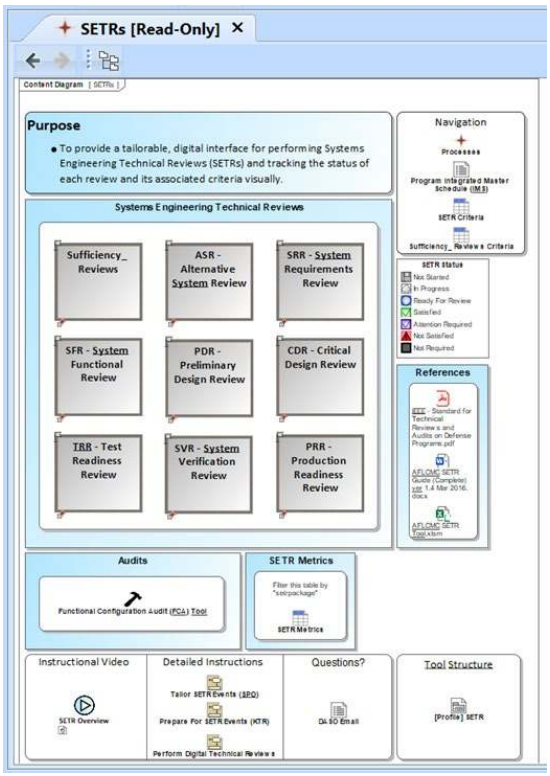


FIGURE 15. DASO weapons GRA SETR home page

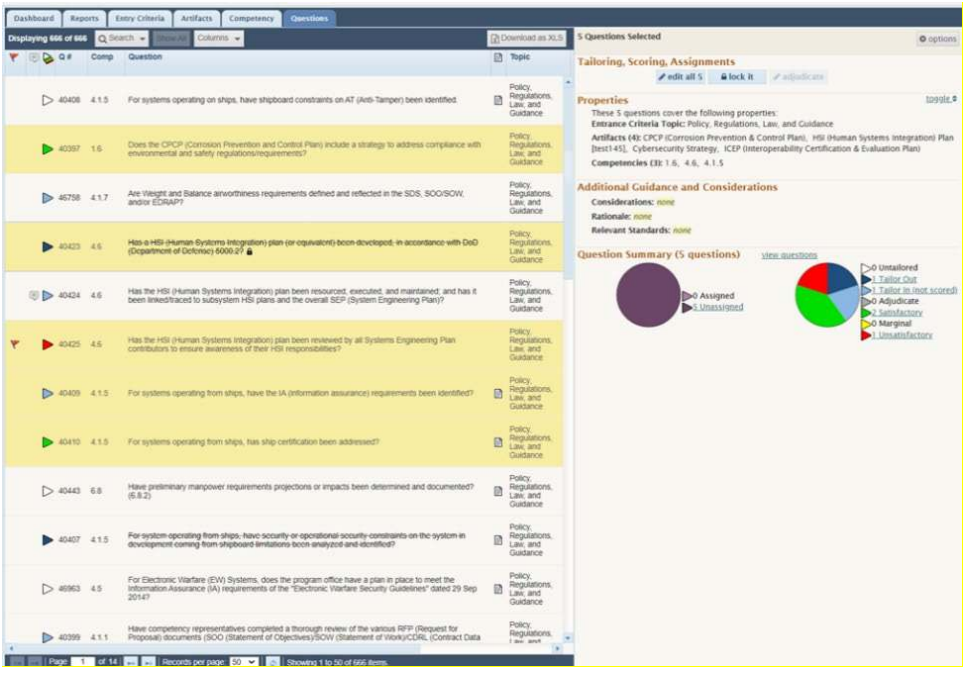


FIGURE 16. Integrated systems engineering environment dashboard.

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8.2. Systems Engineering Plan (SEP) considerations.

8.2.1. The SEP and DE. As specified in the SEP Outline, the SEP is a planning and management tool, specific to the program and tailored to meet program needs. The SEP defines the methods for implementing all system requirements having technical content, technical staffing, and technical management. The SEP should also describe how software, such as DTR tools, will be incorporated into the program-level technical review and audit process. Specifically, for system-level technical reviews, audits, and technical baselines, the SEP should describe how DE activities will be integrated into the program-level SE processes and acquisition documents and models. DE-specific information will be in the Digital Engineering Implementation Plan as Appendix E to the SEP. This plan will summarize the program's DE strategy, goals, objectives, and overall approach. It should identify the individuals responsible for DE activities as part of its SE process. The plan should detail the program's approach to establishing, maintaining, revising, communicating, controlling, and using models within a continuous DEE. It should list the automated tools used and their purposes, and it should include tools for modeling and simulation, system design, building, testing, maintaining an ASoT, and ensuring system security and survivability.

8.2.2. The SEP and Systems Engineering Management Plan (SEMP). The SEP outlines the acquirer's strategy for how it intends to conduct technical reviews. If contractually specified as a deliverable, the supplier then develops a SEMP to align with the acquirer's SEP. These plans identify minimum requirements and define DTR metrics for proper execution. The SEP should document entrance and exit criteria as well as when the acquirer will approve the critical work elements, products, and maturity metrics associated with the reviews. Detailed guidance for developing criteria for each review event can be found in the SE Guidebook and in the ISO/IEC/IEEE 24748-8. The SEP should identify participating members of the DTR event by duty title. DTR event criteria and artifacts should be properly tailored to the specific event and should be documented and approved in the program's SEP before moving to the next phase of the technical review. For more information regarding the SEP, see the SEP Outline and the SE Guidebook.

8.3. Hybrid technical review considerations. As of this writing, the technical review process is in transition from a traditional document-based approach to a model-based standard. During this phase, it is common for technical reviews to be executed using a hybrid approach, using both traditional document-based and model-based DTR tools. Although the goal is to conduct reviews in a fully digital, model-based environment, it may be necessary to incorporate traditional methods to address gaps in the model-based format. The focus should be on meeting all deliverables and exit and entrance criteria, regardless of the methods used. Over time, the review process will evolve to be conducted entirely using digital tools.

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8.4. Digital technical review process. The DTR process is a form of SETR that includes the integration of MBSE tools, methodologies, and models. Incorporating digital capabilities is the element that distinguishes the DTR process from the SETR. As specified in DoDI 5000.88, SETRs provide a venue to establish technical baselines, assess the system's technical maturity, and review and assess technical risks. At each technical review, the program manager will, to the extent practicable, use information from the digital ASoT to assess identified risks, issues, opportunities, and mitigation plans to understand cost, schedule, and performance implications. As an example, the F-35 Program Office approach to conducting reviews is shown at https://www.waru.edu/sites/default/files/2024-06/DAU%20Let%27s%20Get%20Digitial%20F-35%2020240627.pdf

8.4.1. Planning and participant identification. DTRs should be maturity driven and conducted only when the system under development meets the agreed-upon entrance criteria and participating parties are identified. Detailed guidance for developing criteria for each review event can be found in the SE Guidebook and ISO/IEC/IEEE 24748-8. These criteria should be documented in the program's SEP. Participating members of the DTR are affiliated with the acquirer or supplier parties. For each DTR, a technical review chair is identified and is responsible for evaluating DTR artifacts and determining whether the criteria are met and action items can be closed. As specified in the SE Guidebook, the technical review chair is designated by the acquirer party and could be the PM, systems engineer, or other subject matter expert (SME) selected according to the Service's guidance. The technical review team, as designated by the acquirer party, is the body responsible for conducting the technical review or audit. Under the direction of the chair, the review team ensures the entrance criteria are met, evaluates the technical products under review, and documents action items as required to correct deficiencies found during the technical review or audit. Additional details about the responsibilities of the technical review team members can be found in the SE Guidebook and ISO/IEC/IEEE 24748-8. The acquirer and supplier parties may both use DTR tools in developing artifacts for the system under review and are responsible for providing access to the relevant models and artifacts that will be subject to review. The minimum normative properties for each technical review and audit to which all parties should agree are:

a. Review or audit name.

b. Purpose (why perform this technical review or audit?).

c. Description (what system properties does it address?).

d. Timing (when in the system life cycle or contract performance does it occur?).

e. Entrance criteria, content, and exit criteria.

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8.4.2. DTR tailoring. The DTR process allows for the tailoring of entrance and exit criteria to fit to the level of design or development maturity necessary for each DTR event. The supplier and acquiring parties should collaborate to determine how these criteria will be tailored to fulfill the goals for each DTR event. Once the criteria and artifacts have been properly tailored to the specific event and documented and approved in the program's SEP, the acquirer and supplier parties can then move into the next phase including artifact and model preparation. The acquirer and supplier parties should ensure the contract includes delivery dates of required contractual artifacts and deliverables well in advance (e.g., 30 days) of all DTR event start dates to allow adequate contractual process time between the supplier and the acquirer to accept the delivery of deliverables before the start of independent reviews of the deliverables for a DTR event.

8.4.3. DTR preparation. Fulfilling the DTR preparation phase involves the development of necessary data, model views, and review artifacts. Detailed listings of these artifacts can be found in the SE Guidebook. Suppliers are responsible for the creation of artifacts and the meeting of model requirements during this phase. The artifacts and model data should be organized into a format that allows reviewers to navigate the model(s) with ease, with relevant views and artifacts readily visible. Acquirers, suppliers and any other relevant DTR reviewers must be connected to the relevant DEE to ensure they have access to the necessary capabilities to conduct the review. As specified in DoDI 5000.97, a DEE may include, but is not limited to, acquirer-to-acquirer, supplier-to-acquirer, and supplier-to-supplier digital collaboration. These collaborative digital environments are key to involving all stakeholders and reviewers in developing models, executing simulations, and performing analysis and optimizations for digital models or digital twins. In some instances, customers, regulators, contractors, suppliers, or operators must be integrated into the DEE to complete the digital thread. Acquirers and suppliers must ensure accessibility to their models by sharing necessary files, uploading the models to shared locations, and granting necessary permissions. Connection to DEEs should be documented in the SEP. A properly prepared DTR should include the following details:

a. Chairperson.

b. Program Management Office (PMO) participants.

c. Anticipated stakeholder.

d. Participant organizations.

e. Purpose statement.

f. Entrance criteria.

g. Exit criteria.

h. Products.

i. Models/review artifacts for each review.

j. Details and examples for each category can be found in table III.

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TABLE III. Technical review and audit details (mandatory) (sample).

XXX Details Area	XXX Review Details *(Fill out tailored criteria for this acquisition phase, etc.)*
Chairperson	Identify the Technical Review Chair.
PMO Participants	Identify Positions/functions/IPTs within the program offices which are anticipated to participate (Engineering Leads; Risk, Logistics, and Configuration Managers; DCMA Rep., and Contracting Officer, etc.).
Anticipated Stakeholder Participant Organizations	Identify representatives (stakeholders) from Service SE and Test, OUSD(R&E) external dependent programs, the User, and participants with sufficient objectivity with respect to satisfying the preestablished review criteria. For ACAT ID programs, ensure that OUSD(R&E) receives invitations to attend subsystem-level reviews as well as the system level reviews (particularly PDR and CDR) to allow for their independent Post-PDR and Post-CDR assessment. Independent Review Team (IRT) (for MDAPs)
Purpose (of the review)	Describe the main purpose of the review and any specific SE goals.
Entry Criteria	Identify tailored Entry Criteria established for conducting an event-driven review. (Criteria should be objective and measurable/observable.)
Exit Criteria	Identify tailored Exit Criteria. (Criteria should be objective and measurable/observable.)
Products/Artifacts (from the review)	List expected products from the technical review (for example): • Established system allocated baseline • Updated risk assessment for Engineering, Manufacturing, and Development • What artifacts constitute the baseline • Assessment of SW development progress • Updated Cost Analysis Requirements Document (CARD) or CARD-like document based on system allocated baseline • Updated program schedule including system and SW critical path drivers • Approved Life-Cycle Sustainment Plan updating program sustainment development efforts and schedules. • Updated list of list of Key Risks, Issues, and Opportunities and accepted Mitigation Plans/Strategies where appropriate.

Source: DoD SEP Outline v4.1, Table 3.2-11.

8.4.4. Review execution. The execution phase of the DTR process involves a thorough analysis of models and artifacts developed during the preparation phase. The review team conducts the technical review or audit and assesses the review products according to the tailored acceptability criteria, in accordance with the focus of the specific technical review. Reviewers should review the digital artifacts before the scheduled technical review meeting and provide comments that refer directly to the artifacts and model(s). Suppliers and acquirers should work together to communicate when artifacts are ready for review. These artifacts should be reviewed beforehand along an agreed upon timeline to ensure that their content can be effectively discussed during the review event.

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8.4.5. Review criteria. Artifacts should be evaluated with emphasis on the event's established entrance and exit criteria. Detailed guidance for developing criteria for each review event can be found in the SE Guidebook and in the ISO/IEC/ IEEE 24748-8. These criteria should be documented in the program's SEP. Acquirers should ensure the model meets the requisite level of detail for the given review stage, and emphasis should be placed on identifying any maturity gaps, issues or risks in the system that arise from the review process. Action items should be identified to address these issues and followed until completion. At any given technical review or audit, the chair oversees and manages the review or audit. The technical review or audit itself is conducted by the technical review team and approved by the chair. Chair approval of any given technical review or audit should include the following:

a. Satisfying the exit criteria agreed upon for the technical review or audit.

b. Approval of the disposition of action items generated during the technical review or audit.

c. Approval of the technical baseline, and cost baseline as applicable, evaluated during the technical review or audit.

d. Confidence that the maturity of the technical baseline (and cost baseline as applicable) is sufficient to proceed to the next stage of the program.

e. Assessment that the risks associated with the system are documented, have resourced and funded mitigation plans, and are at an acceptable level to support completing the technical review or audit.

f. Chair approval of the formal closure documentation.

8.4.6. Metrics. The specific metrics for what constitutes the completion of a DTR will vary depending on the type of event and its tailored criteria. As part of technical review and audit reporting, the acquirer should:

a. Ensure that the technical review or audit closure and approval (including the applicable content) are documented.

b. Ensure that the technical review or audit summary report is approved and distributed.

Further details on approval metrics and technical review closure actions can be found in ISO/IEC/IEEE 24748-8.

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9. PRODUCT LIFECYCLE MANAGEMENT (PLM)

9.1. General overview. Weapons Systems acquisition PLM is a comprehensive strategic business approach focused on collecting and reusing the data generated throughout the life cycle of a weapon system (figure 17). PLM involves the integration of people, processes, business systems, and information across the entire product line and organization to maximize efficiency and productivity. The system:

a. Manages the data, drawings, and models.

b. Allows for collaboration and communication between users.

c. Includes engineering and design tools.

d. Automates workflow and information.

e. Tracks configuration, facilitates configuration management.

f. Manages manufacturing and quality.

g. Manages suppliers, facilitates supply chain management.

h. Generates manufacturing information.

i. Generates operation and support requirements, manuals, and schedules.

j. Analyzes and reports key performance indicators and identifies where operational performance is not meeting design parameters.

k. Provides security to the data and compliance to required standards.

l. Is scalable and flexible to handle the program across the Department.

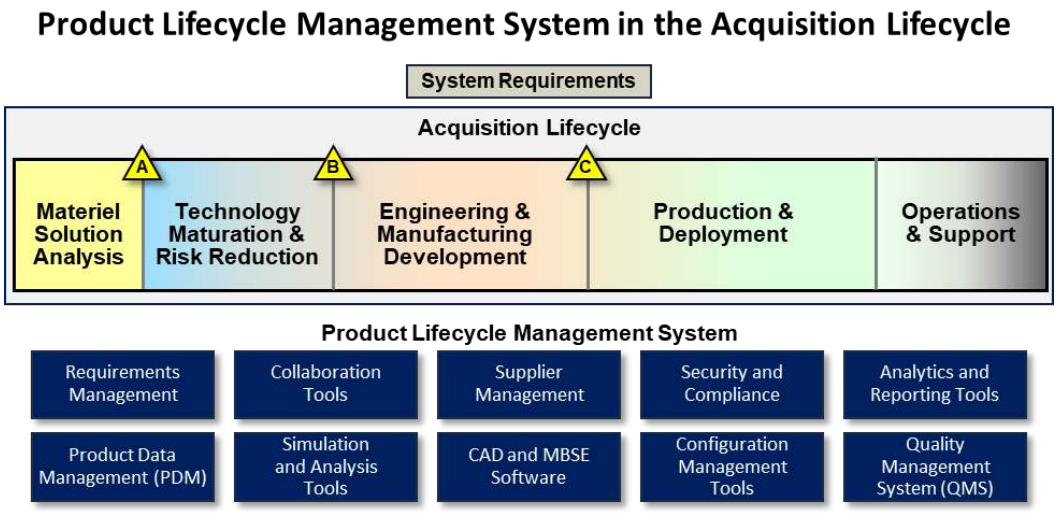


FIGURE 17. Product lifecycle management system in the acquisition life cycle.

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9.1.1. PLM and data reuse. PLM is a powerful tool for improving efficiency, collaboration, and innovation throughout the acquisition life cycle, leading to better products and improved operations and support. PLM data reuse can preclude the need for reverse engineering later in the life cycle, saving money, time, and reducing the chance of introducing variations into the system.

9.1.2. PLM digital transformation. PLM can be managed in a single PLM System (PLMS) or within a larger connected IDE or DEE. PLM provides an enterprise data solution to Government and industry programs, enabling a PM to work with program stakeholders in an integrated fashion to enable life-cycle management of weapon systems. The digital transformation of data and consolidation into a PLM approach involves modernizing the product data management system, including configuration management, provisioning, readiness, modeling, technical data management support, and incorporation of commercial capabilities. In addition, MOSA should be considered to drive sustainment efficiencies.

9.1.3. Benefits of PLM. The benefits of using PLM are numerous and will be realized throughout the system life cycle. Benefits include improved collaboration, efficient data management, faster time-to-market, cost savings, better product quality, enhanced innovation, end-to-end visibility, enhanced data analysis, informed decisions, and improved warfighter satisfaction (figure 18).

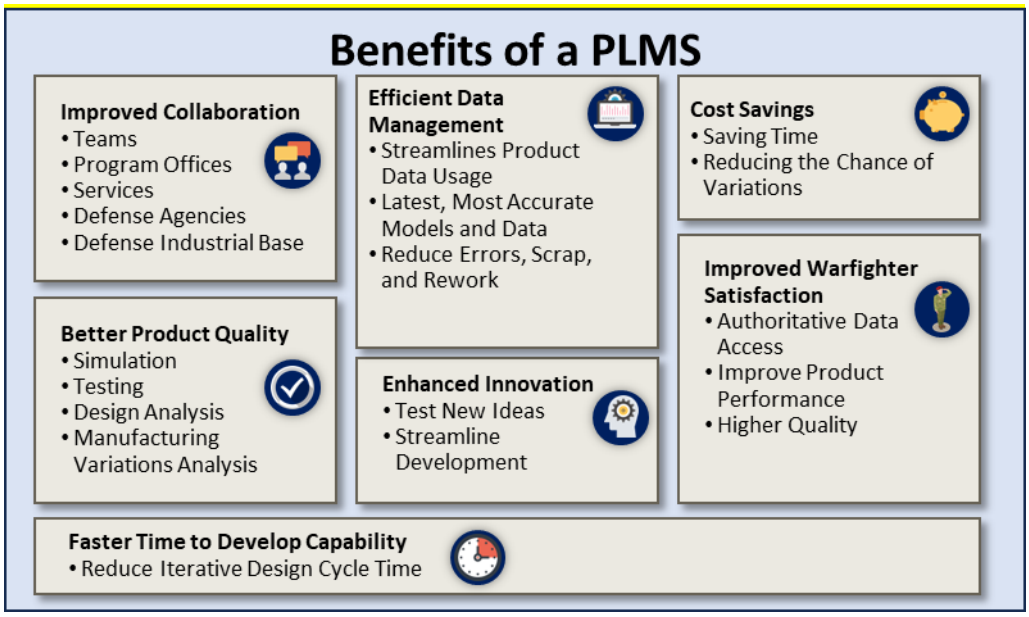


FIGURE 18. Benefits of a product lifecycle management system.

9.1.4. Components of PLM. The key components to a PLM may include:

a. Requirements management.

b. Product data management.

c. Collaboration tools.

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d. CAD and modeling software.

e. Simulation and analysis tools.

f. Workflow management.

g. Configuration management.

h. Supplier management.

i. Manufacturing execution system.

j. Quality management.

k. Analytics and reporting tools.

l. Security and compliance.

Figure 19 illustrates the relationship among PLM components.

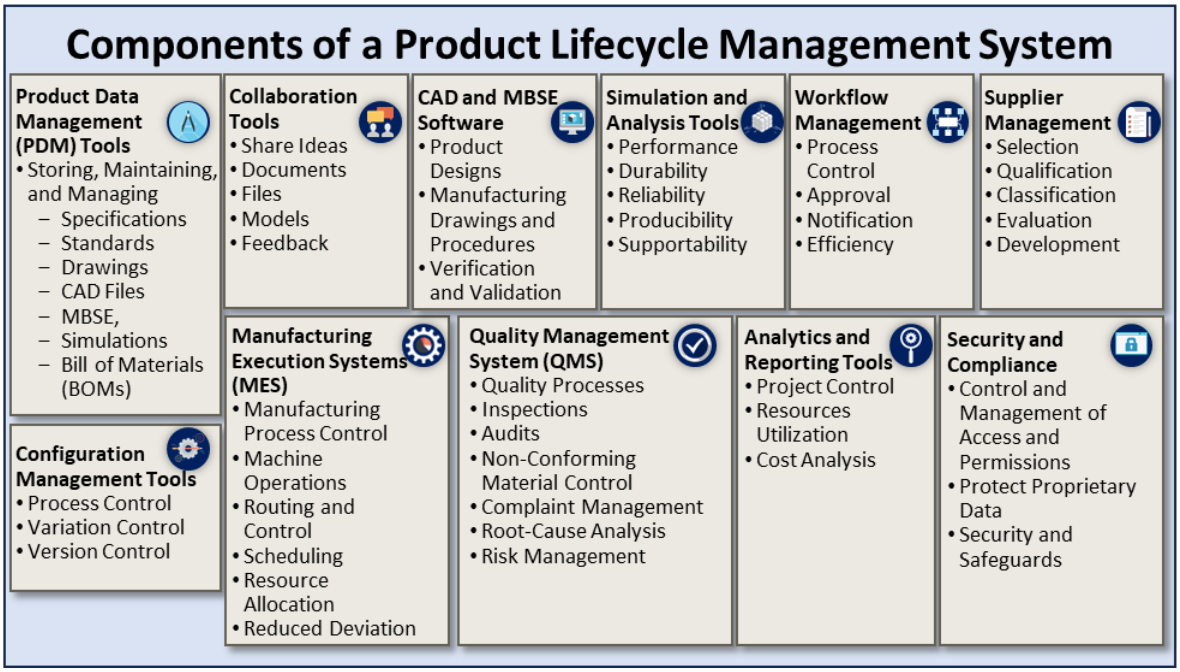


FIGURE 19. Components of a product lifecycle management system.

9.1.5. Product data management (PDM) tools. At the heart of the PLMS are the PDM tools used for storing, maintaining, and managing all product-related data and documentation generated from design, engineering, manufacturing, operation, and support. The data includes specifications, standards, drawings, CAD files, model-based DE, simulations, configuration management, BOM lists, and change orders, as well as other data as applicable. The data is available to all the stakeholders from the original equipment manufacturer (OEM), to program, maintenance, operational, and support offices.

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9.1.5.1. Collaboration tools. Collaboration tools allow users to communicate and collaborate among team members to share ideas, documents, files, and models, enabling real-time feedback.

9.1.5.2. CAD and model-based systems engineering (MBSE) software. CAD and 3D parametric solid modeling software are used for creating and modifying the product designs, generating manufacturing drawings, and developing procedures. Integration with the PDM system will ensure the design data is being managed and that production is using the latest design revision. Details of MBSE are delineated in section 6.

9.1.5.3. Simulation and analysis tools. These tools allow designers and engineers to subject models to simulated conditions and analyze product fit, performance, durability, reliability, producibility and supportability. The results will influence design and operation decisions to optimize product performance.

9.1.5.4. Workflow management. Workflow tools automate and streamline tasks to ensure work is directed to the appropriate team members, approvals are recorded, notifications are sent, and documents are routed to the proper offices. These tools support collaboration, increase efficiency, and ensure consistent execution.

9.1.5.5. Configuration management tools. PLMs can manage the configuration baselines of each individual item in a product or system throughout the life cycle of the item. The configuration baseline digital threads will maintain proper linkages to their defining data and relevance to the system's design and operational platform. A digital configuration management process provides the ability to capture, record, retrieve, and report status and performance information about the product. This process allows current information to be accessible to support other impact analyses and capture the comments and history of each change. The resulting audit trail facilitates verification testing and analysis of other products and in making manufacturing decisions for enterprise integration efforts. Verification and audit activities are performed with data sets to maintain consistency between the end-product and its configuration information throughout the life cycle. This supports the design integrity and ensures the end-product configuration information is complete, accurate, and current to produce the end-product including applicable operation, maintenance instructions, training, and repair parts. In the end, the PLMs configuration management application will provide for traceability of mission capability to system requirements to performance and execution metrics.

9.1.5.6. Analytics and reporting tools. These tools will provide insight into the product development process, including project statuses, resource use, and cost analysis.

9.1.6. Supplier management. The system will manage the technical requirements for the suppliers, including supplier selection, qualification, classification, evaluation, and supplier development. Supplier management will have improved transparency and accuracy with improved communication between manufacturers and suppliers. Product specifications will be the latest revision as controlled by the configuration management. Production costs will more accurately be calculated, including supplies, materials, and manufacturing costs. Sourcing management and traceability will be improved. The risk of errors and delays will be reduced.

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9.1.7. Manufacturing execution systems (MES). The MES manages the manufacturing process, including machine operations, routing and control, scheduling, resource allocation, quality control, and finishing processes. Where PLM provides the "what," the MES provides the "how" in the manufacturing process. MES ensures that products are manufactured efficiently and effectively according to specifications and integrates the shop floor equipment and processes. The system collects and analyzes real-time production data to improve efficiency, quality, and compliance.

9.1.8. Quality management system (QMS). A QMS is used to manage and track quality-related processes, such as inspections, audits, and non-conformance reporting. It helps ensure that products meet quality standards and regulatory requirements. Integrating the QMS with the PLM provides a unified, comprehensive view of the entire product development process with incorporated quality management practices. This allows stakeholders to understand the product life cycle holistically from design to manufacture to sustainment. Process insights are also improved as automated workflows, data analytics, and communication allow increased collaboration across departments and functions. The QMS allows non-conformance, complaint management, root-cause analysis, and risk management to be visible across the organization. An organization will be able to determine the cost of quality (COQ) and the cost of poor quality (COPQ). Risks can be identified and addressed throughout the product life cycle and minimize the impact of potential issues.

9.1.9. Security and compliance features. The system must control and manage the access and permissions of all users in the system. Much of the data will be sensitive to the product and the life-cycle operation of the overall platform. The system must be able to protect data that is proprietary and have robust security features and relevant safeguards.

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10. DIGITAL PRODUCT SUPPORT

10.1. Digital product support overview. Digital product support is the use of digital engineering methods and digital data and system models to implement the product support strategy (PSS), enable data-driven decision making, and deliver effective and efficient product support outcomes throughout the system life cycle.

10.2. Product support management. Product support management is the development and implementation of product support strategies to deliver the package of product support functions required to field and maintain the readiness and operational capabilities of weapon systems, subsystems, and components. It optimizes performance outcomes, minimizes operating and support costs, and includes the enterprise integration of the 12 Integrated Product Support (IPS) elements. For digital programs, the Product Support Manager (PSM), or lead life cycle logistician (LCL) if a PSM is not assigned, collaborates with systems engineers in developing and implementing the program's Digital Engineering Implementation Plan (Appendix E to the Systems Engineering Plan), including identifying Digital Product Support activities and documenting those activities in the program's Life Cycle Sustainment Plan (LCSP). These activities include, for example, model-based simulations and analyses used to develop and implement the PSS. This applies to systems that are "born digital" (designed using modern DE tools), digitally engineered modifications to legacy systems, and digitally enabled sustainment activities.

10.3. Digital product support considerations. Digital transformation requires a modern DEE or IDE and interoperable logistics product data systems that support cross-functional activities such as configuration management, readiness modeling, and technical data management. Incorporation of COTS DEE or IDE capabilities and MOSA should be considered to drive product support efficiencies. The 12 IPS Elements that make up the product support package can each benefit, in many cases substantially, from use of DSMs and data, particularly 3D CAD models. Table IV identifies example activities and considerations for each element.

TABLE IV. Digital product support considerations for the 12 IPS elements

IPS Element	Example Digital Product Support Activities
Product Support Management	Use of model-based Supportability Analysis (Product Support Analysis) and modeling and simulation tools to support development of the Reliability, Availability, Maintainability and Cost (RAM-C) Rationale Report informing development of warfighter requirements documents and sustainment performance attributes.
Design Interface	Use of a model-based Human Engineering Design Approach Document (HEDAD)-Maintainer to evaluate maintainability (e.g., access, component removal and installation, servicing) for the primary system and support equipment. Use of model-based systems engineering and technical reviews (e.g., PDR or CDR) leveraging digital system models and CAD models such as "fly-through" visualizations of key systems and supportability features.
Sustaining Engineering	Use of Digital Twin/Digital Thread for tracking defects (e.g., corrosion and cracking) and new failure modes. Model-based deficiency report

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IPS Element	Example Digital Product Support Activities
	analysis, structural and fluid dynamics analyses, etc. to design and optimize solutions such as component modifications or updated repair or servicing procedures. Model-based technical assistance such as telemaintenance.
Supply Support	Model-based provisioning and cataloging (e.g. use of model-based Engineering Data for Provisioning (EDFP)), DMSMS solutions, obsolescence management, and alternate part sourcing, including through Additive Manufacturing. Model-based supply chain illumination and analysis, Supply Chain Risk Management (SCRM), and design of resilient supply chains.
Maintenance Planning and Management	Model-based Condition Based Maintenance Plus (CBM+) implementation including predictive analytics tools using digital twins, artificial intelligence and machine learning (AI/ML), and onboard and offboard collection of performance and maintenance data to optimize the maintenance plan.
Packaging, Handling, Storage and Transportation (PHS&T)	Use of model-based PHS&T solutions such as shipping container design, warehouse design, and air/sealift load-planning optimization.
Technical Data	A model-based Technical Data Package including DSM serves as the nexus of the ASoT. Technical manuals are directly derived from the ASoT and include model-based instructions, such as Interactive Electronic Technical Manuals with embedded rotatable models and interactive wire trace illumination to increase maintenance efficiency and effectiveness.
Support Equipment	Use of model-based Support Equipment Recommendation Data (SERD), virtual test and verification, and other benefits similar to the prime system.
Training and Training Support	Use of model-based simulation for operator and maintenance training using Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR), including just-in-time training for remote telemaintenance.
Manpower and Personnel	Use of modeling and simulation, such as Discrete Event Simulation analyses, to optimize maintenance manpower based on system design and various cost and performance factors to construct an effective and affordable PSS.
Facilities and Infrastructure	Model-based identification of facility physical and utility requirements and interfaces. Condition-based monitoring of critical support facilities.
Information Technology Systems Continuous Support	Application of digital principles to Agile software continuous integration/continuous development. Use of model-based System Integration Lab and Software Development Lab capabilities.

10.3.1. Service initiatives enabling digital product support. As DoW adopts DE transformation strategies, the benefits from a modernized logistics portfolio with elevated capabilities should be considered to meet evolving threats. Modern digital solutions that increase effective mission outcomes are needed to transform efforts across the enterprise.

a. Department of the Navy. One current naval tool that supports the *Navy and Marine Corps Digital Systems Engineering Transformation Strategy* and integrates the PLM and sustainment activities using an MBSE approach is Navy-PLM (N-PLM). Various DoW efforts to support logistics provide a single source of truth and product data analytics. N-PLM is being

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developed as the singular PLM solution for all Navy programs, merging maritime PLM requirements from the Naval Air Systems Command (NAVAIR) and Naval Sea Systems Command (NAVSEA) into a unified solution. As of this writing, N-PLM includes two integrated components: Aviation PLM (AVPLM) and Model-Based Product Support (MBPS). N-PLM allows enterprise DE integration that will drive PLM cost reductions. The U.S. Marine Corps utilizes a different PLM software solution suite hosted on the Naval-Lift Platform.

c. Department of the Air Force. The *Department of the Air Force Digital Product Support Vision* emphasizes adoption of digital tools and processes to accelerate development and delivery of new capabilities providing a more lethal, agile, and ready force. Air Force Materiel Command's *Digital Materiel Management* (DMM) calls for "shatter[ing] existing paradigms and adopt[ing] DMM capabilities to radically accelerate our fielding, sustainment, and modernization." DMM includes six key initiatives and is multidisciplinary and cross-functional. The *U.S. Space Force Vision for a Digital Service* is an interconnected, innovative, digitally dominant force. This includes a DEE that establishes the interconnected foundation and analytic underpinning for fluid, flexible, and frictionless force design, capability planning, development, test, delivery, operations, and sustainment across mission areas at the speed of need. Air Force PLM (AF-PLM) enables the Digital Enterprise by creating the linkages and synchronization needed between derived technical data types throughout the life cycle as well as enabling other external tools and processes requiring data across the logistics and engineering communities. AF-PLM is foundational to enabling DMM.

e. Department of the Army. The *Army Digital Transformation Strategy* is the overarching framework that will set the vision, establish lines of effort, and implement strategic digital transformation initiatives prioritized and resourced as required to achieve the end state of a more ready, lethal, and modern force. Army Directive 2024-03, *Army Digital Engineering,* provides four DE tenets and identifies focus areas, interoperability and implementation guidance, pathfinder programs, and talent and expertise development guidance.

10.3.2. Product changes within a DE environment. Most DoW systems require product upgrades, technical refresh, repair, and maintenance to ensure they remain relevant in their current environment and meet operational requirements. As of this writing, there is no comprehensive, overarching DoW policy, plan, or standards to synchronize all the multidisciplinary, cross-functional DE efforts. If individual programs do not develop such a plan, the process is at risk for gaps and inefficiencies, which may result in failing to realize benefits of investments in DE and digital product support. Modifications result in updates to the ASoT (e.g., DSMs and logistics product data), which may require the program office and product support integrators and product support providers (e.g., shipyard or depot) to update their DEE or IDE with additional DE tools and capabilities.

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10.3.3. Defense Logistics Agency (DLA) and sustainment support. DLA is the DoW's primary supply chain manager for common parts and consumables. DLA also provides tailored acquisition support for specified repairables. During acquisition, DLA provides provisioning and cataloging support and assists in demand planning and procurement of initial spares. Except for Contractor Logistics Support weapon systems, DLA is the lead agency for common supply support during the sustainment phase of weapon system life cycles. Other DoW and industry product support providers, such as depots, shipyards, and arsenals, as well as other supply support providers, require access to the ASoT or derivative products to provide required products and services for weapon system sustainment.

10.3.4. Digital data in sustainment. Weapon system sustainment requirements for use of digital data should be synchronized with DLA's data management processes and repositories. DLA's technical and quality assurance activities are based on the authoritative data and master data, controlled through the cognitive engineering services. DLA currently has a dual role in support of the DoW:

a. To maintain the consumables supply chains for the Department, across multiple classes of supply.

b. To maintain DoW's shared Joint Additive Manufacturing Model Exchange (JAMMEX), as shown in figure 20.

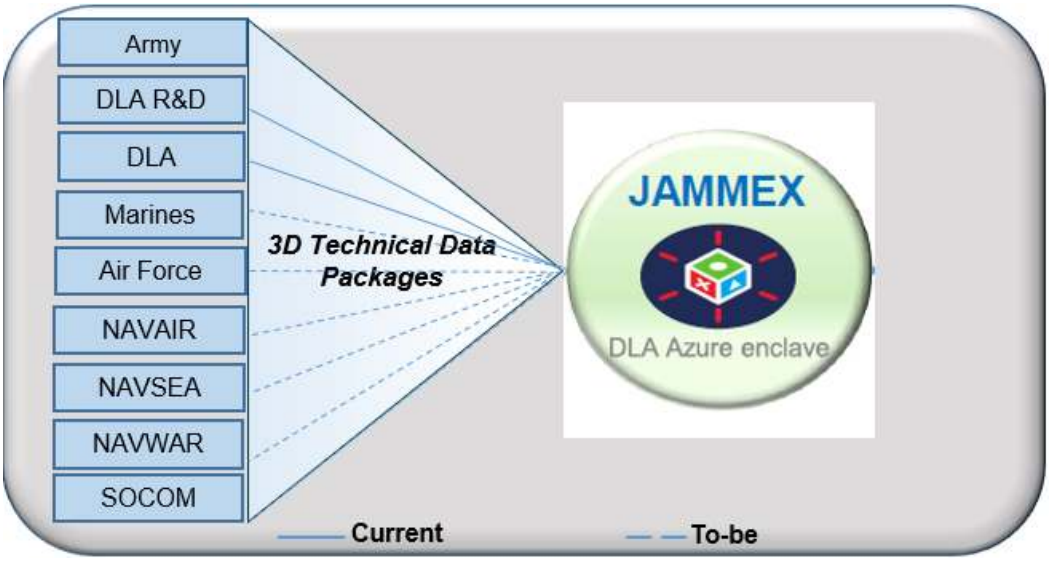


FIGURE 20. Joint Additive Manufacturing Model Exchange (JAMMEX).

10.3.5. Systems modernization. DLA is embarking on an enterprise IT modernization that will include expansion of the technical data management system. DLA accesses product data and item performance requirements from the Military Services' repositories, adding quality assurance and other provisions to the technical requirements in DLA contracts. DLA's IT modernization will expand PDM capabilities and repositories to include 3D and other modeling formats. DLA is also exploring digital thread environments that may enhance supply chain processes, to include secure sharing stakeholder information (customers, DLA, and suppliers) and enabling more integrated and timely decisions.

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10.3.6. Digital data sharing. As specified in DoDI 5000.93, DLA has been authorized to manage a 3D modeling platform, known as JAMMEX, with emphasis on sharing additive manufacturing (AM) models across the DoW. JAMMEX is a secure inter-Service platform within the DoW that provides capabilities to store, manage, and collaboratively share 3D AM models and associated metadata. The AM data models are approved by the engineering organizations for sharing in JAMMEX to fulfill immediate needs. JAMMEX uses the common DoW AM data framework that enables DoW entities to securely access and share data as required. The external systems are the authoritative source.

10.3.7. JAMMEX interface with Military Services' repositories. The JAMMEX system will interface with various Military Services' authoritative data repositories and systems containing 3D AM models. The system allows users to search for and view the 3D models and download the associated files to manufacture the component. This capability will give users the critical ability to manufacture parts on an as-needed basis, in locations where resupply is difficult or in situations where time is critical. JAMMEX currently has repositories for: Army, Navy, Marine Corps, U.S. Special Operations Command, Air Force, and DLA. JAMMEX provides a flexible platform that can interface with the various military services' system repositories.

10.3.8. User access to JAMMEX. JAMMEX provides access to Common Access Card (CAC)-authenticated users. JAMMEX uses a modular structure with role-based security access control. Users request access through the Account Management and Provisioning System (AMPS). Users are assigned to the repository for their Service branch. Service-designated approvers provide the final approval for the roles and permissions of individuals. These roles and permissions determine what level of access and capability each user has. Users not associated with a Service branch can be assigned access to view or download any models in JAMMEX that have been approved by the applicable Service as viewable by anyone. Generally, a user can view models/metadata and download files. These permissions can be more or less restrictive depending on their approved access.

10.3.9. Uploading data into JAMMEX. Models/metadata can be loaded into JAMMEX in several ways. Data can be loaded from an external database through an automated interface and periodically updated with scheduled jobs. Data can also be loaded into JAMMEX in a one-time data upload from an external system. Where no external system/connection is possible (e.g., files are all stored locally), the system provides the capability for a user to manually upload models into JAMMEX. JAMMEX is designed to be agnostic and support all file formats (PDF, STL, STEP, x3D, etc.). The external system owners will determine what data is stored/accessible from JAMMEX.

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11. NOTES

11.1. Intended use. This handbook is intended for use by program management staff and discusses how DE and modeling activities can be used across all functional areas within the acquisition life cycle. The handbook also provides guidance to assist users in transitioning systems engineering and modeling activities from document-based activities to digital-based processes.

11.2. Subject term (key word) listing.

a. Model-based Enterprise (MBE)

b. Digital engineering ecosystem (DEE)

c. Model-based systems engineering (MBSE)

d. Digital thread

e. Digital twin

11.3. Proprietary notice. Object Management Group®, OMG®, Systems Modeling Language™, SysML®, UAF®, and UML® are either registered trademarks or trademarks of Object Management Group, Inc., in the United States and/or other countries.

11.4. Change notations. The margins of this handbook are marked with vertical lines to indicate modifications generated by this change. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations.

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CONTRACT DATA REQUIREMENTS LIST (CDRL) FOR 3D DATA DELIVERY

A.1 SCOPE

A.1.1 Scope. The CDRL provides a structured format to describe to the contractor what data they are to provide and the format of that data. The contract will reference in Section C or an attached Statement of Work/Statement of Objectives (SOW/SOO), what data the Government wants by citing "IAW CDRL XXXX." The "XXXX" is the data item number and will be listed in an attachment to the contract. Each of these will reference a Government standard Data Item Description (DID) that provides the format and details.

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A.2 SAMPLE CDRL

CONTRACT DATA REQUIREMENTS LIST *(1 Data Item)*				*Form Approved* *OMB No. 0704-0188*

Public reporting for this collection of information is estimated to average 110 hours per response including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information including suggestions for reducing this burden to Department of Defense, Washington Headquarters Services, Directorate for Information Operations and Reports, 1215 Jefferson Davis Highway, Suite 1204, Arlington, VA 22202-4302, and to the Office of Management and Budget, Paperwork Reduction Project (0704-0188), Washington, DC 20503. Please DO NOT RETURN your form to either of these addresses. Send completed form to the Government Issuing Contracting Officer for the Contract/PR No. listed in Block E.

A. CONTRACT LINE ITEM NO. XXXX	**B. EXHIBIT** A	**C. CATEGORY:** TDP _X__ TM ___ OTHER
D. SYSTEM/ITEM DoW Acquisition Example	**E. CONTRACT/PR NO.** N000XX-22-N-XXXX	**F. CONTRACTOR** Prime Contractor

1. DATA ITEM NO. A00X	2. TITLE OF DATA ITEM Product Drawings/Models and Associated Lists	3. SUBTITLE Solid Model Data – DoW System
4. AUTHORITY (Data Acquisition Document No.) OT-XXXX-XXXXX	5. CONTRACT REFERENCE Section C, Part X	6. REQURING OFFICE DoW Program Office

7. DD 250 REQ	9. DIST STATEMENT REQUIRED	10. FREQUENCY	12. DATE OF FIRST SUBMISSION	14, DISTRIBUTION
DD	"D" See Block 16	ASREQ	See Block 16	
8. APP CODE A		11. AS OF DATE N/A	13. DATE OF SUBSEQUENT SUBMISSION See Block 16	

16. REMARKS

Block 9 – Use the following*: (Insert appropriate distribution statement, export warning, and destruction notice in accordance with DoD Directive 5230.24 or DD Form 254.)*

Blocks 12 & 13 – No later than 60 days after contract award the contractor/developer shall submit native electronic Solid Model Data to the designated Cloud environment. Product drawings (assembly, control, detail, etc.) and lists (parts, data, and index) will be furnished in the final drawing package. The Government will review the submitted data to ensure compliance with contractual requirements. All other drawings developed under this contract shall adhere to the format approved by the Government.

In-process technical review: The contractor/developer shall submit a copy of the completed drawing package 75 days before the PCA. When the contractor/developer has received comments resulting from the Government in-process technical review, the contractor/developer shall review the Government comments, make corrections to the entire package and resubmit all new and revised drawings within 30 days of comment receipt. Government comments, whether directed against a specific drawing, or made as a general comment, may apply to other drawings that have not been reviewed by the Government. The Government will subject resubmitted drawings and other drawings not previously reviewed to another technical review. Any additional comments generated as a result of subsequent technical reviews shall also be furnished to the contractor/developer for action. This process shall be repeated as necessary. Response time for each submittal thereafter shall be 15 days.

Final delivery: The contractor/developer shall submit the final drawing package within 30 days of Government request.

Block 14 – Data shall be delivered as native Solid Model Data or in digital format in compliance with the attached TDP Option Selection Worksheet. Native Solid Model Data delivered to the designated Cloud environment shall be considered as the data item submittal to Addressees. Additional product drawings and lists delivered to the designated electronic repository shall be considered as the data item submitted to Addressees. Addressee notification of data availability shall be accomplished through an automated notification.

Final submittals shall be by DD Form 250, as indicated in Block 7.

a. ADDRESSEE	b. COPIES		
	Draft	Final	
		Reg	Repro
In-Process Format Review:			
DoW Program Office (Code XXX)	0	1	0
DoW Acquisition Agent (Code XXX)	1	1	0
DoW ISEA (Code XXX)	0	1	0
DoW Repository (Code XXX)	0	1	0
Other DoW Activity (Code XXX)	0	1	0
See Block 16			
15. TOTAL	1	5	0

G. PREPARED BY	**H. DATE**	**I. APPROVED BY**	**J. DATE**

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A.3 DATA ITEM DESCRIPTIONS

ASSIST is the DoW repository for DIDs. DIDs can be located and retrieved via the "Basic Search" screen by typing "DI" in the Document ID field and search keywords in the Find Term field. The "Advanced Search" screen provides additional search parameters.

These current DIDs define requirements that can be used for DE data. This sample list is current as of August 2025.

DID	Title
DI-ADMIN-81306	Program Protection Implementation Plan (PPIP)
DI-SESS-80463	Engineering Release Record
DI-SESS-80792	Validation Report
DI-EMCS-80199	Electromagnetic Interference Control Procedures (EMICP)
DI-EMCS-80200	Electromagnetic Interference Test Report (EMITR)
DI-EMCS-80201	Electromagnetic Interference Test Procedures (EMITP)
DI-EMCS-81540	EM Environmental Effects (E3) Integration & Analysis Report (E31AR)
DI-EMCS-81541	Electromagnetic Environmental Effects (E3) Verification Procedures (E3VP)
DI-EMCS-81542	Electromagnetic Environmental Effects (E3) Verification Report (E3VR)
DI-EMCS-81846	HEMP Test Plan for System Verification Test
DI-EMCS-81847	HEMP Test Report for System Verification Test
DI-FACR-81216	Trainer Engineering Report
DI-FNCL-80331	Funds and Man-Hours Expenditure Report
DI-FNCL-81765	Contractor Business Data Report
DI-FNCL-82162	Cost and Hour Report (Flexfile)
DI-HFAC-80745	Human Engineering Systems Analysis Report (HESAR)
DI-HFAC-80746	Human Engineering Design Approach Document-Operator
DI-HFAC-80747	Human Engineering Design Approach Document-Maintainer
DI-HFAC-81399	Critical Task Analysis Report
DI-HFAC-81743	Human Systems Integration (HSI) Program Plan (HSIPP)
DI-HFAC-81833	Human Systems Integration Report (HSIR)
DI-ILSS-80868	Special Equipment Tools and Test Equipment
DI-IPSC-81427	Software Development Plan (SDP)
DI-IPSC-81429	Software Transition Plan (STrP)
DI-IPSC-81431	System/Subsystem Specification (SSS)
DI-IPSC-81432	System/Subsystem Design Description (S/SDD)
DI-IPSC-81433	Software Requirements Specification (SRS)
DI-IPSC-81434	Interface Requirements Specification (IRS)
DI-IPSC-81435	System Software Design Document
DI-IPSC-81436	Interface Design Description (IDD)
DI-IPSC-81438	Software Test Plan
DI-IPSC-81439	Software Test Description (STD)
DI-IPSC-81440	Software Test Report (STR)/ Software/Systems Integration Laboratory (SIL) Test Report
DI-IPSC-81441	Software Product Specification (SPS)

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DI-IPSC-81442	Software Version Description (SVD)
DI-MFFP-81403	Prevention and Control Plan (CPCP)
DI-MGMT-80368	Status Report
DI-MGMT-80441	Government Property Inventory Report
DI-MGMT-80503	Report of Shipping (Item) and Packaging Discrepancy
DI-MGMT-80920	List of Items Delivered During the Terms of the Contract
DI-MGMT-81398	Hazardous Materials Management Program - Plan
DI-MGMT-81453	Data Accession List (DAL)
DI-MGMT-81468	Contract Funds Status Report (CFSR)
DI-MGMT-81605	Briefing Material
DI-MGMT-81797	Program Management Plan
DI-MGMT-81803	Item Unique Identification (IUID) Marking Plan
DI-MGMT-81809	Risk Management Status Report
DI-MGMT-81845	Information Assurance Design Review Information Package (DRIP)
DI-MGMT-81861	Integrated Program Management Data Analysis Report
DI-MGMT-81889	Manufacturing Plan
DI-MGMT-81910	Repair Procedures
DI-MGMT-81911	Work Management Plan
DI-MGMT-81948	DMSMS Management Plan
DI-MGMT-81954	Personnel Qualification Standard (PQS)
DI-MGMT-81987	Statistical Process Control (SPC) Plan
DI-MGMT-81994	Priced MBOM
DI-MGMT-82001	DOD Risk Management Framework (RMF) Package Deliverables
DI-MGMT-82035	Software Resources Data Reporting
DI-MGMT-82133	Requirements Traceability Verification Matrix (RTVM)
DI-MGMT-82144	Program Protection Implementation Plan (PPIP)
DI-MGMT-82163	Maintenance and Repair Parts Data Report
DI-MGMT-82164	Quantity Report
DI-MGMT-82165	Technical Data Report
DI-MGMT-82191	Cybersecurity Vulnerability Report
DI-MGMT-82247	Systems Security Plan (SSP) and Associated Plans of Action & Milestones (POA&M)
DI-MGMT-82274	DMSMS Life Cycle Management Data
DI-MISC-80048	Scientific and Technical Reports Summary
DI-MISC-80508	Technical Report - Study/Services
DI-MISC-80711	Scientific/Technical Report
DI-MISC-80875	Welding Procedures
DI-MISC-80876	Welding Procedure Qualification Test Report
DI-MISC-81321	Equipment Installation Instructions
DI-MISC-81356	Certificate of Compliance
DI-MISC-81397	Hazardous Materials Management Program - Report
DI-MISC-81807	Software/Firmware Change Request
DI-MISC-81832	Counterfeit Prevention Program Plan

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DI-MISC-81889	Manufacturing Plan
DI-NDTI-80566	Test Plan
DI-NDTI-80809	Test/Inspection Report
DI-NUOR-80156	Nuclear Survivability Program Plan
DI-NUOR-80926	Nuclear Survivability Assurance Plan
DI-NUOR-80927	Nuclear Survivability Design Parameters Report
DI-NUOR-80928	Nuclear Survivability Test Plan
DI-NUOR-80929	Nuclear Survivability Test Report
DI-PACK-80121	Special Packaging Instructions
DI-PACK-80457	Packaging Test Report
DI-PACK-80683	Container Design Retrieval System (CDRS) Search Request
DI-PACK-80880	Transportability Report
DI-PACK-81581	Equipment Preservation Data Sheet (EPDS)
DI-PSSS-80134	Proposed Spare Parts List
DI-PSSS-81656	Bill of Materials (BOM) for Logistics and Supply Chain Risk Management
DI-PSSS-81872	Level of Repair Analysis (LORA)
DI-PSSS-81873	Level of Repair Analysis (LORA) Input Data
DI-PSSS-82046	Software License Requirements List (SLRL)
DI-PSSS-82114	Reliability Centered Maintenance (RCM) Analysis Report
DI-PSSS-82354	Maintenance Training Materials
DI-QCIC-81068	Preparation of Final Inspection Record (PFIR)
DI-QCIC-81794	Quality Assurance Program Plan (QAPP)
DI-QCIC-81795	Software Quality Assurance Report (SQAR)
DI-QCIC-81890	Qualification Test Report (QTR)
DI-SAFT-80102	Safety Assessment Report (SAR)
DI-SAFT-80970	Critical Safety Item (CSI), Characteristics and Critical Defect Report (CDR)
DI-SAFT-81124	Threat Hazard Assessment (THA)
DI-SAFT-81125	Threat Hazard Assessment Test Report
DI-SAFT-81626	System Safety Program Plan (SSPP)
DI-SAFT-82085	Hazard Tracking System (HTS) Data
DI-SCRE-82258	Contractors Record of Tier 1 Level Suppliers Receiving/Developing Covered Defense Information
DI-SESS 81758	Logistics Product Data
DI-SESS-80255	Failure Summary and Analysis Report
DI-SESS-80639	Engineering Change Proposal
DI-SESS-80640	Request For Variance (RFV)
DI-SESS-80642	Notice of Revision
DI-SESS-80643	Specification Change Notice (SCN)
DI-SESS-80776	Technical Data Package
DI-SESS-80792	Validation Report
DI-SESS-80858	Supplier's Configuration Management Plan (CMP)
DI-SESS-81000	Product Engineering Design Data and Associated Lists
DI-SESS-81022	Configuration Audit Summary Report and Certification

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DI-SESS-81121	Baseline Description Document
DI-SESS-81245	Installation Complete Notification (ICN)
DI-SESS-81248	Interface Control Document
DI-SESS-81254	Request for Nomenclature
DI-SESS-81253	Configuration Status Accounting (CSA) Information
DI-SESS-81359	Parts List
DI-SESS-81495	Failure Mode, Effects, and Criticality Analysis Report
DI-SESS-81496	System Reliability & Maintainability (R&M) Block Diagrams and Math Models Report
DI-SESS-81497	Reliability and Maintainability Prediction Report
DI-SESS-81613	Reliability and Maintainability (R&M) Program Plan
DI-SESS-81646	Configuration Audit Plan
DI-SESS-81704	Test Plans and Test Procedures
DI-SESS-81758	Logistics Product Data
DI-SESS-81759	Logistics Product Data Summaries
DI-SESS-81856	Contractual Baseline Report
DI-SESS-81874	Engineering Data for Provisioning (EDFP)
DI-SESS-81938	Performance Analysis Report
DI-SESS-81965	Provisioning Technical Documentation Submission Schedule
DI-SESS-82019	Product Baseline Index
DI-SESS-82043	Software Architecture Document (SAD)
DI-SESS-82168	Technical Review Presentation Package
DI-SESS-82177	Specification Tree
DI-SESS-82322	Built-In-Test Fault Tree
DI-SESS-82328	Computational Aerodynamics Analysis Results
DI-SESS-82330	Computational Aerodynamic Model (CAM)
DI-SESS-82364	Digital System Model
DI-SESS-82368	Simulated Tactical Environment Algorithm Descriptions (STEAD)
DI-SESS-82400	System Architecture Model (SAM)
DI-SESS-82426	Model-Based Engineering Failure Modes, Effects, and Criticality Analysis Profile (SYSML Version)
DI-SESS-82433	Cybersecurity Supply Chain Risk Management (C-SCRM) Software Bill of Material (SBOM) Requirements
DI-SESS-82436	Cybersecurity Supply Chain Risk Management Hardware Bill of Materials Requirement
DI-SESS-82438	System/Design Trade Study Report
DI-SESS-82448	Data Management Plan
DI-SESS-82466	Software Systems Product Design (SSPD)
DI-SESS-82483	Digital Twin Description (DTD)
DI-TMSS-80527	Commercial Off the Shelf (COTS) Manual(s) & Associated Supplemental Data
DI-TMSS-81817	Technical Manual Quality Assurance Program Plan
DI-TMSS-81818	Technical Manual Validation Plan
DI-TMSS-81819	Technical Manual Validation Certificate

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A.4 SAMPLE TDP OPTION SELECTION WORKSHEET

TDP OPTION SELECTION WORKSHEET			
SYSTEM: Sample Navy Program Contract Example			DATE PREPARED:
A.CONTRACT NO. As Applicable	B. EXHIBIT/ATTACHMENT As Applicable	C. CLIN As Applicable	D. CDRL DATA ITEM NO. As Applicable

1.TDP Lifecycle Level (X and complete as applicable.)	
A. ☐ CONCEPTUAL LEVEL ☐ DEVELOPMENTAL LEVEL ☒ PRODUCTION LEVEL	B. REMARKS:

2.TYPE AND FORMAT (X all that apply and complete as applicable)	
A.	B.
	☒ NATIVE CAD (Native file format the data was created in. Specify level of annotation.):
☒ TYPE 2D: 2D DRAWINGS	☐ ISO 10303 STEP FORMAT (SPECIFY):
☐ TYPE 3D: 3D MODEL ONLY	☒ ISO 32000 PORTABLE DOCUMENT FORMAT (PDF)
	☒ OTHER ELECTRONIC FORMAT (SPECIFY TYPE)
☒ TYPE 3D: 3D MODELS WITH ASSOCIATED 2D DRAWINGS	MODEL ORGANIZATIONAL SCHEMA IAW ASME Y14.47 (Specify Appendix B) NEUTRAL CAD FORMAT (Specify e.g. ISO 10303, AP203, AP214, AP216, or AP242) (Specify e.g. 3D PDF, JT) PDF required for 2D; 3D PDF format for full MBD – Model only. (Specify Microsoft EXCEL, CSV, or XML)
	☒ HARDCOPY
See Block 9	REMARKS: Cloud environment defined in contract

3. CAGE Code AND DOCUMENT NUMBERS	A.	D.
	☒ CONTRACTOR CAGE AND DOCUMENT NUMBERS – See Block 9 ☒ GOVERNMENT CAGE (COMPLETE 3B & 3C OR 3D) – See Block 9	To Be Assigned By: PMA-XXX As specified in the contract
B. USE CAGE CODE: 30003	C. USE DOCUMENT NUMBERS: See Block 9	

4. DRAWING FORMATS (X one and complete as applicable)
☒ CONTRACTOR FORMAT.
☐ GOVERNMENT FORMAT.
REMARKS: _____ CONTRACTOR CAGE CODES SHALL BE USED FOR COMPANY STANDARDS

5. TDP ELEMENTS REQUIRED (X all that apply)

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☐ ELEMENTS REQUIRED TO BE DETERMINED BY CONTRACTOR or THE FOLLOWING ARE REQUIRED:

☐ CONCEPTUAL DRAWINGS/MODELS AND ASSOCIATED LISTS

☐ DEVELOPMENTAL DESIGN DRAWINGS/MODELS AND ASSOCIATED LISTS

☒ PRODUCT DRAWINGS/MODELS AND ASSOCIATED LIST – CDRL XXXX

☐ COMMERCIAL DRAWINGS/MODELS AND ASSOCIATED LIST

☐ QUALITY ASSURANCE PROVISIONS

☒ SPECIAL INSPECTION EQUIPMENT (SIE) DRAWINGS/MODELS AND ASSOCIATED LISTS – CDRL XXXX

☐ SPECIFICATIONS

☒ SOFTWARE DOCUMENTATION – CDRL XXXX

☒ QUALITY ASSURANCE PROVISIONS / METADATA / SUPPLEMENTARY TECHNICAL DATA – CDRL XXXX
For all items requiring Acceptance Testing (DoD-STD-2101).
SOURCE CONTROL DRAWING (SOCD) APPROVAL REQUEST – CDRL XXXX

6. ASSOCIATED LISTS (X and complete as applicable)

☒	A. PARTS LIST (X ONE)	☐	(1) INTEGRAL	☒	(2) REQUIRED (CONTRACTOR SELECT)
☒	B. DATA LIST (X ONE)	☐	(1) NOT REQUIRED	☒	(2) REQUIRED (Lowest level of repair/indenture)
☒	C. INDEX LIST (X ONE)	☐	(1) NOT REQUIRED	☒	(2) REQUIRED (Lowest level of repair/indenture)
☒	D. WIRING LIST (X ONE)	☐	(1) NOT REQUIRED	☒	(2) REQUIRED (Lowest level of repair/indenture)
☒	E. INDENTURE DATA LIST (X ONE)	☒	(1) NOT REQUIRED	☐	(2) REQUIRED (SPECIFY LEVELS OF ASSEMBLY)
☒	F. APPLICATION LISTS (X ONE)	☐	(1) NOT REQUIRED	☒	(2) REQUIRED (CONTRACTOR SELECT)

7. APPLICABLILTY OF STANDARDS. The following Standards apply: (X as applicable)

☒ ASME Y14.100 ENGINEERING DRAWING PRACTICES WITH APPENDICES	☒ ASME Y14.24 TYPES AND APPLICATIONS OF ENGINEERING DRAWINGS	☐ OTHER STANDARDS APPLY AS DESCRIBED:
	☒ ASME Y14.34 ASSOCIATED LIST	
	☒ ASME Y14.35 REVISION OF ENGINEERING DRAWINGS AND ASSOCIATED LIST	COMPANY STANDARDS PERMITTED ☐ YES ☒ NO
☒ B ☒ C ☒ D ☒ E	☒ ASME Y14.41 DIGITAL PRODUCT DATA PRACTICES	
	☒ ASME Y14.5 DIMENSIONING AND TOLERANCES	

8. OTHER TAILORING

Blocks 2A and 2B – The data delivery format requirements are as follows and shall be used by the contractor as required for the 2D and 3D data being delivered as part of the TDP:

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Block 2A – 2D drawings – will be delivered in their native file format and with ISO 32000 PDF files for non-CAD users. If native file format is not available, then the current source file format and PDF will be submitted to the government.

Block 2B – 3D Models "only" – or full Model Based Definition (MBD) submittals will be fully annotated such that the model serves as the master source data for all geometry, dimensions, tolerances, notes, annotations, attributes, and product and manufacturing information (PMI) necessary to describe a product. 3D Model only submittals will be delivered in native CAD format and will include a validated STEP file and a viewable format such as 3D PDF or ISO 14739 PRC compliant 3D viewable. When delivering 3D Model only submittals a Model Organizational Schema IAW ASME Y14.47 shall be included with the delivery to ensure the government has complete understanding of how the models are structured and organized.

3D Models with Associated 2D Drawings – 3D Models used to generate 2D drawings will be submitted in native CAD format. The native CAD files will be a complete submittal with model and drawing files. The data delivery will include ISO 32000 PDF files for 2D drawings.

Block 2C – TDP Metadata provides the administrative data required for TDP management. Refer to CDRL XXXX TDP Metadata and One-Time DID XXXX for specific metadata format requirements. Metadata to be delivered in Microsoft Excel file format, CSV format, or XML format. Coordinate with PMA-XXX for valid field values to be used. TDP Metadata will be provided for the following TDP Elements checked in Block 5: Product Drawings/Models and Associated Lists (CDRL XXXX) Special Inspection Equipment Drawings, Models, and Associated Lists (CDRL XXXX) Special Tooling Drawings, Models, and Associated Lists (CDRL XXXX)

Block 2D – Contractor required to provide hardcopies of Associated Lists to support In-Process reviews and Physical Configuration Audits (PCAs) as requested by the Government. Not required for Final TDP Delivery.

Block 2E – Supplemental Technical Data shall include engineering analysis to include finite element models and finite element analysis (e.g., airframe loads models, wing analysis/geometric analysis/load sets, internal fatigue models, damage tolerance analysis, strength/fatigue analysis, engineering reports, and modeling and simulation data). Include embedded metadata to the model file.

Blocks 3A, 3B, 3C, 3D – Government CAGE and Drawing numbers required for all new drawings generated as part of this effort. Existing drawings, or revisions to existing drawings may retain Contractor drawing number and CAGE except when a new item identification is being established. A new NAVSEA drawing, model, or associated list will be created when creating new item identification (e.g., when creating a new “dash” configuration, or when identifying a new item). Contact Program XXX, for issuance of NAVSEA engineering drawing numbers (XXXAS0100-XXXXAS9999). Contractor is required to maintain tracking of NAVSEA engineering drawing number usage.

Block 8 – Company standards shall not be referenced in NAVSEA drawings, models, associated lists and other TDP products. If a company standard is referenced in a contractor drawing, model, or associated list that is being delivered as part of the government TDP, a copy of the company standard will be delivered with the final TDP delivery in order to meet design disclosure requirements.

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ELECTRONIC DELIVERY OF CDRLS

B.1 SCOPE

B.1.1 Scope. This appendix is an example of a contract clause. It recommends specific data formats in which TDP documents (drawings, specifications, and reports) and data on procedures, systems, and applications should be delivered. The contract also can specify preferences with data formats to drive data exchange efficiencies.

B.2 SAMPLE REQUIREMENTS FOR DRAWINGS AND 3D MODELS

ELECTRONIC DELIVERY OF CONTRACT DATA REQUIREMENTS LISTS (CDRLs)

The following shall be incorporated into the Contract Data Requirements List (CDRL) Special Instructions. Unique requirements that differ from the information defined below shall be specified in the contract.

1. Drawings shall be delivered in accordance with:
   a. Viewing orientation may be 0, 90, 180 or 270.
   b. Image sizes, including over scanning shall be per MIL-PRF-28002 for drawing sizes "A" through "F" and "A4" through "A0" as defined in ASME Y14.1.
   c. Drawings are reviewed in native or PDF formats but delivered in native format.
   d. Preferred Solid Model Data native formats to avoid data translation are; Creo 5.0 and above, CATIA V4 or above, Siemens NX/Unigraphics NX, SolidWorks, Autodesk Inventor. These platforms allow improved productively in a Multi-CAD environment.
2. Other Technical Data Package (TDP) documents (i.e. "A" size drawings, specifications and reports) and other data items shall be delivered on electronic media (through a World Wide Web page, or electronic mail as agreed to per individual contract. Windows or Macintosh format to be identified for each contract. Use of software applications other than those listed below are to be coordinated between Government and Contractor before delivery.
   a. All text shall be in Microsoft Word and/or PDF format.
   b. All graphs and charts shall be in Microsoft Project/Power Point.
   c. Microsoft Excel may be used where applicable.
   d. Microsoft Power Point/Project may be used where applicable.
   e. When the above listed software programs do not meet the formatting required for a particular application, the software program shall be saved in a Microsoft Word/Microsoft Office document in a readable and correctable format.

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f. Where the text is a part of the document, Microsoft Word shall be the preferred deliverable, although PDF formats are acceptable. A document shall not require further assembly or manipulation before print and distribution.

3. Solid Model Data or 3D Computer-Aided Design (CAD) Models, when required, shall be delivered in accordance with the following order of preference:

   a. Native 3D CAD data files shall be used as the primary method of data file transfer using a preferred Solid Model Data native data format.

   b. If a preferred Solid Model Data native data format is unavailable, the ISO 10303, Standard for the Exchange of Product Model Data (STEP), Application Protocol (AP) Part(s) 242 (See *NOTE) which is compatible with Government and Contractor systems and applications shall be used if native data is unavailable.

   c. If native data or compatibility with STEP is unavailable, an IGES file which is compatible with Government and Contractor systems and applications and has been shown to be effective in the translation of data files through test shall be used. Data files will be created using MIL-PRF-28000, Class II as guidance.

4. When Government and Contractor procedures, systems and applications are data files may be delivered using an established on-line Continuous Acquisition and Life Cycle Support (CALS)/Contractor Integrated Technical Information System (CITIS) approach.

*NOTE: ISO 10303-XXX AP: Application Protocol: Documents that specify the format for representing product data within a set of related processes or activities that can be added by contract.

AP210: AP for Electronic Assembly, Interconnect and Packaging Design

AP215: AP for Ship Arrangement
AP216: AP for Ship Molded Forms
AP218: AP for Ship Structure
AP227ed2: AP for Plant Spatial Configuration
AP239: AP for Product Life Cycle Support
AP242: AP for Managed Model-Based 3D Engineering

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SAMPLE ONE-TIME DATA ITEM DESCRIPTION

C.1 SCOPE

C.1.1 Scope. The one-time DID provides specific data format requirements. Most program offices will need a customized approach using a one-time DID. The DID should include a standardized list of data used with interfacing subsystems and systems, as well as program specific data requirements.

C.2 SAMPLE ONE-TIME DID

Title: TECHNICAL DATA PACKAGE (TDP) METADATA REQUIREMENTS

Number: OT-1234-56789	**Approval Date:** XXXXXXXXX
AMSC Number: N/A	Limitation:
DTIC Applicable: No	**GIDEP Applicable:**
Preparing Activity:	**Project Number:**
Applicable Forms:	

Use, Relationships: Technical Data Package (TDP) metadata is used to store, manage, manipulate and provide access to TDP elements stored in the Program Authoritative Source repository. The metadata required is the data needed by the information management system, used by the Procuring Activity to perform these functions. The metadata needs to be delivered in a format that the information management system can process. TDP metadata includes all relations, parameters and system information used in managing the TDP.

This DID contain the format, content, and intended use information for the data product resulting from the work task described in the Statement of Work (SOW). This DID is for one-time use for Contract number (*insert Contract Number for TDP Procurement).*

Requirements

1. References: An example of the TDP Metadata format can be obtained from the following address: (*insert the Program address and Point of Contact with phone number – should be the TDP Life Cycle Management (LCM) for the Program).*

2. TDP Metadata shall be delivered with all TDP data file deliveries and the delivery schedule for TDP Metadata shall coincide with the delivery requirements of the appropriate TDP Element Data delivery.

3. Format: The TDP Metadata shall be submitted in a comma separated value (csv), XML, or Microsoft Excel file format. Required data elements and metadata format requirements are detailed below. The TDP Metadata deliverable is separated into 3 tables: TDP Drawings, TDP Parts, and TDP Valid value types:

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SAMPLE TDP DRAWING METADATA FORMAT

D.1 SCOPE

D.1.1. Scope. A Data Item Description (DID) will typically be required and will need additional format definitions to support the program specific data requirements. A sample supplemental formatting matrix is included in D.2.1.

D.2 SAMPLE TDP DRAWINGS METADATA FORMAT (AIRCRAFT CARRIER EXAMPLE)

D.2.1. TDP drawings metadata format. An example of the TDP drawings metadata format can be seen in table D-I below.

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TABLE D-I. TDP drawings metadata format matrix example.

Field Name	Business Definition	Alternative Names	Unique Value	Data Type	Length	Valid Values	Default Value	Calculation	Required	Business Rule	User Role	Owner Role	Admin Role
Drawing Number	The number used to uniquely identify a TDP element	DWG No	Yes	Alpha numeric	32	May contain only "-" (dash) special character, alpha characters, or numbers	Null	N/A	Yes	Must be in accordance with ASME Y14.100-2013, Section 6	View	View, Edit	View, Edit, Delete
Drawing Revision	The current revision of a document, drawing, or list	DWG Rev	No	Alpha numeric	3	May contain only "-" (dash) special character or alpha characters	Null	N/A	Yes	The revision assigned to the drawing record must be in accordance with ASME Y14.35	View	View, Edit	View, Edit, Delete
Current Rev	Is this the Current Rev?			Alpha	1	Y,N	Null	N/A	Yes		View		
Drawing Type	The type of drawing, document, or list	DWG Type	No	Alpha numeric	50	May contain only alpha characters, numbers, and special characters	Null	N/A	Yes	Must be a valid drawing type in accordance with ASME Y14.24-2012,Table A-1	View	View, Edit	View, Edit, Delete
Title	The title of the drawing, document, or list	Nomenclature	No	Alpha numeric	99	May contain only alpha characters and numbers	Null	N/A	Yes	Must be in accordance with ASME Y14.100-2013, Section 5	View	View, Edit	View, Edit, Delete
Drawing CAGE	The CAGE code identified on the drawing, document, or list (Design Activity Identifier)	DAI	No	Alpha numeric	5	May contain only alpha characters and numbers	Null	N/A	Yes	Must be a valid CAGE	View	View, Edit	View, Edit, Delete
System	The system the item is used on	N/A	No	Alpha numeric	32	May contain only "-" (dash) special character, alpha characters, or numbers	Null	N/A	Yes	Must be a valid System	View	View, Edit	View, Edit, Delete
Release Date	The date the item was authorized for release	N/A	No	Date	12	DD- MMM- YYYY	Null	N/A	No	If left empty the date imported will be assigned as the release date	View	View, Edit	View, Edit, Delete
Contract	Contract listed on the drawing, document, or list	N/A	No	Alpha numeric	25	May contain only alpha characters and numbers	Null	N/A	Yes	Contract under which the item was developed and the data was created	View	View, Edit	View, Edit, Delete
Security	A value to denote the security classification value	N/A	No	Alpha numeric	1	May contain only alpha characters	Null	N/A	Yes	Valid Values are UNCLASSIFIED (U), CONFIDENTIAL (C), SECRET (S), or TOP SECRET (T)	View	View, Edit	View, Edit, Delete
Next Higher Drawing	The next higher assembly	NHA	No	Alpha numeric	32	May contain only "-" (dash) special character, alpha characters, or numbers	Null	N/A	Yes	Must be a valid drawing number	View	View, Edit	View, Edit, Delete
Next Higher Drawing Type	Drawing type of the next higher assembly	NHA DT	No	Alpha numeric	20	May contain only alpha characters, numbers, and special characters	Null	N/A	Yes	Must be a valid drawing type in accordance with ASME Y14.2 4-2012, Table A-1	View	View, Edit	View, Edit, Delete
Distribution Statement	Distribution statement of the drawing/ document	N/A	No	Alpha	1	A, B, C, D, E, F, X	no default	N/A	Yes	Must be in accordance with DoDI 5230.24 and NAVAIRINST 5230.24	View	View, Edit	View, Edit, Delete
Drawing Number	The number used to uniquely identify a TDP element	DWG No	Yes	Alpha numeric	32	May contain only "-" (dash) special character, alpha characters,	Null	N/A	Yes	Must be in accordance with ASME Y14.100-2013, Section 6	View	View, Edit	View, Edit, Delete
Drawing Revision	The current revision of a document, drawing, or list	DWG Rev	No	Alpha numeric	3	May contain only "-" (dash) special character or alpha characters	Null	N/A	Yes	The revision assigned to the drawing record must be in accordance with ASME Y14.35	View	View, Edit	View, Edit, Delete
WBS	WBS breakdown	N/A	Yes	Alpha numeric	25	May contain only alpha characters and numbers	Null	N/A	No	Required for new acquisitions. Must be in accordance with MIL-STD-881	View	View, Edit	View, Edit, Delete

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TABLE D-I. TDP parts metadata format matrix example – continued.

Field Name	Business Definition	Alternative Names	Unique Value	Data Type	Length	Valid Values	Default Value	Calculation	Required	Business Rule	User Role	Owner Role	Admin Role
CAGE	The CAGE code identified on the drawing,	DAI	No	Alpha numeric	5	May contain only alpha characters and numbers	Null	N/A	Yes	Must be a valid CAGE	View	View, Edit	View, Edit, Delete
Ref Des	Reference designator of an item in a system	Reference Designator	Yes	Alpha numeric	12	May contain only alpha characters and numbers	Null	N/A	No	Must be a reference designation in accordance with ASME Y14.44	View	View, Edit	View, Edit, Delete
Part No	Part number(s) of item being defined by the drawing, model, or list	PN, Reference No., Ref No.	No	Alpha numeric	32	May contain only alpha characters, special characters, and numbers	Null	N/A	Yes	Part number of item being defined in drawing, model, or list. If multiple part numbers are defined, then a single entry must be made for each part number	View	View, Edit	View, Edit, Delete
Part Revision	If part revisioning is used, then indicate revision associated with part	N/A	No	Alpha numeric	15	May contain only alpha characters and numbers	Null	N/A	No	Only required when a revision is part of the part identification number	View	View, Edit	View, Edit, Delete
Part CAGE	CAGE code for the source of supply for the part. This may not be the same as the Design Activity CAGE	Vendor CAGE	No	Alpha numeric	5	May contain only alpha characters and numbers	Null	N/A	Yes	Must be a valid CAGE	View	View, Edit	View, Edit, Delete
Description	Item name or nomenclature (i.e., screw, bolt, cable transmitter)	N/A	No	Alpha numeric	99	May contain only alpha characters and numbers	Null	N/A	Yes	Must be a common noun name, title, or nomenclature	View	View, Edit	View, Edit, Delete
Data Rights	Data Rights category asserted by data developer	DR	No	Alpha	4	May select from the following values: "LR" – Limited Rights; "GPR" – Govt Purpose Rights; "SNLR" –	Null	N/A	Yes	Select from the following: LR, GPR, SNLR, UL, COTS, PGR, or SBIR	View	View, Edit	View, Edit, Delete
Drawing Number	The number used to uniquely identify a TDP element	DWG No	Yes	Alpha numeric	32	May contain only "-" (dash) special character, alpha characters, or numbers	Null	N/A	Yes	Must be in accordance with ASME Y14.100- 2013, Section 6	View	View, Edit	View, Edit, Delete
Drawing Revision	The current revision of a document, drawing, or list	DWG Rev	No	Alpha numeric	3	May contain only "-" (dash) special character or alpha characters	Null	N/A	Yes	The revision assigned to the drawing record must be in accordance with ASME Y14.35	View	View, Edit	View, Edit, Delete
CAGE	The CAGE code identified on the drawing,	DAI	No	Alpha numeric	5	May contain only alpha characters and numbers	Null	N/A	Yes	Must be a valid CAGE	View	View, Edit	View, Edit, Delete
Filename	Drawing, model, list, or document filenames(s). Complete filenames(s) with file extension.	N/A	Yes	Alpha numeric	99	May contain alpha characters, special characters, and numbers	Null	N/A	Yes	Filename for data file of the drawing, model, associated list, or document. Filename for native file defining the item with file extension.	View	View, Edit	View, Edit, Delete
Frame Number	The number of the frame	N/A	No	Numeric	4	May contain only numbers	Null	N/A	No	When drawing or list is cut into multiple frames, this should be used to identify the frame	View	View, Edit	View, Edit, Delete

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PRODUCT MODEL SAMPLE VIEWS

E.1 SCOPE

E.1.1 Scope. 3D product data provides many potential benefits over 2D product data. The 3D model is used in many applications within the engineering community. Some of the benefits include the following:

a. Provides enhanced clarity on product drawings.

b. Supports training requirements and publications.

c. Simplifies and improve effectiveness of technical manuals.

d. Supports shipyard and depot maintenance, repairs, and process illustrations.

e. Improves communication within technical briefs.

Examples of various 3D product models are included in this appendix in figures E-1 through E-5.

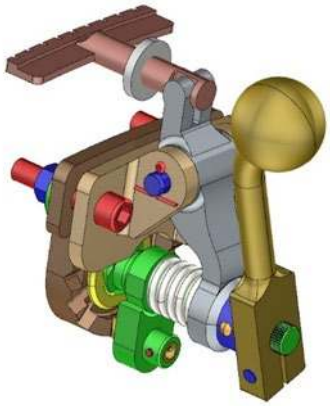


FIGURE E-1. Example of an assembled view 3D model.

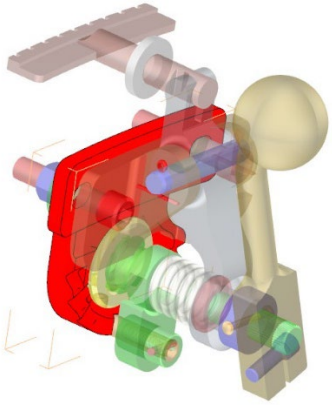


FIGURE E-2. Example of a transparent view 3D model.

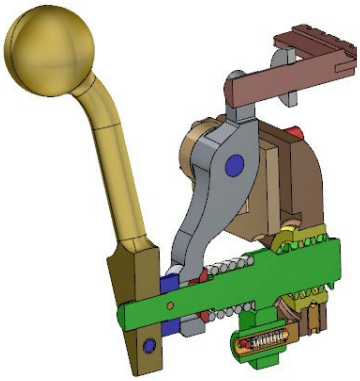


FIGURE E-3. Example of a sectioned view 3D model.

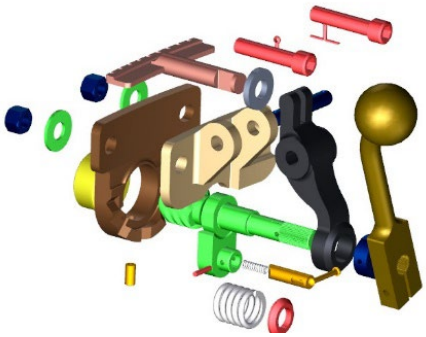


FIGURE E-4. Example of an exploded view 3D model..

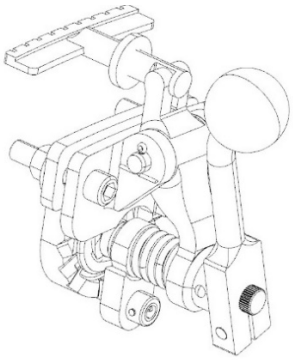


FIGURE E-5. Example of line art view 3D model.

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MODEL-BASED ENTERPRISE GUIDANCE

F.1. SCOPE

F.1.1 Scope. This appendix provides guidance in the model-based enterprise (MBE).

F.2. GUIDANCE

F.2.1 Model-based enterprise. The MBE is a collection of all models that may be used to perform analysis and prediction of various aspects of the acquisition life cycle. The lower-level model-based domains (referred to generically as MBx's), taken collectively form the MBE, which is foundational to implementing DE. Figure F-1 shows the first level decomposition of the MBE.

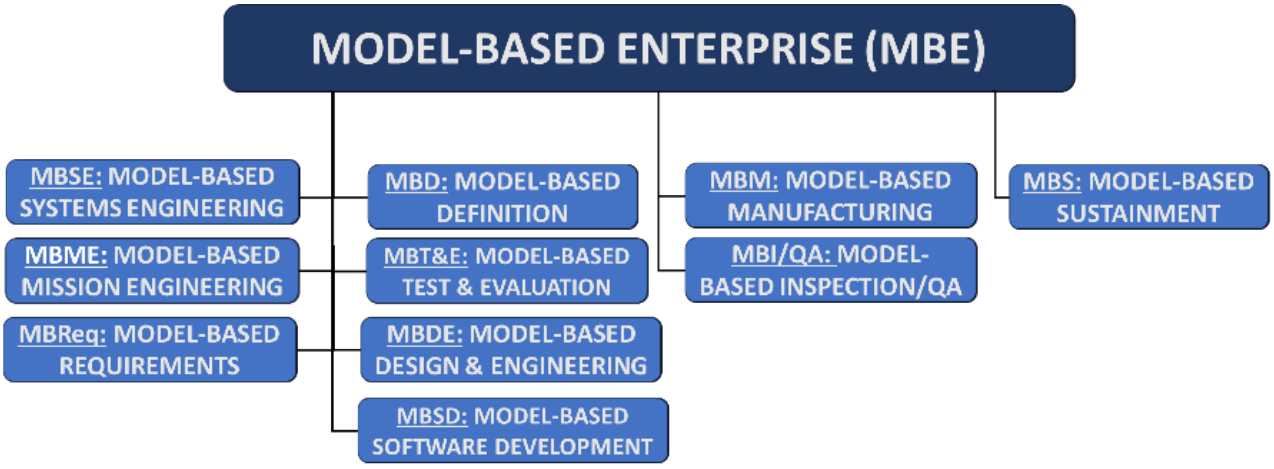


FIGURE F-1. Model-based enterprise.

F.2.2 MBE decomposition and example applications. Table F-I is a further breakdown of the MBE including examples of software applications in each of the MBx areas. This breakdown is notional and not prescriptive. Each organization will need to tailor this MBE breakdown to fit their unique situation.

F.2.2.1 Intended use. The intended use of the MBE decomposition in table F-I is for each organization to identify those MBx areas for which they have primary responsibility. Then perform an analysis of the effectiveness of that MBx capability in their organization. (Use of the template in table F-II and example in table F-III may be used as a guide.)

F.2.2.2 Action. Once gaps in the MBE have been identified, a plan to address those gaps should be developed and executed.

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TABLE F-I. Model-based enterprise decomposition.

Model Based Enterprise Decomposition				
21-Feb-24				
First Level MBx	Second Level MBx	Lower Level MBx	Software App Examples*	Notes
Model Based Enterprise	Model Based Mission Engineering	Model Based Combat Analysis Simulation	CASTFOREM; COMBATXXI; IBM Tational LC; SystemLink	
	Model Based Requirements		CAMEO; PTCModeler	
	Model Based Systems Engineering	Model Based Mission Analysis	CAMEO; PTCModeler	
		Model Based Requirements Analysis	CAMEO; PTCModeler	
		Model Based System Architecture	CAMEO; PTCModeler	
		Model Based Digital Integration		
	Model Based Software Development			
	Model Based Design & Engineering Analysis	Model Based Tolerance Analysis	3DCS Variation Analysis; Sigmetrix VSA; Siemens NX Tolerance Analysis; CETOL 6 Sigma	
		Model Based Stress Analysis/FEA	Ansys; Thermo Analytics;	
		Model Based Computation Fluid Dynamic	Ansys Fluent; SimScale CFD;	
		Model Based Thermal Analysis	Ansys; Thermo Analytics;	
		Model Base Vibration Analysis	VibrationVIEW;enDAQ LAB; FEMtools	
		Model Based Cost Analysis	aPriori	
		Model Based Producibility	DFMPro	
		Model Based Generative Design	CREO Generative Design Extension; Autodesk FUSION360; ANSYS Discovery; APEX Generative Design;	
	Model Based Test & Evaluation		LabVIEW;	
	Model Based Definition		SolidWorks; PTC CREO; Siemens NX; CATIA; SolidEdge; AutoDesk Inventor	
	Model Based Manufacturing	Model Based Process Planning		
		Model Based NC	Vericut; NX CAM	
		Model Based Injection Molding Anaysis	MOLDEX3D; Siemens Femap;	
		Model Based Additive Mfg. Analysis	ANSYS Additive Suite;	
	Model Based Inspection/QA		MBDVidia	
	Model Based Sustainment	Model Based Level of Repair Analysis (LORA)	LORA; BQR apmOptimizer;	
		Model Based Reliability Centered Maintenance (RCM)	ReliaSoft RCM++	
		Model Based Reliability Availability Maintainability Analysis	PANDA	
		Model Based Inventory Analysis	BQR apmOptimizer; OPUS Suite	
		Model Based Life Cycle Cost (LCC)	Cost Analysis Strategy Assessment (CASA);	
		Model Based Supply Chain Analysis	Siemens Supply Chain Suite; Supply Chain Explorer	
* The list of example software applications in no way implies DOD endorsement or recommendation of these tools or the use of these tools in any application.				

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TABLE F-II. Model-based enterprise gap analysis template.

MODEL BASED ENTERPRISE MBx GAP ANALYSIS TEMPLATE			
ORGANIZATION:		**DATE:**	**POC:**
MBx Category:	**RATING** GREEN AMBER RED	**SUB-ORG WITH PRIMARY RESPONSIBILITY FOR THIS MBx:** **COMMENTS:**	
1. Does your organization have modeling software application(s) installed to fulfill this MBx function?			
2. Is the modeling software application widely available to the applicable workforce?			
3. Is there a cadre of subject matter experts in your organization on this software application?			
4. Is there sufficient training available in your organization on this software application?			
5. Are there adequate policies and procedures on the use of this software application(s)?			
6. Are the output artifacts of this modeling software application stored, maintained and managed for reference and future use in a controlled database or PLM system?			
7. Is this software application generally applied in all situations and programs in your organization in cases in which it is appropriate?			

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TABLE F-III. Model-based enterprise gap analysis example.

MODEL BASED ENTERPRISE MBx GAP ANALYSIS TEMPLATE			
ORGANIZATION:		**DATE:**	**POC:**
US ARMY DEVCOM-Armaments Center		March 11, 2024	John Doe, Modeling Chief Coordinator
MBx Category:	**RATING**	**SUB-ORG WITH PRIMARY RESPONSIBILITY FOR THIS MBx:**	
Model Based Defininition using PTC CREO	GREEN	DEVCOM -Armaments Center 3D- Infrastructure Team, Product Data Mgt Division	
	AMBER		
	RED	**COMMENTS:**	
1. Does your organization have modeling software application(s) installed to fulfill this MBx function?		Yes: PTC Creo is the primary Model Based Definition software used in DEVCOM-AC.	
2. Is the modeling software application widely available to the applicable workforce?		Partial: CREO is widely available to the workforce, however lack of license availability hinders many projects.	
3. Is there a cadre of subject matter experts in your organization on this software application?		Yes: PDM division has a 3D Modeling Infrastructure Branch responsible for this software. In addition, throughout DEVCOM-AC there are experts in various aspects of CREO.	
4. Is there sufficient training available in your organization on this software application?		No: Recent loss of training contract means no training is available in basic CREO, Advance Assembly Modeling, Modeling Standards, or Sheetmetal Design.	
5. Are there adequate policies and procedures on the use of this software application(s)?		Yes: There are published policies and procedures on modeling practices.	
6. Are the output artifacts of this modeling software application stored, maintained and managed for reference and future use in a controlled database or PLM system?		Yes: PTC Windchill EDGE system is the primary PLM system for creation, storage and maintenance of CREO related files. (Moving to Windchill ePDM in the near future.)	
7. Is this software application generally applied in all situations and programs in your organization in cases in which it is appropriate?		CREO used extensively for TDP development and model-based definition across most PM customers and programs. However, many legacy products still defined with 2D drawings only (no models).	

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DATA MANAGEMENT EVOLUTION

G.1 SCOPE

G.1.1. Scope. This appendix illustrates the different stages of data management evolution within the DEE.

G.2 DE DATA EVOLUTION OVERVIEW

G.2.1 The six stages. Figure G-1 summarizes the six basic stages of incremental DE data management. These six stages show the progression of moving from paper documents to a full digital environment. Stage 1 is a prime contractor-controlled Technical Data Package (TDP) with 2D paper driven outputs as shown in figure G-2, and the final stage (stage 6) is a Government-controlled TDP with 3D model-based data ownership, as shown in figure G-7 (and as specified in DoDI 5000.93). As a program office progresses through these stages, it can achieve process efficiencies and cost savings with data management. For each subsequent stage in the DE progression, the Government establishes the data type and data format requirements to acquire more authority to view, access, and modify the master data. Each program office should review which stage of digital data provides the best value for their program. Solutions will need to be tailored based on infrastructure and stakeholders, as well as cost and schedule constraints. The data includes the TDP, as specified in MIL-STD-31000, data analytics and simulation, and the development of acquisition data products (e.g., data item descriptions [DID], CDRLs), as specified in MIL-STD-963.

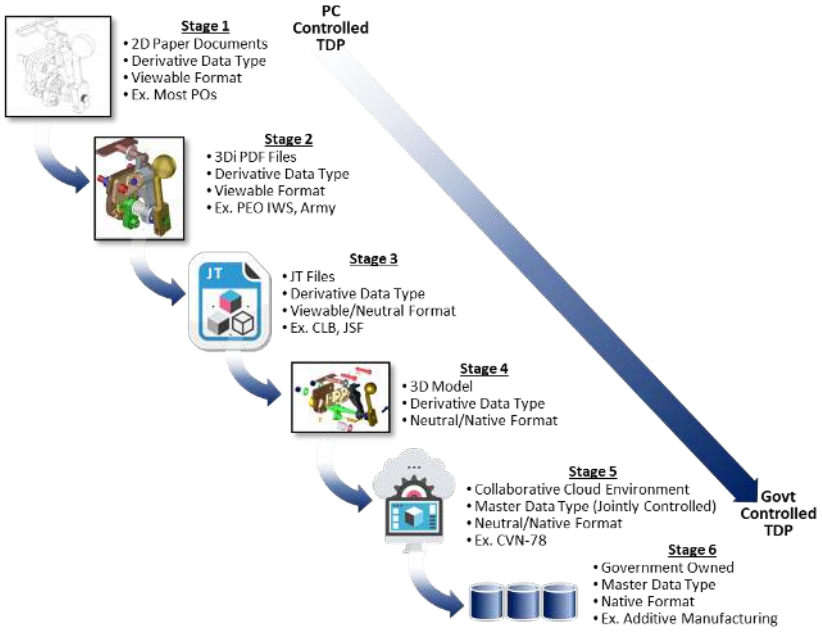


FIGURE G-1. Example of DE data evolution stages.

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G.2.2 Stage assessment. Each program office should periodically assess and identify their current data management approach and DE stage. Incorporation of advanced DE stages has resulted in quantifiable savings to program teams. Cost savings include increased overall program office agility and efficiencies along with more efficient maintenance guidance, development process sheets or production illustrations, and maintenance cards.

G.3 DE DATA EVOLUTION STAGE DETAIL

G.3.1 Data progression. The various stages will have product drawing classifications as specified in ASME Y14.100 Appendix F, Classification Codes for Drawings and Data Sets. Stage 1 refers to a classification code 2 data set with a model and drawing graphics sheet. The 2D drawing will contain master data elements. Stages 2 through 4 refer to a classification code 3 data set with a model and simplified drawing graphics sheet. The 3D model becomes the master source for geometric part attributes. Stages 5 through 6 refer to a classification code 5 data set with a model and no drawing graphic sheet. The 3D model becomes the master source of product data. The applicability of the various data attributes by progressive evolution stages is shown in table G-I.

TABLE G-I. DE data progression by evolution stage.

Data Attribute	Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6
2D Drawings Delivered to Govt.	X	X	X	X	---	---
3D Viewable Drawings Delivered to Govt.	---	X	X	X	X	---
3D Neutral Models Delivered to Govt.	---	---	X	X	X	X
3D Native Models Delivered to Govt.	---	---	---	X	X	X
Master 3D Model Data Managed by the Contractor	X	X	X	X	Partial	---
Master 3D Model Data Jointly Controlled by the Contractor and Govt. Collaboratively	---	---	---	---	X	---
Master 3D Model Data Managed by the Govt.	---	---	---	---	Partial	X
Master 3D Model Data Developed and Managed by the Govt.	---	---	---	---	---	X
ASME Y14.100, Appendix F, Classification Codes and Data Sets	Code 2	Code 3	Code 3	Code 3	Code 5	Code 5

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G.3.2. Stage 1 – data access of traditional 2D documents. Traditional product data approaches have used stage 1. Data delivery of legacy 2D drawings, as seen on figure G-2, is typically contractually invoked between the prime contractor and Government program office. The prime contractor typically has ownership of and manages the master data, while the Government has access as detailed in the data rights section of the agreed contract. The program office should determine the program data rights in the best interest of the Government to maintain configuration and data management.

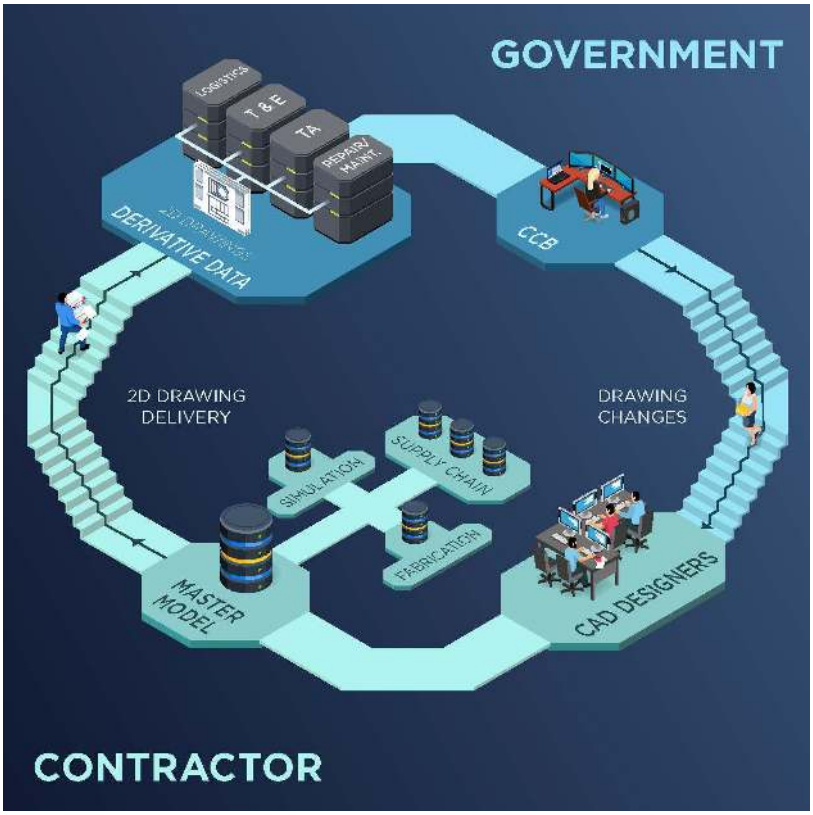


FIGURE G-2. Stage 1 – Data access of traditional 2D documents.

G.3.3 Stage 1 master model. The master model is typically controlled by the prime contractor in their digital environment. To create 3D models configurationally controlled by the prime contractor, derivative data sets pulled from the authoritative product model are made available to the Government in the form of 2D paper drawings. Traditional 2D paper documents, such as 2D assemblies or shells or 2D product illustrations, are transferred to various Government repositories owned by various program offices and Government end users to support logistics and in-service entities. This stage is common in most Government program offices. Requested changes from either side can be built into the existing 3D master model after approval of the configuration control board (CCB). Any proposed changes from the Government side are sent to the Government CCB for approval, and then to the prime contractor CCB. If the change is approved, a prime contractor CAD designer will make the change in the master model. Proposed changes from the prime contractor are either sent to the Government CCB for approval or directed straight to the prime contractor CAD designer, depending on where the technical authority to approve the change lies.

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G.3.4. Stage 2 – data access with 3-dimensional intelligent (3Di) data delivery. Stage 2 includes the delivery of 3Di models that represent the native 3D models that can be viewed on a personal computer. The 3Di models provide a greater capability to the user over the traditional 2D drawings, as shown in figure G-3. The file format used can vary from program office to program office. Derivative viewable 3Di portable document formats (PDF) are generated from a prime contractor managed 3D model. For example, sections of Program Executive Office Integrated Warfare Systems and the Army have transitioned to using 3Di PDFs. Other formats such as the hypertext markup language can also be used.

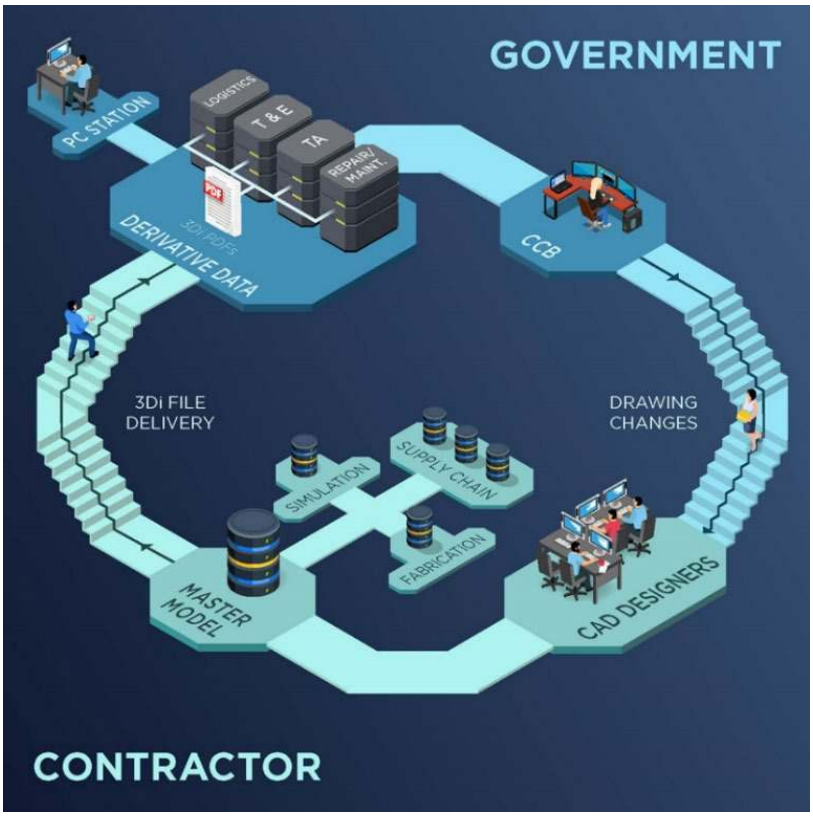


FIGURE G-3. Stage 2 – Data access with 3Di data delivery.

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G.3.5 Stage 3 – data delivery of Jupiter Tessellation (JT) files and 3D models. Stage 3 includes the data delivery of JT files and the 3D models as shown on figure G-4. The 3D model formats are contractually invoked with the prime contractor. Depending on the structure of the contract, the prime contractor may still own and manage the master data. The overall drawing requirements for Government products will be determined by the corresponding drawing deliverable approval procedure. The DID will define the data format of deliverables from the prime contractor to the Government end users, as specified in DoDM 4120.24. Derivative JT files provide data in a viewable and neutral format. Similar to a combination of a 3Di model and a neutral STEP file, JT files are delivered as a 3D model with other details included in a PDF. The JT file is viewable, lightweight, and can also be used in downstream manufacturing.

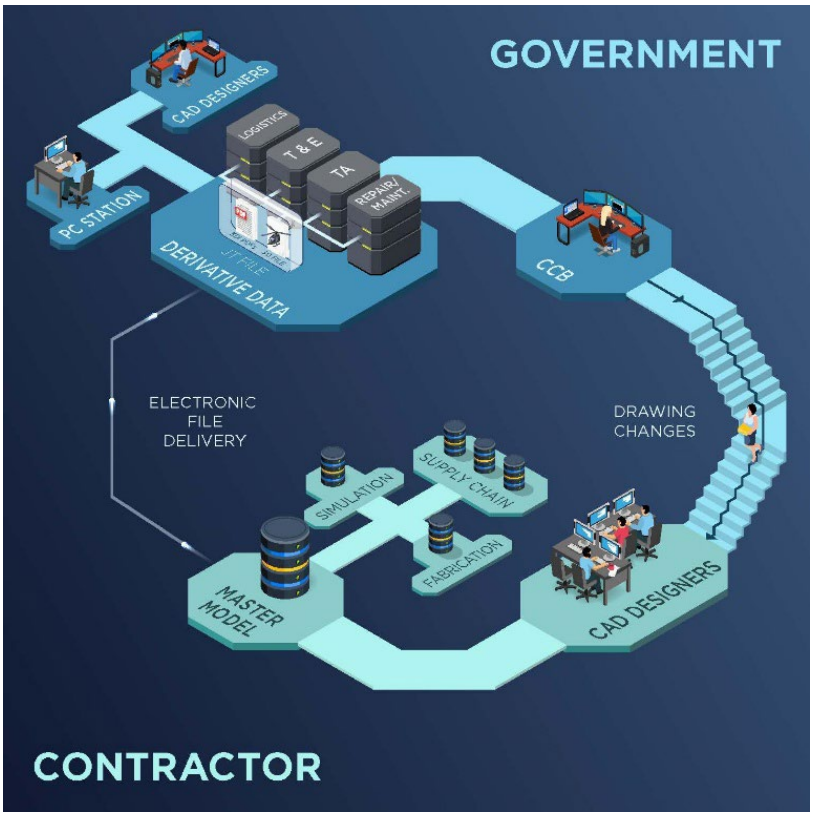


FIGURE G-4. Stage 3 – Data delivery of JT files and 3D models.

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G.3.6 Stage 4 – data delivery of native 3D models. Stage 4 includes the data delivery of the 3D models as shown on figure G-5. The derivative 3D models can be in a neutral or native format. For the Government to manage a master data set, the 3D models should be in a native format.

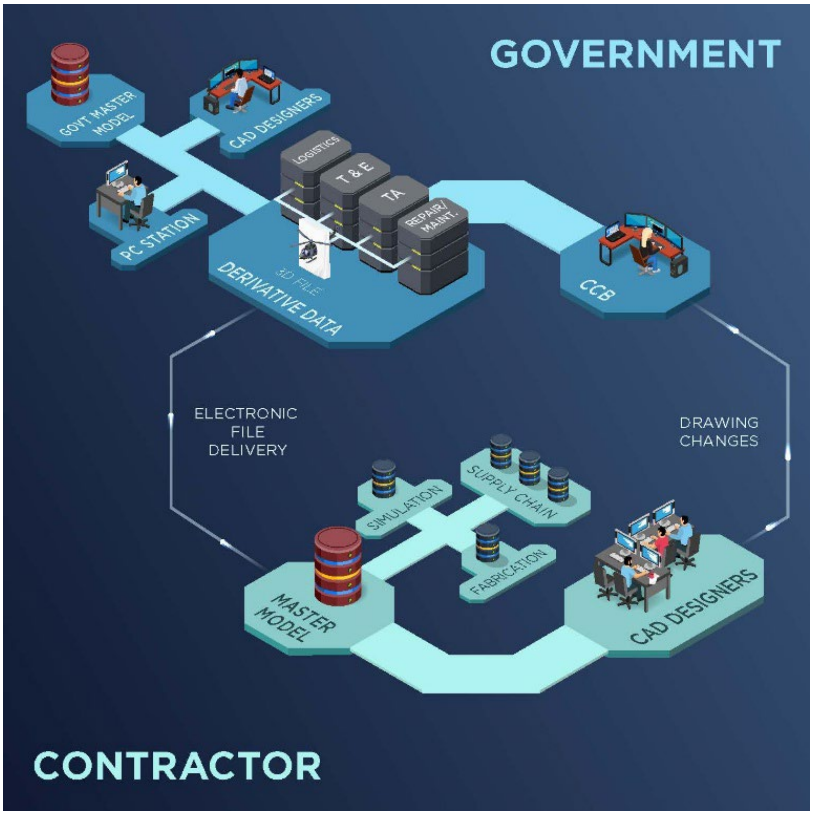


FIGURE G-5. Stage 4 – Data delivery of native 3D models.

G.3.6.1 Master model data sharing. The 3Di PDF, JT, and 3D approach of sharing data from the master model saves costs in comparison to the traditional 2D approach. The production of these model files incurs more upfront costs, but once it is produced it can be used for every function that requires derivative data produced from the 3D model. Use of exclusively 2D data is more expensive because each source will require their own copy produced specifically to their standards. Any changes are then relayed and repeated through several different documents. Drawings are also more frequently lost in the 2D format. By the time each of the sources obtains a copy of the 2D data using the traditional method and all changes have been made, the total time and money spent far exceeds the cost of using the 3Di PDF, JT, or 3D model. Requested changes are still ultimately made by a prime contractor CAD designer.

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G.3.7 Stage 5 – collaborative cloud environment. In stage 5, the master data and 3D models previously owned by the prime contractor are stored and modified in a jointly controlled cloud environment, as shown on figure G-6. Ownership of the cloud environment could vary, but most likely belongs to a third party. Both the Government and the prime contractor have their own CAD designers. The Government CAD designer or Technical Warrant Holder can recommend and redline changes while the prime contractor integration CAD designer can make changes. As an example, Ford class aircraft carriers use this management approach. Changes recommended by Government entities are reviewed by the applicable Technical Warrant Holder, redlined, and put into the collaborative cloud environment, but are not yet incorporated into the master model. The changes are viewed in the cloud environment by the prime contractor CCB. If approved, the prime contractor integration CAD designer builds the changes into the master model.

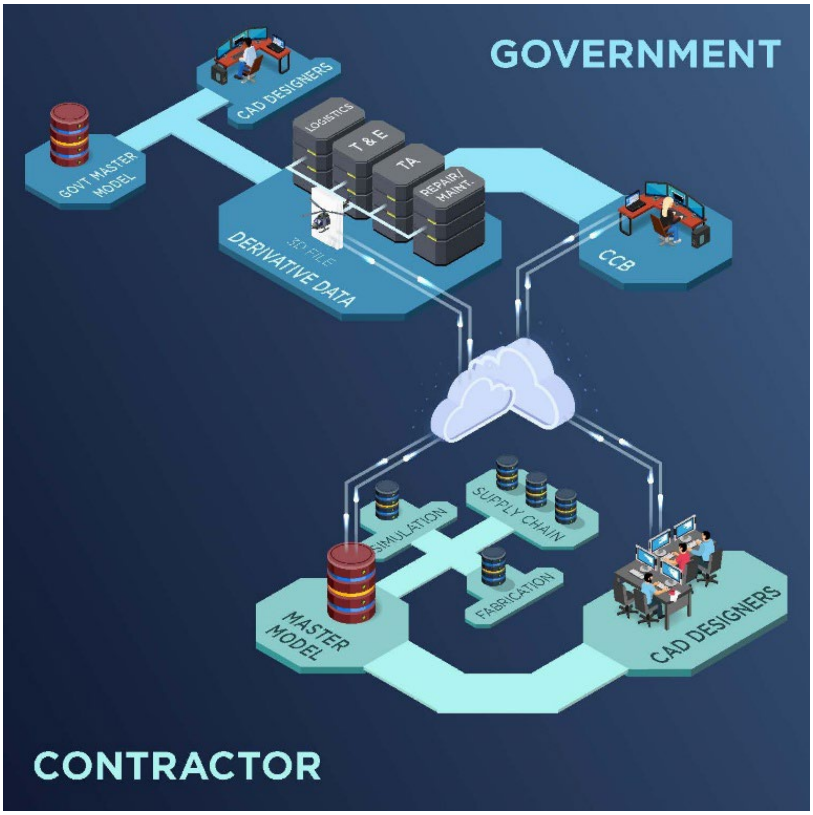


FIGURE G-6. Stage 5 – Collaborative cloud environment.

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G.3.8 Stage 6 – Government-owned master data. In stage 6, the Government ideally would have complete control of program integrated logistics support activities, as seen in figure G-7 below, and the Government owns the master data set. Government CAD designers would manage all master product changes and directly support all Government requirements using a Government repository. This stage offers the greatest benefit to DoW with PLM and sustainment requirements. The increased control and access to product data reduces PLM contracting costs and allows the DoW to respond quickly with evolving threats.

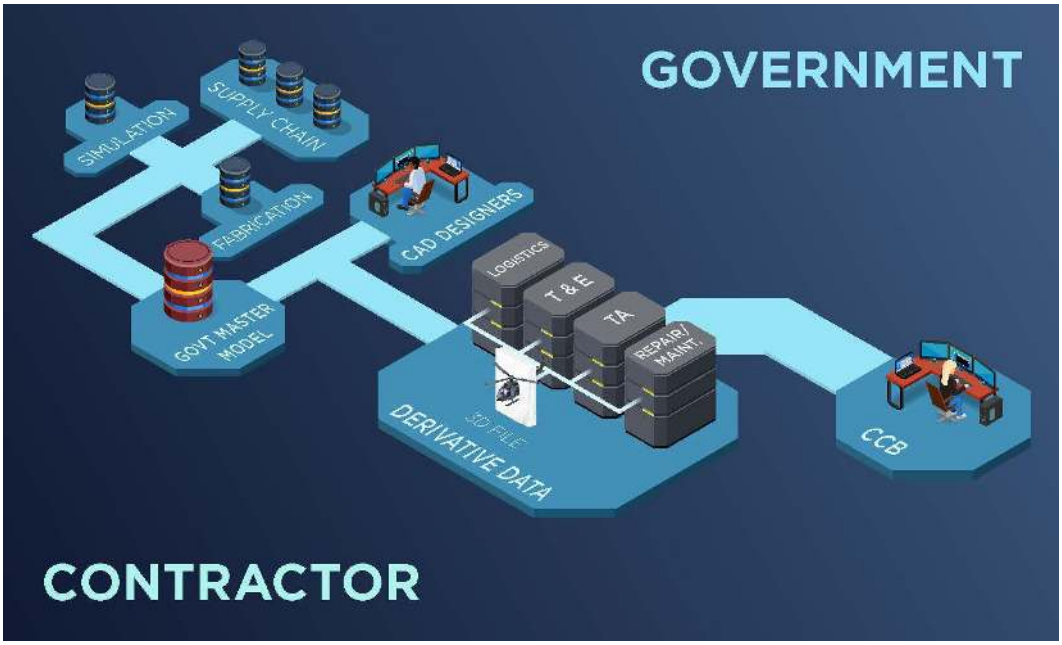


FIGURE G-7. Stage 6 – Government-owned master data.

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APPENDIX H

DATA MANAGEMENT TEMPLATE

H.1 SCOPE.

H.1.1 Scope. This Appendix contains a data management template that can be used to manage data and the data life cycle. Amplifying information is provided in the template introduction section.

Instructions to use this template:

a) Rename attachment to correct filename.

b) Complete header information.

c) Items in << brackets >>replace with project specific information.

d) Paragraphs in *italics* are instructions. Fill in the appropriate information. The final text should be regular font, not italicized.

e) Add classification markings and distribution statements as necessary.

f) Update Table of Contents (Entire Table) and Revisions Table.

g) Delete this instruction box as well as the About This Template section so the Revisions Table starts on the second and the Table of Contents starts on the third page, respectively. The first page should be a title page.

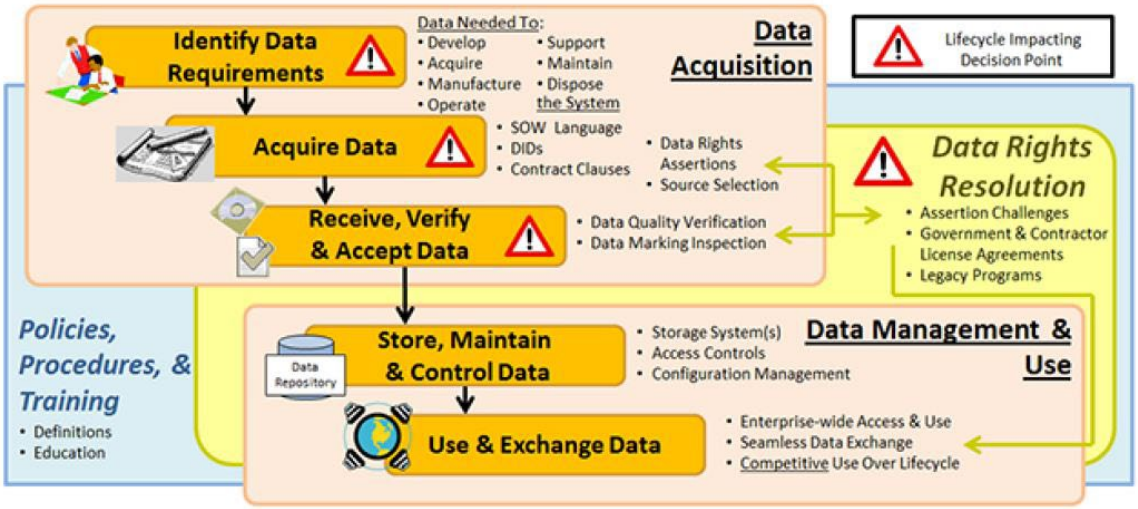
*Note this template provides recommended items that one should consider when putting together a data management plan for your project. Your project may need to have more (or less) items depending on its circumstance.

Revision	Effective Date	Summary
0	DD MMM YYYY	Original issue of the Data Management Plan.

H.1.2. About This Template. This template provides a method, based off the DoD-approved Systems Engineering Guidebook (SEG) December 2021, that contains guidance about

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data and the data life cycle that should be considered. Not all sections of the template will apply to everyone, and as each project is a little different, it is intended for people to at least consider the different sections and use those that apply to their project. If the template is missing an element, the program team can add it to the appropriate section. Every piece of data should be addressed from cradle to grave to result in the best outcomes for traceability and reuse. The flow of this template follows the DoD Systems Engineering Guide outline of data management activities (figure H-1).



Source: DoD SE Guide, Figure 4-12.

FIGURE H-1. Data management activities.

The terms “project” and “program” are interchangeable. If you are using this for a specific project, then use the term project. If you are using this for a family of projects, then use the term program. Also, this template should tailored to the level of your project. If it is a larger project, then it will likely have its own instructions. For a smaller project follow your work center’s guidance. The numbered sections are required.

This template was created in 2024 by a data consortium from the NAVSEA Warfare Centers.

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Data Management Template

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1. Introduction

Provide a brief background, description, status, and any special features of your project related to Data Management.

1.1. Purpose

Briefly describe your project's rationale and requirement for data management activities. Do you have a hard requirement? Are you doing this as a best practice? What do you hope to accomplish? Establish uniform requirements for the project's application of data management principles for data that is developed, acquired, or supported. Briefly describe the project's processes for documenting the approach for data management.

1.2. Scope

Briefly describe the work to be performed and how it will be accomplished and how data management is applied to your project.

1.3. Roles and Responsibilities

Provide a brief description of Data Management responsibilities for project members and other agencies/stakeholders (as applicable on your team). How are they supposed to work together? What resources will you need for Data Management? For example:

- *Project Manager*
- *Technical Authority*
- *Data Manager*
- *Lead Systems Engineer*
- *Configuration Manager*
- *etc.*

1.4. VAULTIS Principles

How does the DMP enact VAULTIS principles of DoD Data Strategy 2020? VAULTIS principles guide how data should be treated guide the further actions taken in the DMP. Many of the questions asked below are based on VAULTIS principles.

- *Make Data Visible – Consumers can locate the needed data.*
    - *Sample statement: Dataset information will be listed in <<insert Approved DoW data catalog link >> i.e. Jupiter data catalog*

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- *Make Data Accessible – Consumers can retrieve the data.*
    - *Sample statement: Dataset request is made through <<insert Approval process link>> i.e. FS form or powerApps*
- *Make Data Understandable – Consumers can recognize the content, context, and applicability.*
    - *Sample statement: Data Dictionary for dataset is found in <<insert data dictionary link>> using in the team's SharePoint folder or shared drive or pull from a contractor SharePoint page.*
- *Make Data Linked – Consumers can exploit data elements through innate relationships.*
    - *Sample statement: Dataset metadata is mapped to <<insert semantic ontology model name>> as defined in <<a knowledge graph or a cameo data modeler>> i.e. Naval Integrated Dictionary*
- *Make Data Trustworthy – Consumers can be confident in all aspects of data for decision-making.*
    - *Sample statement: Dataset is validated through <<insert data curation process information link>> or a data processing workflow diagram*
- *Make Data Interoperable – Consumers have a common representation/ comprehension of data.*
    - *Sample statement: Dataset metadata is connected to Common data model in the following <<insert common data model rules link>> i.e. Signed Common Data Models | Report Catalog | Collibra Data Governance Center (Link: https://catalog.advana.data.mil/domain/840a3221-fef5-4865-b509-b46d4b0c7c14?view=19fb12bc-ff4f-469c-8984-9acc1792bccc)*
- *Make Data Secure – Consumer data is protected from unauthorized use/manipulation.*
    - *Sample statement: Dataset will store in a zero-trust environment <<insert environment name>> and will be encrypted at transit*

1.5. References

List all references mentioned in this Data Management Plan. These can include not only larger DoW and Navy instructions but include documents that are generated by the project such as contract (SOW, DIDs, Data Rights, etc.), configuration management plans, etc. Include any document that says why you are collecting your data.

- *Examples include* DoD Data Strategy 2020, Security Classification Guides, Configuration Management Plan, Project Acquisition Strategy

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2. Acquire and Receive Data

Summarize how is your project going to acquire the data? Are you going to collect it on your own? Are you going to rely on a contractor or another mechanism?

2.1. Data Collection Plan

- *What file formats would the data be generated in? Is it numeric, image, text, modeling, etc. What naming conventions are going to be used?*
    - *If the file format is proprietary, does Dow have the necessary software tool to read the information? Provide instruction if necessary.*
- *From where the source data is obtained? Is there a DID that identifies what a contractor is supposed to provide?*
- *How often is the source data updated?*
- *Does your organization become the data owner after the data is collected of some or all the data?*
- *How will the data be collected/acquired or created? Include the source and, if scientifically pertinent, the time and location of data acquisition.*
- *Does provenance and/or pedigree of the data need to be maintained?*
- *Does the data have to be physically shipped or transferred on networks? How is that going to happen? Can the networks beings used handle the volume of data and reasonable speeds?*
- *What is the anticipated volume of data to be collected, acquired, and/or generated?*
- *[MUST BE ADDRESSED] What is the classification of data that will be collected (e.g., public, CUI, classified, secret, or top secret) and/or will the data have additional handling restrictions (e.g., PII, PHI, or business-sensitive data)? What does the SCG the project is using say?*
- *How will the data be protected in transit? Are you using encryption such as Data at Rest or network?*
- *How will you verify the authenticity of your data upon delivery?*
- *What documentation and metadata will accompany the data? When or how often will these documentations be provided?*
- *What standards will be applied to the metadata format and content?*

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- *Describe any quality assurance and quality control measures to be applied during the collection, analysis, and processing of the data and other assets (documents, samples, etc.).*

- *How are you going to check to make sure that data marking is appropriate? Audit? Spot check?*

- *Have you generated a data card for your data? (Data Cards are structured summaries of essential facts about various aspects of ML datasets needed by stakeholders across a project's life cycle for responsible AI development)*

3. Store, Maintain and Control Data

Now that you have the data, where is it going to go?

3.1. Data Governance

This data governance section serves as a foundational guide for managing and protecting your data holdings. It should outline the policies, standards, and procedures that ensure data is accurate, consistent, secure, and accessible. This section establishes roles and responsibilities, delineates data owner and stewardship practices, and provides a framework for maintaining data integrity throughout its life cycle.

3.2. Data Access and Distribution

- *Which data will be shared?*

- *How will you share the data?*

- *Are any data-sharing restrictions required?*

- *Are there any data sharing agreement between any organizations?*

- *Would release of the data lead to any national security or controlled unclassified information concerns? (***Always include a statement that the data cannot be made available to the public when there are national security or controlled unclassified information concerns. ***)*

- *How are you going to make sure that access controls limit the data to only those who are allowed to see it?*

- *How are you going to protect sensitive data from unauthorized access and sharing?*

- *How will you ensure discoverability of the data by those who may need to use it?*

- *How will access of the data be managed?*

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- *Have you talked with contracts on what language to include in the SOW or agreements so that contractors and their subs, partners, collaborators, or other groups (including other Government groups) protect data? What about NDAs? Or terms of service/agreement?*

- *Will there be any transformation or quality check made to the data before data is accessible?*

3.3. Data Storage, Scalability, Resiliency, Security

- *Describe the enterprise Storage/Backup solutions appropriate for the data classification level. If not using an enterprise solution, provide justification for local storage.*

- *How much data are you going to store?*

- *Can you scale up your storage capacities?*

- *How will data be stored and backed up during the project (e.g., when, where, and who)?*

- *How will security of the data be managed?*

- *Are separate classified and unclassified systems needed?*

- *Will data need to cross classification boundaries? Will a cross-domain solution / guard be needed?*

- *Will there be specific roles for access?*

- *What are the different levels of access people would need?*

3.4. Data Retention

- *Which data is required be retained (i.e., storing data to meet regulatory and recordkeeping obligations) and/or preserved (i.e., the safekeeping of electronically stored information (ESI) for any anticipated legal matter)?*

- *What, if any, are the applicable records management policies?*

- *What is the long-term preservation plan (after the project/program/task period of performance) for the dataset and does this meet the requirements for the type of data?*

- *What is the expected retention duration for data? (or what is the expected data life cycle duration)?*

- *Who is responsible for managing archived data?*

- *Provide justification for the decisions re: preservation such as balancing preservation and associated costs*

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3.5. Disposal of Data

- *How will the data be disposed of? When and how will data be disposed? Are there physical items that must be destroyed?*
- *Who makes the decision to dispose the data? What are the criteria?*

4. Use and Exchange Data

What are you going to do with the data? How will the data be used? Can you create the products you want?

4.1. Data Integration

- *How is the data going to be integrated to so that it is accessible from other systems to create deliverables?*
- *Is there an architecture set up to analyze, visualize, or create deliverables from other systems?*

4.2. Data Representation

Describe: business terms, logical, physical, serialization format(s) (e.g. JSON, XML, CSV), modeling standards, model management, etc.

- *Will the data align with one of the Common Data Model? Provide link to the CDM if available. See Jupiter Data Management Module for example.*
- *Does the metadata match with one or more of the entity types of information within a DoW approved semantic ontology, taxonomy, or glossary? i.e. Naval Data Dictionary System List V1.0. Cite document or link if available.*
- *Are you using Data Identifiers for your data that contains the proper labels in accordance with your local data management policy?*
    - *ID:*
    - *Project Name:*
    - *Project Description/Purpose:*
    - *Author:*
    - *Affiliation:*
    - *Funder:*
    - *Funding Reference Number:*

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- *Project Start date:*
- *Project End date*
- *Collaborating Agencies*
- *Accessibility / Website:*
- *Configuration Management Data (versioning/curation status)*

4.3. Data Aggregation

- *After the data is aggregated and analyzed, what classification (e.g., public, CUI, classified, secret, or top secret) and/or restrictions (e.g., PII, PHI, or business-sensitive data) will the output data include? What does the SCG the project is using say?*

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APPENDIX I

WARFIGHTING ACQUISITION UNIVERSITY LEARNING RESOURCES

I.1 SCOPE

I.1.1 Scope. This appendix provides web links to the learning resources offered by the Warfighting Acquisition University (WarU).

I.1.2 References. Table I-1 includes a web link to the digital acquisition and digital engineering learning assets available from WarU along with links to frequently visited WarU website landing pages including those referenced in this appendix.

I.2. WARU

I.2.1 Background. In late 2025 the DoW transitioned the Defense Acquisition University's name to Warfighting Acquisition University. The WarU is the DoW's primary institution for training and educating acquisition professionals. WarU provides learning opportunities, resources, and tools that empower the acquisition workforce to deliver effective and affordable solutions to the warfighter.

I.2.2 Eligibility. WarU courses and other activities are available at no cost to military personnel and government employees, and, with some limitations, to government contractors supporting the DoW. Participants earn Continuous Learning Points upon successful completion of eligible courses and other activities.

I.2.3. WarU user account. Users must establish a WarU account to gain access to most learning content. See https://www.waru.edu/help-center/faq/user-account-setup.

I.2.4 Format. Most WarU courses are offered online through the https://www.waru.edu/ (see also table I-1).

I.3 WARU WEBSITES

1.3.1 WarU website. WarU offers learning resources to support all facets of acquisition. See also Table I-1.

I.3.2. Digital Acquisition learning assets website. WarU provides an overview of WarU digital acquisition and digital engineering courses, credentials, playlists, workshops, and labs. Each entry includes a link to the WarU registration site for that activity. See: https://www.waru.edu/digital-acquisition

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TABLE I-1. WarU Learning Resource Web Pages

Learning Area	Page	URL
WarU account	User Account Setup	https://www.waru.edu/help-center/faq/user-account-setup
Articulate	WarU Digital Acquisition Credentials, Courses, Workshops, Playlists and Guided Projects/Labs	https://www.waru.edu/digital-acquisition
Course catalog	WarU iCatalog	https://icatalog.waru.edu/
Credentials	Credentials Home	https://icatalog.waru.edu/onlinecatalog/tabnav_credentials.aspx.
Credentials	Defense Acquisition Credentials	https://www.waru.edu/credentials.
DAWIA certification	DAWIA Career Field Certifications	https://www.waru.edu/help-center/faq/dawia-career-field-certifications
DAWIA certification	iCatalog Courses and Schedules	https://icatalog.waru.edu/onlinecatalog/CareerLvl.aspx
Commercial course providers	Commercial Learning Opportunities	https://www.waru.edu/partnerships/commercial-learning-opportunities
Games and simulations	Games and Simulations (G&S)	https://www.waru.edu/playlists/digital-acquisition-guided-projects/labs-playlist-0
Playlists	Learning Playlists	https://www.waru.edu/playlists
Webinars and web events	WarU Web and Other Events	https://www.waru.edu/events
Workshops	Mission Assistance Workshops	https://icatalog.waru.edu/onlinecatalog/targeted_training.aspx .

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CONCLUDING MATERIAL

Custodians

Army – AR

Navy – SH

Air Force – 01

DLA – DH

OSD – SE

Preparing activity

Army - AR

(Project SESS-2026-022)

Review activities

Army – AC, AV, CR, MI, PT, SY, TE, TM

Navy – AS, CG, CH, EC, MC, OS, SA, NM

Air Force – 08, 10, 11, 16, 19, 24, 33, 70, 84, 94

DLA – CC, CQ, CT, GS

DISA – DC1, DC5

MDA – MDA

NSA – NS

OSD – DMS, MA, RS, SO

USTC – USTC

NOTE: The activities listed above were interested in this document as of the date of publication. To confirm the latest version, please see the ASSIST online database at https://assist.dla.mil.

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